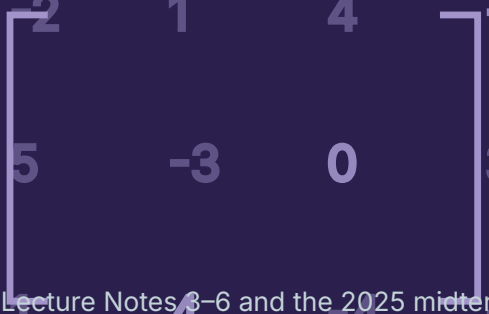
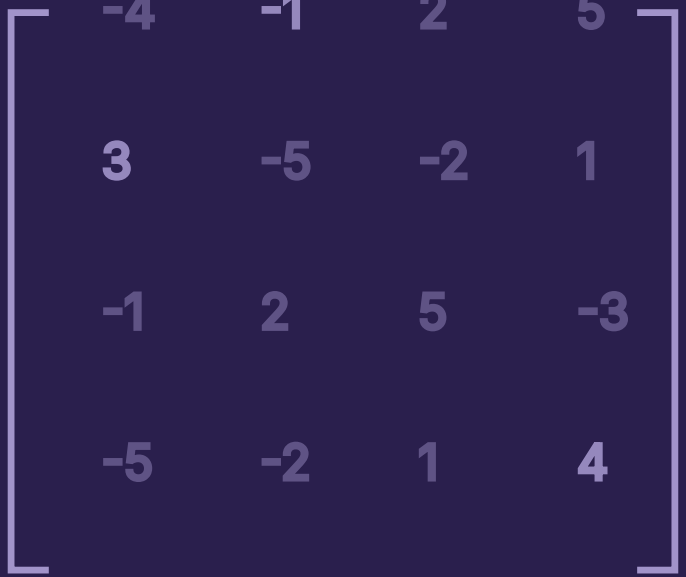
MAT 2037 · EXAM GUIDE

Linear Algebra, row by row

Linear systems, Gaussian elimination, rank, matrix operations and inverses explained step by step, with the 2025 midterm fully solved, a cheat sheet and practice tests.







Covers Lecture Notes 3–6 and the 2025 midterm · Practise online: brtzl.github.io/study-shelf

What's inside

1 Linear equations and their solutions 5
What a linear equation in n variables is, what it looks like geometrically (a hyperplane), and what it means to 'solve' a system: find every solution and write them all in one parametric formula.

2 Elementary row operations 9
The three moves you are allowed to make on a system without changing its solutions, a full worked example with four variables, free and pivot variables, and Theorem 1 with its proof.

3 Matrix notation and the augmented matrix 13
Stop copying x's: write a system as Ax = b, keep only the numbers in an augmented matrix [A|b], and do the row operations on that. Includes the slide 28 example fully solved.

4 Echelon and reduced echelon forms 18
The 'staircase' shape that makes a system easy to read, its stricter cousin the reduced echelon form, and how to spot pivots, pivot variables and free variables. Every 'Why?' and every dotted blank from the slides is answered here.

5 Gaussian elimination 24
One recipe that solves every linear system: reduce the augmented matrix to echelon form, then read off 'one solution', 'infinitely many' (in parametric vector form) or 'none'. Ends with the exam's favourite use of it: is b a linear combination of u, v, w?

6 Two ways to read Ax = b 30
The same system can be read row by row (intersecting hyperplanes) or column by column (a linear combination). After this chapter you can turn a 'is b in the span?' question into a system and settle it with one row reduction.

7 Consistent systems and constraint equations 35
When does Ax = b have a solution? Row-reduce once with a symbolic b, and the zero rows hand you the answer as equations in b1, ..., bm. This is midterm Question 3, and after this chapter you can do it on autopilot.

8 The rank of a matrix 44
One number, the rank, counts the pivots. Compare the rank of A with the rank of [A|b] and you know whether a system has a solution without solving it (the Rouché–Capelli theorem).

9 Homogeneous and inhomogeneous systems 48
Every system Ax = b has a twin with zero on the right, Ax = 0. Solve the twin, find one solution of the original, add them, and you have every solution. You will also learn to spot a homogeneous system in one glance.

10 Uniqueness, and how to classify any system 53
None, exactly one, or infinitely many: three numbers decide it, rank A, rank [A|b] and n. This chapter gives you the one-glance table and a fixed four-line answer for midterm Question 2, plus the nonsingular-matrix theorem.

11 Matrices, sums and scalar multiples 59

What an m×n matrix is, how to name its entries, rows and columns, the special square shapes (diagonal, upper and lower triangular), and the two easy operations: adding matrices and multiplying them by a number.

12 Matrix multiplication 65
The one operation that is not entry by entry. Learn the row-times-column rule, the size check that tells you in two seconds whether a product exists, the identity matrix, the rules that do hold, and the classic traps (AB is not BA, and friends).

13 Matrices as functions 72
Every m×n matrix A gives a function that sends a vector x with n entries to the vector Ax with m entries. Multiplying matrices is then just doing one function after another. That one idea explains associativity and solves the slide exercises, and it leads straight to midterm Q4(b): when is this function onto?

14 Right and left inverses 77
For numbers, 5 has the inverse 1/5. Matrices can be rectangular and AB is usually not BA, so there are two one-sided kinds of inverse. After this chapter you can tell them apart, use them to solve Ax = b, and say exactly what each one guarantees.

15 Invertible matrices 83
A square matrix is invertible when one matrix B undoes it from both sides. You will learn the definition word for word, why invertible and nonsingular mean the same thing (Theorems T1 and T2), and the short proofs the exam asks for, such as $BA = I \Rightarrow B = A^{-1}$ and $(AB)^{-1} = B^{-1}A^{-1}$.

16 Computing A^{-1} by Gauss–Jordan elimination 89
One procedure does everything: write $[A \mid I]$, row-reduce until the left block is I, and read A^{-1} off the right. If a zero row shows up on the left instead, A is not invertible. At the end, the same idea finds right inverses of wide matrices.

17 The exam playbook: five question types 100
The 2025 midterm had five questions, and each one is a type that can come back with new numbers. For each type you get what it looks like, a fixed recipe, a fresh worked example, the answer layout to copy onto your paper, and the usual mistakes. The last part shows how to write a complete proof at this level.

18 The 2025 midterm, fully solved 110
The real MAT 2037 midterm of 05.11.2025: five questions, 20 points each. For each one you get how to recognise it, a clean solution that follows the official typed solution, the gaps in that solution filled in, and the traps that cost points.

19 Cheat sheet 120
Two dense pages to print and review before the exam

20 Self-test 123
30 multiple-choice questions and one full mock midterm

21 Answers 130
With a short explanation for every question

How to use this guide

Each chapter follows the lecture notes in order: the idea in plain words, the definition or theorem, the lecturer's examples worked step by step (with every blank on the slides filled in), then the traps.

Colored box	What it tells you
! Red	A trap: where students lose points.
★ Yellow	A tip or memory trick.
→ Teal	A key idea worth pausing on.
✓ Purple	Learn this: very likely on the exam.
i Grey	A side note, including places where the slides or official solutions have a small slip.

The midterm, in one picture

The 2025 midterm had 5 questions × 20 points, one per question type below. Chapter 17 ("The exam playbook") teaches a fixed recipe for each type, and Chapter 18 solves the real exam.

Question	Type	Chapters
1	Is b in the span? Find the coefficients	3–5
2	Classify systems: rank, pivots, number of solutions	8–10
3	Constraint equation for consistency	7
4	Short proofs (inverses, onto)	13–15
5	Inverse by Gauss–Jordan	16

A plan if you have a week

Day	Read	Then do
1	Chapters 1–5	Practice: Linear equations, Row operations, Echelon forms.
2	Chapters 6–8	Find two constraint equations by hand.
3	Chapters 9–10	Practice: Rank, Homogeneous, Uniqueness.
4	Chapters 11–13	Multiply three pairs of matrices by hand.
5	Chapters 14–16	Invert a 3×3 by Gauss–Jordan, then check both products.
6	Chapter 17 + cheat sheet	Mock midterm 1 on the web page, timed.
7	Chapter 18	Real 2025 midterm, timed, then compare. Then mock midterms 2–4.

→ **The web page: brtzl.github.io/study-shelf**

Every question here plus more, timed mock midterms with step-by-step solutions, and flashcards are at **https://brtzl.github.io/study-shelf/**. It works on a phone and saves your progress.

CHAPTER 1

Linear equations and their solutions

What a linear equation in n variables is, what it looks like geometrically (a hyperplane), and what it means to 'solve' a system: find every solution and write them all in one parametric formula.

From Lecture 3, pages 3–8

One linear equation

A linear equation is the simplest kind of equation there is: every variable appears only to the first power, multiplied by a fixed number. No x_1x_2, no x_1^2, no $\sin x_3$.

Linear equation in n variables, p.3

A **linear equation** in n variables $x_1, x_2, \ldots, x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where the coefficients $a_1, a_2, \ldots, a_n, b$ are fixed real numbers.

The left side is exactly a dot product. If we collect the coefficients into one vector and the variables into another, the equation becomes very short:

$$\mathbf{a} \cdot \mathbf{x} = b, \qquad \mathbf{a} = (a_1, a_2, \ldots, a_n), \quad \mathbf{x} = (x_1, x_2, \ldots, x_n)$$

VECTOR FORM OF ONE LINEAR EQUATION (P.3)

Geometry. This equation is the equation of a **hyperplane in** $\mathbb{R}^n$. A vector $\mathbf{x}$ is a solution of the equation if and only if the point $\mathbf{x}$ lies on that hyperplane.

- In $\mathbb{R}^2$ a "hyperplane" is a **line**: $x_1 + x_2 = 3$.
- In $\mathbb{R}^3$ it is an ordinary **plane**: $x_1 + 2x_2 - x_3 = 2$.
- In $\mathbb{R}^4$ and higher we cannot draw it, but the word "hyperplane" still means "the solution set of one linear equation".

The normal vector

The coefficient vector $\mathbf{a}$ is perpendicular (normal) to the hyperplane $\mathbf{a} \cdot \mathbf{x} = b$. This is why, in Ex. 1 below, the direction of the line of intersection is orthogonal to both coefficient vectors.

A system of m equations in n variables

A **system of m linear equations in n variables** is just several such equations that must hold at the same time (p.4):

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned}$$

Read the double index as an address: a_{ij} is the coefficient in **equation** i of the **variable** x_j. So a_{23} is the coefficient of x_3 in the second equation. The notation looks heavy, but it is the standard way to talk about every system at once.

Solution of a system, p.5

A **solution** $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is an n-tuple of real numbers that satisfies **all** m **equations** of the system.

Each equation is a hyperplane, and a solution must lie on every one of them. Therefore **a solution corresponds to a point in the intersection of the m hyperplanes** defined by the equations.

All equations, not one

A point that satisfies the first equation but not the second is **not** a solution. When you check an answer, substitute it into *every* equation.

The goal: the general solution

Solving a system does not mean "find one solution". It means describing **all** of them.

General solution, p.7

To solve a system of linear equations, our goal is to provide a complete **parametric description** of the solutions. This description is called the **general solution** of the system.

EX. 1 · LECTURE 3 P.6 (THE DOTTED LINES FILLED IN) Find a parametric description of the line of intersection of the two planes in $\mathbb{R}^3$:

$$x_1 + 2x_2 - x_3 = 2 \qquad \text{and} \qquad x_1 - x_2 + 2x_3 = 5.$$

1. Eliminate x_1: subtract the first equation from the second. $(x_1 - x_1 = 0)$

$$(x_1 - x_2 + 2x_3) - (x_1 + 2x_2 - x_3) = 5 - 2 \implies -3x_2 + 3x_3 = 3.$$

2. Divide by -3: $x_2 - x_3 = -1$, so $x_2 = -1 + x_3$.
3. Put this into the first equation: $x_1 = 2 - 2x_2 + x_3 = 2 - 2(-1 + x_3) + x_3 = 4 - x_3$.

4. Nothing restricts x_3: it can be any real number. It is the **free parameter**.
5. Collect everything into one vector and split it into "constant part + x_3·(direction)":

$$\mathbf{x} = (x_1, x_2, x_3) = (4 - x_3,\ -1 + x_3,\ x_3) = (4, -1, 0) + x_3(-1, 1, 1).$$

6. Check the geometry: $(-1,1,1)\cdot(1,2,-1) = -1+2-1 = 0$ and $(-1,1,1)\cdot(1,-1,2) = -1-1+2 = 0$. (the direction of the line is orthogonal to both normal vectors $a = (1,2,-1)$, $b = (1,-1,2)$, as the slide remarks)

ANSWER
$\mathbf{x} = (4,-1,0) + x_3(-1,1,1)$, $x_3 \in \mathbb{R}$ a free parameter: a line through $(4,-1,0)$ with direction $(-1,1,1)$.

Always check one point
Put $x_3 = 0$: $(4,-1,0)$. First equation: $4-2-0 = 2$ ✓. Second: $4+1+0 = 5$ ✓. Ten seconds, and you know your constant part is right.

EX. 2 · LECTURE 3 P.8 (THE DOTTED LINES FILLED IN) Solve $x_1 + x_2 = 3, \quad 2x_1 - x_2 = 1.$

1. Add the two equations: $(x_1 + x_2) + (2x_1 - x_2) = 3 + 1$, so $3x_1 = 4$. (x_2 cancels)
2. $x_1 = \frac{4}{3}$.
3. From the first equation: $x_2 = 3 - x_1 = 3 - \frac{4}{3} = \frac{5}{3}$.
4. Check in the second: $2\cdot\frac{4}{3} - \frac{5}{3} = \frac{8}{3} - \frac{5}{3} = 1$ ✓.

ANSWER
The unique solution is $\mathbf{x} = \left(\frac{4}{3}, \frac{5}{3}\right)$, the point of intersection of the two lines defined by these equations.

Three possible pictures
Two lines in the plane can meet in one point (Ex. 2), be the same line (infinitely many solutions) or be parallel (no solution). Two planes in $\mathbb{R}^3$ usually meet in a line (Ex. 1). The same three outcomes, **exactly one, infinitely many, none**, are the only ones possible for any linear system; Chapter 5 shows how to tell which one you have.

In one minute

- Linear equation: $a_1x_1 + \cdots + a_nx_n = b$ with fixed real a_i, b; in vector form $\mathbf{a}\cdot\mathbf{x} = b$.
- Its solution set is a hyperplane in $\mathbb{R}^n$ (line in $\mathbb{R}^2$, plane in $\mathbb{R}^3$); $\mathbf{a}$ is normal to it.
- A solution of a system satisfies **all** m equations, i.e. lies in the intersection of the m hyperplanes.
- a_{ij} = coefficient of x_j in equation i.
- General solution = complete parametric description of all solutions.

- Ex. 1: $\mathbf{x} = (4, -1, 0) + x_3(-1, 1, 1)$, x_3 free. Ex. 2: unique solution $\left(\frac{4}{3}, \frac{5}{3}\right)$.

CHAPTER 2

Elementary row operations

The three moves you are allowed to make on a system without changing its solutions, a full worked example with four variables, free and pivot variables, and Theorem 1 with its proof.

From Lecture 3, pages 9–14 and 29–32

The three allowed moves

When you solved Ex. 2 in Chapter 1 you added two equations together. That is one of only three "safe" moves. Safe means: the new system has **exactly the same solutions** as the old one.

Elementary row operations, p.9

There are three basic algebraic operations that can be performed on a system without affecting the solution set. They are called **elementary row operations**:

- (i) **Interchange** (or swap) any pair of equations.
- (ii) **Multiply** any equation by a **nonzero** real number.
- (iii) **Replace** any equation by its sum with a multiple of another equation.

In this guide we write them in the usual short notation (R_i = row/equation i):

Type	Notation	Example
(i) swap	$R_i \leftrightarrow R_j$	$R_1 \leftrightarrow R_2$
(ii) scale	$R_i \to rR_i,\ r \neq 0$	$R_1 \to \frac{1}{2}R_1$
(iii) replace	$R_i \to R_i + rR_j,\ j \neq i$	$R_2 \to R_2 - 3R_1$

Two classic slips

- Multiplying by 0 is **not** allowed: it would turn an equation into $0 = 0$ and throw information away.
- In (iii) the row you **replace** is the one on the left of the arrow. $R_2 \to R_2 - 3R_1$ changes row 2; row 1 stays exactly as it was. Never write $R_2 \to 3R_1 - R_2$ and call it one step of type (iii) (that is a (ii) and a (iii) together).

Worked example: two equations, four variables

EXAMPLE · LECTURE 3 P.10–13 (BLANK ON P.12 FILLED IN) Solve

$$3x_1 - 2x_2 + 2x_3 + 9x_4 = 4$$
$$2x_1 + 2x_2 - 2x_3 - 4x_4 = 6$$

This system has **four variables** but only **two equations**.

1. Operation (i), swap the equations: (the second one has even coefficients, easy to halve)

$$2x_1 + 2x_2 - 2x_3 - 4x_4 = 6$$
$$3x_1 - 2x_2 + 2x_3 + 9x_4 = 4$$

2. Operation (ii), multiply the first equation by $\frac{1}{2}$:

$$x_1 + x_2 - x_3 - 2x_4 = 3$$
$$3x_1 - 2x_2 + 2x_3 + 9x_4 = 4$$

3. Operation (iii), add -3 times the first equation to the second: ($3x_1 - 3x_1 = 0$; $-2 - 3 = -5$; $2 + 3 = 5$; $9 + 6 = 15$; $4 - 9 = -5$)

$$x_1 + x_2 - x_3 - 2x_4 = 3$$
$$-5x_2 + 5x_3 + 15x_4 = -5$$

4. Operation **(ii)**, multiply the second equation by $-\frac{1}{5}$: (this is the "(...)" on p.12: multiplying by a nonzero number is type (ii))

$$x_1 + x_2 - x_3 - 2x_4 = 3$$
$$x_2 - x_3 - 3x_4 = 1$$

5. Operation (iii), add -1 times the second equation to the first:

$$x_1 + x_4 = 2$$
$$x_2 - x_3 - 3x_4 = 1$$

6. Solve each equation for its first variable and let the others be themselves:

$$x_1 = 2 - x_4, \quad x_2 = 1 + x_3 + 3x_4, \quad x_3 = x_3, \quad x_4 = x_4.$$

7. Write as a vector and split by x_3 and x_4:

$$\mathbf{x} = \begin{bmatrix} 2 - x_4 \\ 1 + x_3 + 3x_4 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} -1 \\ 3 \\ 0 \\ 1 \end{bmatrix}.$$

ANSWER

$\mathbf{x} = (2, 1, 0, 0) + x_3(0, 1, 1, 0) + x_4(-1, 3, 0, 1)$, $x_3, x_4 \in \mathbb{R}$: a plane in $\mathbb{R}^4$, with x_3 and x_4 **free parameters**.

How to split into vectors without mistakes

Write the column of unknowns x_1, x_2, x_3, x_4 with each pivot variable replaced by its formula. Then read it three times: once keeping only the plain numbers, once keeping only the x_3-coefficients, once only the x_4-coefficients. A free variable gets a 1 in its own slot and 0 in the other free slots.

Free and pivot variables

Free and pivot variables, p.14

In a system of linear equations, the variables that are allowed to vary freely (as parameters) are called **free variables**, because they can take any real value. The remaining variables, which can be expressed in terms of the free variables, are called **pivot variables**, as they depend on the free variables.

For example (the "..." on p.14): in the example above, x_1 and x_2 are the **pivot variables** ($x_1 = 2 - x_4$, $x_2 = 1 + x_3 + 3x_4$) and x_3, x_4 are the **free variables**. Two free variables give two parameters, which is why the solution set is a plane.

Count the parameters

Number of free variables = number of parameters in the general solution. 0 free: a single point. 1 free: a line. 2 free: a plane.

Theorem 1: row operations do not change the solutions

Theorem 1, p.29

If a system of equations $A\mathbf{x} = \mathbf{b}$ is transformed into a new system $C\mathbf{x} = \mathbf{d}$ by elementary row operations (i), (ii), (iii), then both systems have **the same set of solutions**.

($A\mathbf{x} = \mathbf{b}$ is the matrix way of writing the system; see Chapter 3. Here you only need: $\mathbf{A}_i$ is the i-th row of coefficients, and equation i reads $\mathbf{A}_i \cdot \mathbf{x} = b_i$.)

Proof (p.29–32). We show that every solution of $A\mathbf{x} = \mathbf{b}$ is also a solution of $C\mathbf{x} = \mathbf{d}$, and vice versa. Let $\mathbf{u}$ be a solution of $A\mathbf{x} = \mathbf{b}$. Denoting the rows of A by $\mathbf{A}_1, \ldots, \mathbf{A}_m$:

$$\mathbf{A}_1 \cdot \mathbf{u} = b_1, \quad \mathbf{A}_2 \cdot \mathbf{u} = b_2, \quad \ldots, \quad \mathbf{A}_m \cdot \mathbf{u} = b_m.$$

- **Case 1, type (i).** Swapping two rows only changes the order of the list. $\mathbf{u}$ still satisfies the same set of equations.
- **Case 2, type (ii).** Multiply the k-th equation by $r \neq 0$. Since $\mathbf{A}_k \cdot \mathbf{u} = b_k$, also $(r\mathbf{A}_k) \cdot \mathbf{u} = r(\mathbf{A}_k \cdot \mathbf{u}) = rb_k$.
- **Case 3, type (iii).** Add r times the k-th equation to the l-th. We know $\mathbf{A}_k \cdot \mathbf{u} = b_k$ and $\mathbf{A}_l \cdot \mathbf{u} = b_l$, so the new l-th equation holds:

$$(r\mathbf{A}_k + \mathbf{A}_l) \cdot \mathbf{u} = r(\mathbf{A}_k \cdot \mathbf{u}) + \mathbf{A}_l \cdot \mathbf{u} = rb_k + b_l.$$

- **Converse.** Suppose $\mathbf{u}$ satisfies $C\mathbf{x} = \mathbf{d}$. The inverse of any elementary row operation is again an elementary row operation, so the reasoning above can be reversed: $\mathbf{u}$ also satisfies $A\mathbf{x} = \mathbf{b}$. ■

Operation	Its inverse (undoes it)	Same type?
$R_i \leftrightarrow R_j$	$R_i \leftrightarrow R_j$ again	(i)
$R_k \to rR_k$, $r \neq 0$	$R_k \to \frac{1}{r}R_k$	(ii)
$R_l \to R_l + rR_k$	$R_l \to R_l - rR_k$	(iii)

Note on p.32: why r must be nonzero

It is important that $r \neq 0$ in an operation of type (ii), as multiplying by zero would trivialize the equation. (And $\frac{1}{r}$, the inverse step, would not exist.)

Writing the proof in the exam

Do not stop after showing "solutions of the old system solve the new one". That is only half. The second half (converse) uses the fact that each operation can be **undone** by an operation of the same kind.

In one minute

- (i) swap two equations; (ii) multiply one by a **nonzero** number; (iii) add a multiple of another equation.
- Notation: $R_1 \leftrightarrow R_2$, $R_1 \to \frac{1}{2}R_1$, $R_2 \to R_2 - 3R_1$.
- Worked example: $\mathbf{x} = (2,1,0,0) + x_3(0,1,1,0) + x_4(-1,3,0,1)$, a plane in $\mathbb{R}^4$.
- Pivot variables are expressed through the free ones; free variables can be any real number.
- Theorem 1: row operations (i)–(iii) keep the solution set the same.
- Proof: check each type for a solution $\mathbf{u}$, then reverse using inverse operations; $r \neq 0$ is essential.

CHAPTER 3

Matrix notation and the augmented matrix

Stop copying x's: write a system as Ax = b, keep only the numbers in an augmented matrix [A|b], and do the row operations on that. Includes the slide 28 example fully solved.

From Lecture 3, pages 15–28

Why matrices?

In Chapter 2 we rewrote x_1, x_2, x_3, x_4 in every single step, although only the **numbers** changed. Matrix notation keeps the numbers and drops the rest. Same system, same operations, much less writing.

Start from the general system of m equations in n unknowns (p.15):

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m. \end{aligned}$$

Row vectors: each equation is a dot product

Put the coefficients of equation i into a **row vector** $A_i = (a_{i1}, a_{i2}, \ldots, a_{in}) \in \mathbb{R}^n$. Then equation i is simply $A_i \cdot x = b_i$, and the system becomes (p.16)

$$A_1 \cdot x = b_1, \quad A_2 \cdot x = b_2, \quad \ldots, \quad A_m \cdot x = b_m, \qquad x = (x_1, \ldots, x_n).$$

The coefficient matrix A

m × n matrix, p.17

Stacking the row vectors gives the $m \times n$ (read "m by n") **matrix**

$$A = \begin{bmatrix} \cdots & A_1 & \cdots \\ \cdots & A_2 & \cdots \\ & \vdots & \\ \cdots & A_m & \cdots \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ a_{21} & \cdots & a_{2n} \\ \vdots & & \vdots \\ a_{m1} & \cdots & a_{mn} \end{bmatrix}.$$

An $m \times n$ matrix has m **rows** and n **columns**. The coefficient a_{ij} in the i-th row and j-th column is called the (i, j)**-entry** of A.

Rows first, always

"$m \times n$" and "a_{ij}" both say **row first, column second**. A system with 3 equations in 4 unknowns has a 3×4 coefficient matrix: one row per equation, one column per variable.

Matrix equality, p.18

Two matrices A and B are **equal** if and only if

- they have **the same shape** (same number of rows and same number of columns), and
- their corresponding entries are equal: $a_{ij} = b_{ij}$ for all i and j.

Same entries is not enough

$\begin{bmatrix} 1 & 2 \end{bmatrix}$ (1×2) and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ (2×1) contain the same numbers but are **not** equal: the shapes differ.

Column vectors and the compact form Ax = b

The unknowns and the right-hand sides are written as **column vectors** (p.19):

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n \qquad \text{and} \qquad b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \in \mathbb{R}^m.$$

$$Ax = \begin{bmatrix} A_1 \cdot x \\ A_2 \cdot x \\ \vdots \\ A_m \cdot x \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ a_{21}x_1 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} = b$$

THE SYSTEM IN COMPACT MATRIX FORM, P.20: ENTRY I OF AX IS ROW I OF A DOTTED WITH X

Note the sizes: A is $m \times n$, $x \in \mathbb{R}^n$ (one entry per **variable**), $b \in \mathbb{R}^m$ (one entry per **equation**).

So (p.21) a solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ is a vector with $\mathbf{A}_i \cdot \mathbf{x} = b_i$ for **all** $i = 1, \ldots, m$.

Geometry (p.21)

Each equation $\mathbf{A}_i \cdot \mathbf{x} = b_i$ is a hyperplane with **normal vector** $\mathbf{A}_i$ at **signed distance** $\dfrac{b_i}{\|\mathbf{A}_i\|}$ from the origin. The system is the intersection of m hyperplanes, and the general solution is a parametric representation of that intersection.

The augmented matrix [A | b]

Augmented matrix, p.22

The **augmented matrix** of the system $A\mathbf{x} = \mathbf{b}$ is

$$[A\,|\,\mathbf{b}] = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ a_{21} & \cdots & a_{2n} & b_2 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array}\right],$$

where A holds **the coefficients of the system** and $\mathbf{b}$ is **the column vector of constants**.

Each **row** of $[A|\mathbf{b}]$ is one equation; each column left of the bar belongs to one variable. The elementary operations on equations become operations on **rows** (hence the name "row operations"). We write $\longrightarrow$ between matrices and put the operation above the arrow (p.23).

Missing variables are zeros

If a variable does not appear in an equation, its entry is 0. $2x_1 + x_3 = 5$ in three variables gives the row $\left[\begin{array}{ccc|c} 2 & 0 & 1 & 5 \end{array}\right]$, not $\left[\begin{array}{cc|c} 2 & 1 & 5 \end{array}\right]$.

REVISITED EXAMPLE · LECTURE 3 P.23–26 Solve $3x_1 - 2x_2 + 2x_3 + 9x_4 = 4,\ \ 2x_1 + 2x_2 - 2x_3 - 4x_4 = 6$ using the augmented matrix.

1. Write the augmented matrix: $\left[\begin{array}{cccc|c} 3 & -2 & 2 & 9 & 4 \\ 2 & 2 & -2 & -4 & 6 \end{array}\right]$.
2. Step 1, operation (i), $R_1 \leftrightarrow R_2$:
$$\left[\begin{array}{cccc|c} 2 & 2 & -2 & -4 & 6 \\ 3 & -2 & 2 & 9 & 4 \end{array}\right]$$
3. Step 2, operation (ii), $R_1 \to \frac{1}{2}R_1$ (make the leading coefficient of row 1 equal to 1):
$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & -2 & 3 \\ 3 & -2 & 2 & 9 & 4 \end{array}\right]$$
4. Step 3, operation (iii), $R_2 \to R_2 - 3R_1$ (eliminate the x_1-term from row 2):
$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & -2 & 3 \\ 0 & -5 & 5 & 15 & -5 \end{array}\right]$$
5. Step 4, operation (ii), $R_2 \to -\frac{1}{5}R_2$ (leading coefficient of row 2 equal to 1):
$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & -2 & 3 \\ 0 & 1 & -1 & -3 & 1 \end{array}\right]$$
6. Step 5, operation (iii), $R_1 \to R_1 - R_2$ (eliminate the x_2-term from row 1):
$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & -1 & -3 & 1 \end{array}\right]$$
7. Read off: $x_1 + x_4 = 2$, $x_2 - x_3 - 3x_4 = 1$, so $x_1 = 2 - x_4$, $x_2 = 1 + x_3 + 3x_4$, $x_3 = x_3$, $x_4 = x_4$.

ANSWER

$\mathbf{x} = (2,1,0,0) + x_3(0,1,1,0) + x_4(-1,3,0,1)$, a plane in $\mathbb{R}^4$: the same answer as in Chapter 2, with far less writing.

EXAMPLE IN THREE VARIABLES · LECTURE 3 P.27–28 (THE BLANK 'SOLUTION' SLIDE, SOLVED)

Solve

$$\begin{aligned} x_1 + 2x_2 - x_3 &= 4 \\ 2x_1 + x_2 + 3x_3 &= 5 \\ 3x_1 - x_2 + 2x_3 &= 7 \end{aligned} \qquad \text{augmented matrix} \qquad \left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 2 & 1 & 3 & 5 \\ 3 & -1 & 2 & 7 \end{array}\right].$$

1. Clear column 1 below the leading 1: $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - 3R_1$:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & -3 & 5 & -3 \\ 0 & -7 & 5 & -5 \end{array}\right]$$

(row 2: $2-2=0,\ 1-4=-3,\ 3+2=5,\ 5-8=-3$; row 3: $3-3=0,\ -1-6=-7,\ 2+3=5,\ 7-12=-5$)

2. $R_2 \to -\frac{1}{3}R_2$ (leading 1 in row 2):

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 1 & -\frac{5}{3} & 1 \\ 0 & -7 & 5 & -5 \end{array}\right]$$

3. $R_3 \to R_3 + 7R_2$ ($5 - \frac{35}{3} = -\frac{20}{3}$, $-5+7=2$):

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 1 & -\frac{5}{3} & 1 \\ 0 & 0 & -\frac{20}{3} & 2 \end{array}\right]$$

4. $R_3 \to -\frac{3}{20}R_3$: the last row says $x_3 = -\frac{3}{10}$. The matrix is now in echelon form (Chapter 4) with a pivot in every column of A: no free variables.
5. Back-substitute. Row 2: $x_2 = 1 + \frac{5}{3}x_3 = 1 + \frac{5}{3}\cdot\left(-\frac{3}{10}\right) = 1 - \frac{1}{2} = \frac{1}{2}$.
6. Row 1: $x_1 = 4 - 2x_2 + x_3 = 4 - 1 - \frac{3}{10} = \frac{27}{10}$.
7. Check in equation 3: $3\cdot\frac{27}{10} - \frac{1}{2} + 2\cdot\left(-\frac{3}{10}\right) = \frac{81-5-6}{10} = 7$ ✓.

ANSWER

Unique solution $\mathbf{x} = (x_1, x_2, x_3) = \left(\frac{27}{10},\ \frac{1}{2},\ -\frac{3}{10}\right)$: the three planes meet in exactly one point.

Fractions are fine

This answer really has fractions; don't panic and restart. If you prefer, you can postpone fractions by a different order of operations, but the final numbers are the same (Theorem 1). Always finish with a check in an equation you did not use last.

In one minute

- A is $m \times n$: m rows (equations), n columns (variables); a_{ij} = (i, j)-entry.
- Equal matrices: same shape **and** $a_{ij} = b_{ij}$ for all i, j.
- $Ax = b$: entry i of Ax is $A_i \cdot x$ (row i dotted with x); $x \in \mathbb{R}^n$, $b \in \mathbb{R}^m$.
- Augmented matrix $[A|\mathbf{b}]$: coefficients, a bar, then the constants. Missing variables are 0.
- Row operations on $[A|\mathbf{b}]$ = equation operations on the system; write them above $\longrightarrow$.
- p.26 answer: $(2, 1, 0, 0) + x_3(0, 1, 1, 0) + x_4(-1, 3, 0, 1)$. p.28 answer: $\left(\frac{27}{10}, \frac{1}{2}, -\frac{3}{10}\right)$.

CHAPTER 4

Echelon and reduced echelon forms

The 'staircase' shape that makes a system easy to read, its stricter cousin the reduced echelon form, and how to spot pivots, pivot variables and free variables. Every 'Why?' and every dotted blank from the slides is answered here.

From Lecture 3, pages 33–44

Leading entries and echelon form

Definition D1, p.33

The first nonzero entry of a row (reading from left to right) is called its **leading entry**. A matrix is in **(row) echelon form** if

- the leading entries move to the right **in successive rows**;
- all entries **below** each leading entry are **zero**;
- all rows consisting entirely of zeros are placed at **the bottom of the matrix**.

The word **echelon** comes from the French *échelle*, "ladder" (p.38). Draw the rungs and you see a **staircase**: each step down moves at least one column to the right.

$$\begin{bmatrix} \boxed{1} & 3 & 1 & 5 \\ 0 & 0 & \boxed{2} & 3 \\ 0 & 0 & 0 & \boxed{2} \end{bmatrix}$$

The staircase matrix of p.38: leading entries boxed. Under every box, only zeros.

Reduced echelon form

Definition D2, p.34

A matrix is in **reduced echelon form** if it is in (row) echelon form and, in addition,

- every leading entry is **1**;
- all entries **above** each leading entry are **zero**.

Echelon vs reduced, in one picture

Echelon form: zeros **below** the leading entries. Reduced echelon form: leading entries are 1, and zeros **below and above** them, so every pivot column looks like a column of the identity matrix. Every reduced echelon matrix is in echelon form; not the other way round.

Pivots, pivot variables, free variables

Definition D3, p.39

If a matrix is in *echelon form*, the leading entry of any nonzero row is called a **pivot**. The columns containing the pivots are called **pivot columns**.

- **Pivot variables** correspond to pivot columns.
- **Free variables** correspond to the remaining columns.

For example (the "..." on p.39): read the staircase matrix of p.38 as the coefficient matrix of a system in x_1, x_2, x_3, x_4. Its pivots $1, 2, 2$ sit in columns 1, 3, 4. So the pivot variables are x_1, x_3, x_4 and the free variable is x_2.

The constants column is not a variable

In an augmented matrix $[A|\mathbf{b}]$ only the columns **left of the bar** belong to variables. Never call the last column a pivot or free "variable". (A pivot in the last column means something else entirely: no solution, see Chapter 5.)

Pivots only after echelon form

D3 defines pivots for a matrix **in echelon form**. In a matrix that is not yet in echelon form, the leading entries are not the pivots. Reduce first, then read.

Key point (Remark, p.40)

When the matrix is **in reduced echelon form**, we can determine the **general solution** by expressing each pivot variable in terms of the free variables.

- If there are no free variables, the solution is **unique**.
- If there are free variables, we express the pivot variables as functions of them, leading to a **parametric solution**.

The slide says 'row echelon form' on p.26

The final matrix of the worked example on p.26, $\left[\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 2 \\ 0 & 1 & -1 & -3 & 1 \end{array}\right]$, is in fact in **reduced** echelon form (leading 1s with zeros above and below). That is exactly why the solution could be read off immediately.

How to classify a matrix (the "Why?" answers)

Check in this order. Stop at the first failure.

1. Are all zero rows at the bottom?
2. Does each leading entry sit strictly to the right of the one above it? (Equivalently: only zeros below every leading entry.)

3. If 1 and 2 pass: **echelon form**. Now are all leading entries 1, with zeros **above** them too? If yes: **reduced echelon form**.

Slides 35–37: three augmented matrices

MATRIX 1 · LECTURE 3 P.35 $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 0 & 1 & 0 & 5 \\ 0 & 0 & 1 & 6 \end{array}\right]$

1. Leading entries: row 1 in column 1, row 2 in column 2, row 3 in column 3. They move to the right. Below each leading entry: zeros. No zero rows.
2. So it is in echelon form. Leading entries are all 1, **but** above the leading 1 of row 2 there is a 2, and above the leading 1 of row 3 there is a 3.
3. Bonus, solve it by back substitution: $x_3 = 6$, $x_2 = 5$, $x_1 = 4 - 2 \cdot 5 - 3 \cdot 6 = -24$.

ANSWER
The augmented matrix is **in echelon form but not in reduced echelon form** (nonzero entries 2 and 3 above leading entries). Its system has the unique solution $(-24, 5, 6)$.

MATRIX 2 · LECTURE 3 P.36 $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array}\right]$

1. Echelon form: leading entries in columns 1, 2, 3, zeros below.
2. Every leading entry is 1 and every entry above a leading entry is 0.
3. So the solution can be read off directly: $x_1 = 3,\ x_2 = 2,\ x_3 = 1$.

ANSWER
The augmented matrix is **in reduced echelon form**; unique solution $\mathbf{x} = (3, 2, 1)$.

MATRIX 3 · LECTURE 3 P.37 $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 4 \\ 1 & 2 & 3 & 5 \\ 0 & 0 & 1 & 6 \end{array}\right]$

1. The leading entry of row 2 is in column 1, the **same** column as the leading entry of row 1. It does not move to the right.
2. Equivalently: there is a nonzero entry (the 1 in position $(2, 1)$) below the leading entry of row 1.
3. Bonus: $R_2 \to R_2 - R_1$ gives the row $\left[\begin{array}{ccc|c} 0 & 0 & 0 & 1 \end{array}\right]$, i.e. $0 = 1$. Rows 1 and 2 ask $x_1 + 2x_2 + 3x_3$ to be both 4 and 5, so this system has no solution.

ANSWER
The augmented matrix is **not in echelon form** (row 2's leading entry is not to the right of row 1's; nonzero entry below a leading entry).

Slides 41–44: echelon, reduced echelon, or neither?

MATRIX 1 · LECTURE 3 P.41 $\left[\begin{array}{cccc|c} 0 & 2 & 1 & 1 & 4 \\ 0 & 0 & 3 & 0 & 2 \\ 0 & 0 & 0 & -1 & 1 \end{array}\right]$

1. Leading entries: 2 (column 2), 3 (column 3), -1 (column 4). They move to the right; only zeros below them; no zero rows. (a first column of zeros is allowed)
2. Not reduced: the leading entries are not 1, and there are nonzero entries above them (the 1s in row 1 above the pivots 3 and -1).
3. Pivot columns: 2, 3, 4. Remaining variable column: 1.

ANSWER
The augmented matrix is **in echelon form** (not reduced). Pivot variables: x_2, x_3, x_4. Free variable: x_1.

MATRIX 2 · LECTURE 3 P.42 $\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 2 \end{array}\right]$

1. Row 2 consists entirely of zeros, but it is **not at the bottom**: row 3 below it is nonzero.
2. Fix: $R_2 \leftrightarrow R_3$ gives $\left[\begin{array}{ccc|c} 1 & 2 & -1 & 4 \\ 0 & 0 & 3 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$, which is in echelon form.

ANSWER
The augmented matrix is **not in echelon form**, because the zero row is not placed at the bottom.

MATRIX 3 · LECTURE 3 P.43 $\left[\begin{array}{cccc|c} 1 & 2 & 1 & 1 & 4 \\ 0 & 0 & 3 & 0 & 2 \\ 0 & 0 & 1 & -1 & 1 \end{array}\right]$

1. Leading entries: row 2 in column 3 and row 3 **also in column 3**. Row 3's leading entry does not move to the right.
2. Equivalently: the entry 1 below the leading entry 3 of row 2 is not zero.
3. Fix: $R_3 \to R_3 - \frac{1}{3}R_2$ turns row 3 into $\left[\begin{array}{cccc|c} 0 & 0 & 0 & -1 & \frac{1}{3} \end{array}\right]$.

ANSWER
The augmented matrix is **not in echelon form** (nonzero entry below the leading entry of row 2).

MATRIX 4 · LECTURE 3 P.43 $\left[\begin{array}{cc|c} 0 & 1 & 1 \\ 1 & 2 & 3 \end{array}\right]$

1. Row 1's leading entry is in column 2, row 2's is in column 1: the leading entries move to the **left**.
2. Fix: $R_1 \leftrightarrow R_2$ gives $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 1 & 1 \end{array}\right]$ (echelon form); then $R_1 \to R_1 - 2R_2$ gives $\left[\begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & 1 \end{array}\right]$ (reduced), so $\mathbf{x} = (1, 1)$.

ANSWER
The augmented matrix is **not in echelon form** (the leading entry of row 2 is to the left of that of row 1).

MATRIX 5 · LECTURE 3 P.44 $\left[\begin{array}{ccccc|c} 1 & 2 & 0 & 0 & 4 & 1 \\ 0 & 0 & 1 & 0 & -2 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{array}\right]$

1. Leading entries: 1 in columns 1, 3, 4. They move to the right; zeros below.
2. All leading entries are 1, and the entries above the leading 1s of columns 3 and 4 are 0. (the 2, 4 and -2 sit in non-pivot columns: allowed)
3. Pivot columns 1, 3, 4; remaining variable columns 2, 5.
4. Bonus (Remark p.40): $x_1 = 1 - 2x_2 - 4x_5$, $x_3 = 2 + 2x_5$, $x_4 = 1 - x_5$, so

$$\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -4 \\ 0 \\ 2 \\ -1 \\ 1 \end{bmatrix}, \quad x_2, x_5 \in \mathbb{R}.$$

ANSWER
The augmented matrix is **in reduced echelon form**. Pivot variables: x_1, x_3, x_4. Free variables: x_2, x_5.

Non-pivot columns may contain anything
"Zeros above and below" is required only in **pivot** columns. In Matrix 5 the column of x_5 is $(4, -2, 1)$, and the matrix is still reduced. Students often call it "not reduced" because they see nonzero numbers above a 1 that is not a leading entry.

In one minute

- Leading entry = first nonzero entry of a row.
- **Echelon (D1):** leading entries move right, zeros below them, zero rows at the bottom.
- **Reduced (D2):** echelon + every leading entry is 1 + zeros above each leading entry.

- **Pivot (D3):** leading entry of a nonzero row of an echelon matrix; pivot columns give pivot variables, the other variable columns give free variables.
- Reduced echelon form: write each pivot variable in terms of the free ones. No free variables means a unique solution.
- Typical "not echelon" reasons: zero row not at the bottom; two leading entries in the same column; leading entries moving left.

CHAPTER 5

Gaussian elimination

One recipe that solves every linear system: reduce the augmented matrix to echelon form, then read off 'one solution', 'infinitely many' (in parametric vector form) or 'none'. Ends with the exam's favourite use of it: is b a linear combination of u, v, w?

From Lecture 3, pages 45–50 (plus the 2025 midterm, Question 1)

The strategy

Gaussian elimination, p.45

Our strategy is to transform the augmented matrix of any system of linear equations into **echelon form** by performing a sequence of **elementary row operations**. This process is known as **Gaussian (Gauss) elimination**.

Why does this work? Theorem 1 (Chapter 2) says row operations never change the solution set. So the echelon system has **exactly the same solutions** as the original, and an echelon system is easy to solve from the bottom up.

The recipe, step by step

Gaussian elimination (to echelon form)

1. Write the augmented matrix $[A|\mathbf{b}]$.
2. Find the leftmost column (left of the bar) that is not all zeros. If its top entry is 0, **swap** a row with a nonzero entry to the top (type (i)). A row whose entry is 1 is the nicest choice.
3. Use type (iii) operations $R_i \to R_i - rR_1$ to make every entry **below** this pivot zero.
4. Cover the pivot's row and repeat steps 2–3 on the rows below, moving one column to the right each time.
5. Stop when the rows below are all zero or you run out of rows: the matrix is in echelon form.
6. **Back substitution:** solve the last nonzero equation for its pivot variable, plug into the

Gauss–Jordan (all the way to reduced form)

1. Do Gaussian elimination first.
2. Scale each nonzero row (type (ii)) so that its pivot becomes 1.
3. Starting from the **last** pivot and working up, use type (iii) to make every entry **above** each pivot zero.
4. Now the matrix is in reduced echelon form: each equation contains exactly one pivot variable, so the general solution can be read off directly (Remark p.40).

Ch. 3's p.23–26 example was done this way: after step 5 it was in reduced echelon form.

one above, and so on upward.

Write every operation

Above each arrow write exactly what you did, e.g. $\xrightarrow{R_2 \to R_2 - 2R_1}$. In the exam this earns method marks even if an arithmetic slip happens, and it lets you find the slip quickly.

Update one row per operation, using the current rows

When you do several operations in one step (e.g. $R_2 \to R_2 - 2R_1$ and $R_3 \to R_3 - 3R_1$), each must use the **unchanged pivot row** R_1. Never use a row you have just modified in the same step to modify another one.

Ex. 3: a 5 × 4 system

EX. 3 · LECTURE 3 P.46–47 ('YOUR TASK', DONE) Use Gaussian elimination to reduce to row echelon form, then solve:

$$\left[\begin{array}{cccc|c} 3 & 6 & 4 & 10 & 27 \\ 2 & 4 & 3 & 7 & 18 \\ 1 & 2 & 1 & 3 & 9 \\ 5 & 10 & 7 & 15 & 45 \\ 4 & 8 & 5 & 12 & 36 \end{array}\right]$$

1. $R_1 \leftrightarrow R_3$ (row 3 starts with a 1: no fractions):

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 9 \\ 2 & 4 & 3 & 7 & 18 \\ 3 & 6 & 4 & 10 & 27 \\ 5 & 10 & 7 & 15 & 45 \\ 4 & 8 & 5 & 12 & 36 \end{array}\right]$$

2. Clear column 1: $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - 3R_1,\ R_4 \to R_4 - 5R_1,\ R_5 \to R_5 - 4R_1$:

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 9 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{array}\right]$$

3. Column 2 is now zero below row 1, so the next pivot is in **column 3** (row 2). Clear below it: $R_3 \to R_3 - R_2,\ R_4 \to R_4 - 2R_2,\ R_5 \to R_5 - R_2$:

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 9 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & -1 & 0 \end{array}\right]$$

4 Zero row must go to the bottom: $R_3 \leftrightarrow R_4$. Then $R_5 \to R_5 - \frac{1}{2}R_3$ $(-1 - \frac{1}{2}(-2) = 0)$:

$$\left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 9 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

This is the row echelon form shown on p.47.

5 Pivots in columns 1, 3, 4: pivot variables x_1, x_3, x_4; free variable x_2. No row $\left[\begin{array}{cccc|c} 0 & 0 & 0 & 0 & c \end{array}\right]$ with $c \neq 0$, so the system is consistent.

6 Back substitution: $-2x_4 = 0 \Rightarrow x_4 = 0$; $x_3 + x_4 = 0 \Rightarrow x_3 = 0$; $x_1 + 2x_2 + x_3 + 3x_4 = 9 \Rightarrow x_1 = 9 - 2x_2$.

ANSWER

Echelon form as on p.47; general solution $\mathbf{x} = (9 - 2x_2,\ x_2,\ 0,\ 0) = (9,0,0,0) + x_2(-2,1,0,0)$, $x_2 \in \mathbb{R}$. (Five equations, but only three were independent: two rows became zero.)

Reading the general solution in parametric vector form

1. Get to echelon (better: reduced echelon) form and check there is no row $[0 \cdots 0 \,|\, c]$ with $c \neq 0$.
2. Circle the pivots. Pivot columns = pivot variables; other variable columns = free variables.
3. Solve each nonzero row for its pivot variable, in terms of the free variables only.
4. Write $\mathbf{x}$ as a column; replace each pivot variable by its formula; each free variable stays as itself.
5. Split: $\mathbf{x} = \mathbf{p} + t_1\mathbf{v}_1 + \cdots + t_k\mathbf{v}_k$, where $\mathbf{p}$ collects the constants and $\mathbf{v}_j$ the coefficients of the j-th free variable.
6. Check: $\mathbf{p}$ must satisfy the original system (all free variables $= 0$).

$$\mathbf{x} = \underbrace{\mathbf{p}}_{\text{one particular solution}} + \underbrace{x_{f_1}\mathbf{v}_1 + \cdots + x_{f_k}\mathbf{v}_k}_{\text{one vector per free variable}}, \qquad x_{f_1}, \ldots, x_{f_k} \in \mathbb{R}$$

SHAPE OF EVERY GENERAL SOLUTION (EX. 1 P.6, P.13, P.26)

When there is no solution

INTERSECTION OF THREE HYPERPLANES IN FOUR DIMENSIONS · LECTURE 3 P.48–50 Find a parametric description of the intersection of the three hyperplanes

$$x_1 - x_2 + 2x_3 + 3x_4 = 2, \qquad 2x_1 + x_2 + x_3 = 1, \qquad x_1 + 2x_2 - x_3 - 3x_4 = 7.$$

1 Augmented matrix, then $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 - R_1$:

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & 3 & 2 \\ 2 & 1 & 1 & 0 & 1 \\ 1 & 2 & -1 & -3 & 7 \end{array}\right] \longrightarrow \left[\begin{array}{cccc|c} 1 & -1 & 2 & 3 & 2 \\ 0 & 3 & -3 & -6 & -3 \\ 0 & 3 & -3 & -6 & 5 \end{array}\right]$$

2. $R_3 \to R_3 - R_2$:

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & 3 & 2 \\ 0 & 3 & -3 & -6 & -3 \\ 0 & 0 & 0 & 0 & 8 \end{array}\right]$$

3. The corresponding system is $x_1 - x_2 + 2x_3 + 3x_4 = 2$, $3x_2 - 3x_3 - 6x_4 = -3$, $\mathbf{0 = 8}$.
4. $0 = 8$ is absurd: no values of x_1, x_2, x_3, x_4 satisfy all three equations.

ANSWER

No solution. The three hyperplanes have no point in common: the system is **inconsistent**.

Definition D4, p.50

A system of linear equations may not have any solutions; in this case it is called **inconsistent**.

0 = 0 is harmless, 0 = 8 is fatal

A row $\left[\begin{array}{ccc|c} 0 & \cdots & 0 & 0 \end{array}\right]$ just means a redundant equation (as in Ex. 3): delete it mentally and carry on. A row $\left[\begin{array}{ccc|c} 0 & \cdots & 0 & c \end{array}\right]$ with $c \neq 0$ means **no solution**, no matter how nice the other rows look. Do not "solve" the other rows and present an answer.

Decision box: how many solutions?

Decision box: none / exactly one / infinitely many

After Gaussian elimination, look at the echelon form of $[A|\mathbf{b}]$ and ask, **in this order**:

1. Is there a row $[0\,0\,\cdots\,0\,|\,c]$ with $c \neq 0$ (a pivot in the constants column)? Then **no solution**: the system is inconsistent (p.48–50, $0 = 8$).
2. Otherwise, does **every** variable column have a pivot (no free variables)? Then **exactly one** solution (p.8, p.27–28, p.36).
3. Otherwise (no bad row, **at least one** free variable): **infinitely many** solutions, one parameter per free variable (p.6, p.13, p.46–47).

Quick consequences

More variables than equations (like the 2×4 system on p.10) means at most as many pivots as rows, so there are free variables: the answer is "infinitely many" **or** "none", never exactly one. The order of the questions matters: first "is there a $[0 \cdots 0|c]$ row?", only then count free variables.

Exam link: is b in Span{u, v, w}?

Question 1 of the 2025 midterm used this lecture's method on vectors. "$\mathbf{b} \in \mathrm{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$" means: there are scalars a, c, d with

$$a\mathbf{u} + c\mathbf{v} + d\mathbf{w} = \mathbf{b}.$$

Comparing entries gives a linear system in a, c, d. Its augmented matrix is simply $[\,\mathbf{u}\ \ \mathbf{v}\ \ \mathbf{w} \mid \mathbf{b}\,]$: the vectors become the **columns**. So $\mathbf{b}$ is in the span **if and only if this system is consistent**, and any solution gives the coefficients.

PRACTICE IN THE STYLE OF MIDTERM 2025 Q1 Let $\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 2 \\ 3 \\ 0 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 4 \\ 6 \\ -5 \end{bmatrix}$. Is $\mathbf{b} \in \mathrm{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$? If so, find a, c, d with $\mathbf{b} = a\mathbf{u} + c\mathbf{v} + d\mathbf{w}$.

1. $a\mathbf{u} + c\mathbf{v} + d\mathbf{w} = \mathbf{b}$ gives the system $a + 2d = 4,\ \ 2a + c + 3d = 6,\ \ -a + 3c = -5$, i.e.
$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 2 & 1 & 3 & 6 \\ -1 & 3 & 0 & -5 \end{array}\right].$$
2. $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 + R_1$:
$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 3 & 2 & -1 \end{array}\right]$$
3. $R_3 \to R_3 - 3R_2$:
$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & -2 \\ 0 & 0 & 5 & 5 \end{array}\right]$$
No $[0\ 0\ 0 \mid c \neq 0]$ row: consistent, so $\mathbf{b}$ is in the span.
4. Back substitution: row 3: $5d = 5 \Rightarrow d = 1$. Row 2: $c - d = -2 \Rightarrow c = -1$. Row 1: $a + 2d = 4 \Rightarrow a = 2$.
5. Check: $2\mathbf{u} - \mathbf{v} + \mathbf{w} = (2 - 0 + 2,\ 4 - 1 + 3,\ -2 - 3 + 0) = (4, 6, -5) = \mathbf{b}$ ✓.

ANSWER
Yes: $\mathbf{b} = 2\mathbf{u} - \mathbf{v} + \mathbf{w}$, so $\mathbf{b} \in \mathrm{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ ($a = 2,\ c = -1,\ d = 1$).

Beyond Lecture 3
The word "span" is not defined in the Lecture 3 slides; it is used here only because the midterm asks it in this form. For this lecture you only need: "$\mathbf{b}$ is in the span" = "the system $[\mathbf{u}\ \mathbf{v}\ \mathbf{w} \mid \mathbf{b}]$ has a solution". If elimination produces a row $[0\ 0\ 0 \mid c]$ with $c \neq 0$, the answer is "no, $\mathbf{b} \notin \mathrm{Span}$". If there are free variables, the coefficients are not unique; any one solution is a valid answer.

In one minute

- Gaussian elimination: row operations on $[A|\mathbf{b}]$ until echelon form, then back substitution. Gauss–Jordan: continue to reduced echelon form and read off.
- Theorem 1 guarantees the solutions never change along the way.
- Ex. 3: echelon form $\left[\begin{array}{cccc|c} 1 & 2 & 1 & 3 & 9 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & -2 & 0 \end{array}\right]$ + two zero rows; $\mathbf{x} = (9,0,0,0) + x_2(-2,1,0,0)$.
- A row $[0 \cdots 0 \,|\, c]$, $c \neq 0$: **inconsistent** (D4), no solution (p.50: $0 = 8$).
- Otherwise: no free variables means one solution; free variables mean infinitely many.
- General solution = particular vector + one direction vector per free variable.
- Span question: columns $\mathbf{u}, \mathbf{v}, \mathbf{w} \,|\, \mathbf{b}$; consistent $\iff \mathbf{b} \in \mathrm{Span}$.

CHAPTER 6

Two ways to read Ax = b

The same system can be read row by row (intersecting hyperplanes) or column by column (a linear combination). After this chapter you can turn a 'is b in the span?' question into a system and settle it with one row reduction.

From Lecture 4, pages 1–15

Where this lecture is going

Until now, Gaussian elimination took a system of equations and produced a **parametric** description of its solutions (free variables, parameters). This lecture runs the machine **the other way round**.

First aim, Lecture 4 p.3

Gaussian elimination is a technique for finding a parametric description of the solutions of a system of linear Cartesian equations. Now we shall see that this same technique allows us to proceed in the opposite direction. **Cartesian Description of a Span:** That is, given vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n \in \mathbb{R}^m$, we would like to find a set of Cartesian equations whose solution is precisely $\mathrm{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$.

Everything in this lecture lives in one object, the **augmented matrix** of a system of m equations in n unknowns, $A\mathbf{x} = \mathbf{b}$ (p.4):

$$[A \,|\, \mathbf{b}] = \left[\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ a_{21} & \cdots & a_{2n} & b_2 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array}\right]$$

A is $m \times n$: m rows = equations, n columns = unknowns. Keep that straight; almost every rule later in the lecture compares the rank with m or with n.

Row view: intersecting hyperplanes

Write $\mathbf{A}_1, \ldots, \mathbf{A}_m$ for the **row vectors** of A. Row i of $A\mathbf{x} = \mathbf{b}$ says exactly $\mathbf{A}_i \cdot \mathbf{x} = b_i$.

Row view, p.5

If $\mathbf{A}_1, \ldots, \mathbf{A}_m$ denote the row vectors of A, then a vector $\mathbf{c}$ is a solution of $A\mathbf{x} = \mathbf{b}$ if and only if

$$\mathbf{A}_1 \cdot \mathbf{c} = b_1, \quad \mathbf{A}_2 \cdot \mathbf{c} = b_2, \quad \ldots, \quad \mathbf{A}_m \cdot \mathbf{c} = b_m.$$

Geometric interpretation: $\mathbf{c}$ is a solution precisely when it lies in the *intersection of all the hyperplanes* defined by these equations.

In $\mathbb{R}^2$ each equation is a line, in $\mathbb{R}^3$ a plane, in $\mathbb{R}^n$ a "hyperplane". Solving the system = finding the points that lie on all of them at once.

Column view: a linear combination

Now cut A into its **columns** $\mathbf{v}_j = \begin{bmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{bmatrix} \in \mathbb{R}^m$, $j = 1, \ldots, n$. Multiplying out $A\mathbf{x}$ and regrouping by x_j gives the key identity:

$$A\mathbf{x} = \begin{bmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ a_{21}x_1 + \cdots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{bmatrix} = x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_n\mathbf{v}_n \qquad (*)$$

MATRIX TIMES VECTOR = COMBINATION OF THE COLUMNS (P.6)

Column view, p.7

Thus, a solution $\mathbf{c} = (c_1, \ldots, c_n)$ of the system $A\mathbf{x} = \mathbf{b}$ provides scalars $c_1, \ldots, c_n$ such that

$$\mathbf{b} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_n\mathbf{v}_n.$$

Geometric interpretation: A solution $\mathbf{c}$ expresses $\mathbf{b}$ as a *linear combination* of the column vectors of A.

One sentence to remember

"$A\mathbf{x} = \mathbf{b}$ has a solution" and "$\mathbf{b}$ is a linear combination of the columns of A" and "$\mathbf{b} \in \mathrm{Span}(\text{columns of } A)$" are **the same statement**. The solution $\mathbf{x}$ is the list of coefficients.

Translating a span question

Asked "is $\mathbf{b}$ in $\mathrm{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$?": put the vectors **as columns** of A, put $\mathbf{b}$ after the bar, row-reduce $[A|\mathbf{b}]$. Solution exists = yes, and the solution gives the coefficients. This is exactly Question 1 of the 2025 midterm.

Example 1: b is in the span

EXAMPLE 1 · LECTURE 4 P.8–13 Let $\mathbf{v}_1 = \begin{bmatrix}1\\2\\0\\1\end{bmatrix}$, $\mathbf{v}_2 = \begin{bmatrix}0\\-1\\1\\2\end{bmatrix}$, $\mathbf{v}_3 = \begin{bmatrix}2\\0\\3\\1\end{bmatrix}$, $\mathbf{b} = \begin{bmatrix}3\\0\\5\\6\end{bmatrix}$. Can $\mathbf{b}$ be written as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$? In other words, is $\mathbf{b} \in \mathrm{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$?

1. **Objective:** find scalars $x_1, x_2, x_3 \in \mathbb{R}$ with $x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + x_3\mathbf{v}_3 = \mathbf{b}$. Comparing the four entries gives the system

$$\begin{aligned} x_1 + 0x_2 + 2x_3 &= 3 \\ 2x_1 - x_2 + 0x_3 &= 0 \\ 0x_1 + x_2 + 3x_3 &= 5 \\ x_1 + 2x_2 + x_3 &= 6 \end{aligned}$$

   (row i of the system = entry i of the vectors)

2. In matrix form $A\mathbf{x} = \mathbf{b}$ with the vectors as the **columns** of A. Augmented matrix:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 2 & -1 & 0 & 0 \\ 0 & 1 & 3 & 5 \\ 1 & 2 & 1 & 6 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & -1 & -4 & -6 \\ 0 & 1 & 3 & 5 \\ 0 & 2 & -1 & 3 \end{array}\right] \xrightarrow{R_2 \leftarrow -R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 6 \\ 0 & 1 & 3 & 5 \\ 0 & 2 & -1 & 3 \end{array}\right]$$

$$\xrightarrow[R_4 \leftarrow R_4 - 2R_2]{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & -1 & -1 \\ 0 & 0 & -9 & -9 \end{array}\right] \xrightarrow{R_3 \leftarrow -R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -9 & -9 \end{array}\right] \xrightarrow{R_4 \leftarrow R_4 + 9R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & 4 & 6 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow[R_1 \leftarrow R_1 - 2R_3]{R_2 \leftarrow R_2 - 4R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

   (the slide only says "elem. row oper. reduce to RREF"; these are the operations. The first five arrows reach echelon form; the last one clears the entries above the pivot of column 3.)

3. Read off the RREF: $x_1 = 1,\ x_2 = 2,\ x_3 = 1$. The last row is $0 = 0$: no contradiction.

4. Check (p.13 says "you can see it is true"): $1 \cdot \mathbf{v}_1 + 2\mathbf{v}_2 + 1 \cdot \mathbf{v}_3 = (1,2,0,1) + (0,-2,2,4) + (2,0,3,1) = (3,0,5,6) = \mathbf{b}$. ✓

ANSWER

$\mathbf{x} = (1,2,1)$, i.e. $\mathbf{b} = 1 \cdot \mathbf{v}_1 + 2 \cdot \mathbf{v}_2 + 1 \cdot \mathbf{v}_3$. **Conclusion:** $\mathbf{b}$ is a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$, i.e. $\mathbf{b} \in \mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

Example 2: $\mathbf{b}'$ is not in the span

EXAMPLE 2 · LECTURE 4 P.14–15 Let $\mathbf{b}' = \begin{bmatrix} 2 \\ 0 \\ 1 \\ 4 \end{bmatrix}$, with the same $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. Determine whether $\mathbf{b}'$ is in $\mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

1. Same coefficient matrix, new right-hand side: $x_1 + 2x_3 = 2,\ 2x_1 - x_2 = 0,\ x_2 + 3x_3 = 1,\ x_1 + 2x_2 + x_3 = 4$.
2. The same row operations as in Example 1 (only the last column changes):

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 2 \\ 2 & -1 & 0 & 0 \\ 0 & 1 & 3 & 1 \\ 1 & 2 & 1 & 4 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 2 \\ 0 & -1 & -4 & -4 \\ 0 & 1 & 3 & 1 \\ 0 & 2 & -1 & 2 \end{array}\right] \xrightarrow{R_2 \leftarrow -R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 2 \\ 0 & 1 & 4 & 4 \\ 0 & 1 & 3 & 1 \\ 0 & 2 & -1 & 2 \end{array}\right]$$

$$\xrightarrow[R_4 \leftarrow R_4 - 2R_2]{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 2 \\ 0 & 1 & 4 & 4 \\ 0 & 0 & -1 & -3 \\ 0 & 0 & -9 & -6 \end{array}\right] \xrightarrow{R_3 \leftarrow -R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 2 \\ 0 & 1 & 4 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & -9 & -6 \end{array}\right] \xrightarrow{R_4 \leftarrow R_4 + 9R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 2 \\ 0 & 1 & 4 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 21 \end{array}\right]$$

(this is the echelon form printed on p.15)

3. The last row means $0x_1 + 0x_2 + 0x_3 = 21$, i.e. $0 = 21$. Impossible.

ANSWER

Conclusion: the system has no solution. Therefore $\mathbf{b}'$ **cannot be written** as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$: $\mathbf{b}' \notin \mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

Reuse your row operations

When only $\mathbf{b}$ changes, the operations stay the same: they are decided by the columns of A alone. That is why, in the next chapter, we can run them **once** with a symbolic $\mathbf{b} = (b_1, b_2, b_3, b_4)$ and answer both examples at the same time.

Columns, not rows

To test "is $\mathbf{b}$ a combination of $\mathbf{v}_1, \ldots, \mathbf{v}_n$?", the vectors go in as **columns**. Putting them in as rows answers a different question and loses points even if the arithmetic is perfect.

'Familiar to Cartesian equation' (p.7)

The slide ends the column view with "Notice that it is familiar to Cartesian equation." The point: deciding which $\mathbf{b}$ are combinations of the columns will produce **equations in** $b_1, \ldots, b_m$, i.e. Cartesian equations of the span. That is exactly the "first aim" of p.3, done in the next chapter.

In one minute

- A is $m \times n$: m equations (rows), n unknowns (columns). Augmented matrix $[A|\mathbf{b}]$.
- **Row view:** $\mathbf{c}$ solves the system $\iff \mathbf{A}_i \cdot \mathbf{c} = b_i$ for every row $\iff \mathbf{c}$ lies in the intersection of the hyperplanes.
- **Column view:** $A\mathbf{x} = x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n$; a solution expresses $\mathbf{b}$ as a linear combination of the columns.
- "$\mathbf{b} \in \text{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$?" = "is $[\mathbf{v}_1 \cdots \mathbf{v}_n \,|\, \mathbf{b}]$ solvable?"
- Example 1: $\mathbf{b} = (3, 0, 5, 6) = \mathbf{v}_1 + 2\mathbf{v}_2 + \mathbf{v}_3$, in the span.
- Example 2: $\mathbf{b}' = (2, 0, 1, 4)$ gives the row $[0\ 0\ 0 \,|\, 21]$: not in the span.

CHAPTER 7

Consistent systems and constraint equations

When does Ax = b have a solution? Row-reduce once with a symbolic b, and the zero rows hand you the answer as equations in b1, ..., bm. This is midterm Question 3, and after this chapter you can do it on autopilot.

From Lecture 4, pages 16–33

Consistent or inconsistent

Definition D1, p.16

If the system of equations $A\mathbf{x} = \mathbf{b}$ has no solutions, the system is said to be **inconsistent**; if it has at least one solution, then it is said to be **consistent**. A system of equations is consistent precisely when a solution exists.

The two blanks on p.16, filled in:

- The system of equations in **Example 1** is **consistent** (it has the solution $x_1 = 1, x_2 = 2, x_3 = 1$).
- The system in **Example 2** is **inconsistent** (its echelon form contains the row $0 = 21$).

"Consistent" says nothing about *how many* solutions: one or infinitely many both count. That question comes in the uniqueness chapter.

How to see it in the echelon form

Recognising an inconsistent system, p.17

A system is **inconsistent** precisely when there is a row in the echelon form of the augmented matrix of the form

$$0x_1 + 0x_2 + \cdots + 0x_n = c, \quad \text{where } c \neq 0.$$

That is, a row where all entries are zero except for the rightmost.

Characterizing consistency, p.18

Let $[U \mid \mathbf{c}]$ be the echelon form of the augmented matrix $[A \mid \mathbf{b}]$. Then the system $A\mathbf{x} = \mathbf{b}$ is **consistent** if and only if every zero row in U corresponds to a zero entry in the vector $\mathbf{c}$. In other words, there must be **no row of the form** $0x_1 + 0x_2 + \cdots + 0x_n = c$, with $c \neq 0$.

→ Only the zero rows of U matter

Rows of U that contain a pivot can always be satisfied (back-substitution solves them). The danger is only in the rows where the A**-part is all zeros**: there the equation reads $0 = c$, fine if $c = 0$, impossible otherwise.

The same fact, seen geometrically (p.19):

- **Row interpretation:** $A\mathbf{x} = \mathbf{b}$ is consistent if and only if the set of hyperplanes $\mathbf{A}_1 \cdot \mathbf{x} = b_1,\ \ldots,\ \mathbf{A}_m \cdot \mathbf{x} = b_m$ has a **nonempty intersection**.
- **Column interpretation:** the system is consistent precisely when $\mathbf{b}$ lies in the span of the column vectors of A, that is $\mathbf{b} \in \mathrm{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$: $\mathbf{b} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$ for some scalars $c_1, \ldots, c_n$.

Example 3: all b at once

Examples 1 and 2 used the same $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ with two different right-hand sides. Instead of redoing the work for every $\mathbf{b}$, keep $\mathbf{b}$ as letters.

EXAMPLE 3 · LECTURE 4 P.20–21 With $\mathbf{v}_1 = (1, 2, 0, 1),\ \mathbf{v}_2 = (0, -1, 1, 2),\ \mathbf{v}_3 = (2, 0, 3, 1)$ as the columns of A, determine for which vectors $\mathbf{b} = (b_1, b_2, b_3, b_4) \in \mathbb{R}^4$ the system $A\mathbf{x} = \mathbf{b}$ has a solution.

1. The augmented matrix and the row reduction (this fills the slide's blank "After row reduction (RREF) ....."; the operations are the same as in Example 1):

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 2 & -1 & 0 & b_2 \\ 0 & 1 & 3 & b_3 \\ 1 & 2 & 1 & b_4 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 0 & -1 & -4 & -2b_1 + b_2 \\ 0 & 1 & 3 & b_3 \\ 0 & 2 & -1 & -b_1 + b_4 \end{array}\right]$$

$$\xrightarrow{R_2 \leftarrow -R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 0 & 1 & 4 & 2b_1 - b_2 \\ 0 & 1 & 3 & b_3 \\ 0 & 2 & -1 & -b_1 + b_4 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 - 2R_2]{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 0 & 1 & 4 & 2b_1 - b_2 \\ 0 & 0 & -1 & -2b_1 + b_2 + b_3 \\ 0 & 0 & -9 & -5b_1 + 2b_2 + b_4 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow -R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 0 & 1 & 4 & 2b_1 - b_2 \\ 0 & 0 & 1 & 2b_1 - b_2 - b_3 \\ 0 & 0 & -9 & -5b_1 + 2b_2 + b_4 \end{array}\right] \xrightarrow{R_4 \leftarrow R_4 + 9R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & b_1 \\ 0 & 1 & 4 & 2b_1 - b_2 \\ 0 & 0 & 1 & 2b_1 - b_2 - b_3 \\ 0 & 0 & 0 & 13b_1 - 7b_2 - 9b_3 + b_4 \end{array}\right]$$

$$\xrightarrow[R_1 \leftarrow R_1 - 2R_3]{R_2 \leftarrow R_2 - 4R_3} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -3b_1 + 2b_2 + 2b_3 \\ 0 & 1 & 0 & -6b_1 + 3b_2 + 4b_3 \\ 0 & 0 & 1 & 2b_1 - b_2 - b_3 \\ 0 & 0 & 0 & 13b_1 - 7b_2 - 9b_3 + b_4 \end{array}\right]$$

2. The A-part of the last row is zero, so that row reads $0 = 13b_1 - 7b_2 - 9b_3 + b_4$. The system is consistent exactly when this is zero. (the slide writes the same expression as $b_4 - 7b_2 - 9b_3 + 13b_1$; only the order differs)

3 Check against the earlier examples. For $\mathbf{b} = (3, 0, 5, 6)$: $39 - 0 - 45 + 6 = 0$, consistent. For $\mathbf{b}' = (2, 0, 1, 4)$: $26 - 0 - 9 + 4 = 21 \neq 0$, inconsistent. That 21 is exactly the 21 in the last row of Example 2.

ANSWER

Constraint equation: $b_4 - 7b_2 - 9b_3 + 13b_1 = 0$. Notice that the vector $\mathbf{b}$ in Example 1 satisfies the constraint equation, whereas $\mathbf{b}'$ in Example 2 does not.

Example 4: the word "constraint equation"

EXAMPLE 4 · LECTURE 4 P.22–24 For which vectors $\mathbf{b} \in \mathbb{R}^4$ does the system $x_1 + x_2 + 2x_3 = b_1,\ x_2 + x_3 = b_2,\ x_1 + x_2 + x_3 = b_3,\ 2x_1 + x_2 + 2x_3 = b_4$ have a solution? In other words: for which $\mathbf{b}$ is there a linear combination of the column vectors of the coefficient matrix that equals $\mathbf{b}$?

1 Row-reduce $[A|\mathbf{b}]$ exactly as the slide does:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 1 & b_2 \\ 1 & 1 & 1 & b_3 \\ 2 & 1 & 2 & b_4 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 - 2R_1]{R_3 \leftarrow R_3 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 1 & b_2 \\ 0 & 0 & -1 & -b_1 + b_3 \\ 0 & -1 & -2 & -2b_1 + b_4 \end{array}\right]$$

$$\xrightarrow[R_3 \leftarrow -R_3]{R_4 \leftarrow R_4 + R_2} \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 1 & b_2 \\ 0 & 0 & 1 & b_1 - b_3 \\ 0 & 0 & -1 & -2b_1 + b_2 + b_4 \end{array}\right] \xrightarrow{R_4 \leftarrow R_4 + R_3} \left[\begin{array}{ccc|c} 1 & 1 & 2 & b_1 \\ 0 & 1 & 1 & b_2 \\ 0 & 0 & 1 & b_1 - b_3 \\ 0 & 0 & 0 & -b_1 + b_2 - b_3 + b_4 \end{array}\right]$$

2 The system is consistent if and only if the last row does not produce a contradiction, i.e. $-b_1 + b_2 - b_3 + b_4 = 0 \quad (**)$.

ANSWER

$\mathbf{b} \in \mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ if and only if $-b_1 + b_2 - b_3 + b_4 = 0$. The set of all such $\mathbf{b}$ is the hyperplane $-x_1 + x_2 - x_3 + x_4 = 0$ in $\mathbb{R}^4$.

Constraint equation, p.24

The vector $\mathbf{b} \in \mathrm{Span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ if and only if it satisfies the condition $(**)$, which is called a **constraint equation** for consistency. Equivalently, the set of all such vectors $\mathbf{b}$ lies in the hyperplane defined by the Cartesian equation $-x_1 + x_2 - x_3 + x_4 = 0$ in $\mathbb{R}^4$.

Example 5: a different matrix

EXAMPLE 5 · LECTURE 4 P.25–27 Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 2 & 5 & 1 \\ 1 & 4 & 2 \end{bmatrix}$. Find the constraint equation (if any) that $\mathbf{b} \in \mathbb{R}^4$ must satisfy in order for $A\mathbf{x} = \mathbf{b}$ to be consistent.

1. Form $[A|\mathbf{b}]$ and row-reduce. The slide's "subtract row 2 from row 3, and eliminate other entries" means: clear column 2, then move the zero row to the bottom and scale:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & b_1 \\ 0 & 1 & 3 & b_2 \\ 2 & 5 & 1 & b_3 \\ 1 & 4 & 2 & b_4 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_3 \leftarrow R_3 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & b_1 \\ 0 & 1 & 3 & b_2 \\ 0 & 1 & 3 & -2b_1 + b_3 \\ 0 & 2 & 3 & -b_1 + b_4 \end{array}\right]$$

$$\xrightarrow[R_4 \leftarrow R_4 - 2R_2]{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 2 & -1 & b_1 \\ 0 & 1 & 3 & b_2 \\ 0 & 0 & 0 & -2b_1 - b_2 + b_3 \\ 0 & 0 & -3 & -b_1 - 2b_2 + b_4 \end{array}\right] \xrightarrow[R_3 \leftarrow -\frac{1}{3}R_3]{R_3 \leftrightarrow R_4} \left[\begin{array}{ccc|c} 1 & 2 & -1 & b_1 \\ 0 & 1 & 3 & b_2 \\ 0 & 0 & 1 & \frac{b_1 + 2b_2 - b_4}{3} \\ 0 & 0 & 0 & -2b_1 - b_2 + b_3 \end{array}\right]$$

2. The zero row of the A-part is the last one. Consistency condition: $b_3 - 2b_1 - b_2 = 0$.

ANSWER

Constraint equation: $b_3 - 2b_1 - b_2 = 0$. The consistent $\mathbf{b}$ form the hyperplane $-2x_1 - x_2 + x_3 = 0$ in $\mathbb{R}^4$, with normal vector $(-2, -1, 1, 0)$.

Sign slip on p.26

The slide's last matrix shows $\dfrac{b_1 + 2b_2 - b_4}{-3}$ in row 3. Dividing $-b_1 - 2b_2 + b_4$ by -3 actually gives $\dfrac{b_1 + 2b_2 - b_4}{3}$ (as above). This entry is not part of the constraint, so the boxed answer $b_3 - 2b_1 - b_2 = 0$ is correct.

Remark, p.27

Since a matrix A can have different echelon forms depending on the row operations used, one may derive *different constraint equations* that describe the same subspace.

Your answer may look different from the key, and still be right

Any nonzero multiple of a constraint is the same constraint: $b_3 - 2b_1 - b_2 = 0$ and $2b_1 + b_2 - b_3 = 0$ are equally correct. With two or more constraints, any set of combinations describing the same equations is also fine. To check your answer, plug in a vector you know is consistent, for example a column of A itself: $\mathbf{v}_1 = (1, 0, 2, 1)$ gives $2 - 2 - 0 = 0$. ✓

From parametric to Cartesian descriptions

Earlier, we described solution sets (lines, planes, hyperplanes) **parametrically**, as linear combinations. The constraint equations go the other way: starting from the column vectors of A, they describe $\mathrm{Span}(A)$ by **Cartesian equations** (p.28). The same trick works for any plane given by a point and two direction vectors.

EXAMPLE 6 · LECTURE 4 P.29–32 Consider the plane $\mathcal{P} \subset \mathbb{R}^3$ given by $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + s\begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + t\begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix}$. Determine the Cartesian equation of $\mathcal{P}$ by finding the constraint condition on x, y, z for which the linear system has a solution (s, t).

1. A point $\mathbf{b} = (b_1, b_2, b_3)$ lies on the plane iff $\begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + s\begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} + t\begin{bmatrix} -2 \\ 1 \\ 4 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$ for some $s, t \in \mathbb{R}$.
2. Move the fixed point to the right: $\begin{bmatrix} 1 & -2 \\ 3 & 1 \\ 2 & 4 \end{bmatrix}\begin{bmatrix} s \\ t \end{bmatrix} = \begin{bmatrix} b_1 - 2 \\ b_2 + 1 \\ b_3 \end{bmatrix}$. (the unknowns are now s, t)
3. Row-reduce:

$$\left[\begin{array}{cc|c} 1 & -2 & b_1 - 2 \\ 3 & 1 & b_2 + 1 \\ 2 & 4 & b_3 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - 3R_1} \left[\begin{array}{cc|c} 1 & -2 & b_1 - 2 \\ 0 & 7 & -3b_1 + b_2 + 7 \\ 0 & 8 & -2b_1 + b_3 + 4 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 - \frac{8}{7}R_2} \left[\begin{array}{cc|c} 1 & -2 & b_1 - 2 \\ 0 & 7 & -3b_1 + b_2 + 7 \\ 0 & 0 & \frac{10b_1 - 8b_2 + 7b_3 - 28}{7} \end{array}\right]$$

(the slide writes the last entry as $(-2b_1 + b_3 + 4) - \frac{8}{7}(-3b_1 + b_2 + 7)$; simplified it is the fraction shown)

4. Consistency: $\dfrac{10b_1 - 8b_2 + 7b_3 - 28}{7} = 0 \iff 10b_1 - 8b_2 + 7b_3 = 28.$

ANSWER

$\mathbf{b}$ lies in the plane iff $10b_1 - 8b_2 + 7b_3 = 28$. A Cartesian equation of the plane: $10x_1 - 8x_2 + 7x_3 = 28$, or equivalently $10x - 8y + 7z = 28$.

Not '= 0' here

A plane through the origin (a span) gives a constraint "$\cdots = 0$". Example 6's plane is shifted by the point $(2, -1, 0)$, so the right side is 28. Quick check: the point $(2, -1, 0)$ itself must satisfy it: $20 + 8 + 0 = 28$. ✓

How many constraints?

The number of constraints, p.33

Given an $m \times n$ matrix A: how many constraints must a vector $\mathbf{b} \in \mathbb{R}^m$ satisfy in order to be a linear combination of the columns of A?

- Each **zero row** in the echelon form of A contributes **one constraint equation**.
- That is, each such row corresponds to a condition that $\mathbf{b}$ must satisfy for the system $A\mathbf{x} = \mathbf{b}$ to be consistent.

The rank chapter turns this into a formula: the number of constraints is $m - r$, where $r = \text{rank}(A)$. Examples 3, 4, 5: $m = 4$, $r = 3$, one constraint each. Example 6: $m = 3$, $r = 2$, one constraint.

The recipe for midterm Question 3

1. Write $[A \mid \mathbf{b}]$ with $\mathbf{b} = (b_1, \ldots, b_m)$ as letters.
2. Row-reduce to echelon form. Name every operation; it acts on the b-column too.
3. Each row whose A-part is all zeros gives one constraint: (its b-entry) $= 0$.
4. Clear fractions, tidy up, box it.
5. Test a given $\mathbf{b}$: substitute. $0 \Rightarrow$ consistent; $\neq 0 \Rightarrow$ no solution.

FIND THE CONSTRAINT EQUATION(S), THEN TEST A VECTOR

- **Pivots are numbers, never b's.** Divide only by numbers. Never divide by an expression like $b_3 - b_1$.
- **No zero row?** Then there is no constraint: the system is consistent for every $\mathbf{b}$. Say so, that is the "(if any)" in the question.
- **Self-check in 10 seconds:** every column of A is a consistent right-hand side, so each column must satisfy your constraint.

EXAM-STYLE · MODELED ON P.25–27 AND THE 2025 MIDTERM Q3 Let $A = \begin{bmatrix} 1 & 1 & 0 \\ -2 & 1 & 1 \\ 1 & 3 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. (a) Determine a constraint equation that b_1, b_2, b_3, b_4 must satisfy for $A\mathbf{x} = \mathbf{b}$ to be consistent. (b) Using (a), decide whether $A\mathbf{x} = \mathbf{b}$ has a solution for $\mathbf{b} = (1, 1, 1, 1)$ and for $\mathbf{b} = (1, 0, 2, 0)$.

1. **Step 1–2.** Row-reduce $[A|\mathbf{b}]$. Row 4 has a 1 in column 2, so swapping it up avoids fractions:

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ -2 & 1 & 1 & b_2 \\ 1 & 3 & 1 & b_3 \\ 0 & 1 & 1 & b_4 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 + 2R_1} \left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 0 & 3 & 1 & 2b_1 + b_2 \\ 0 & 2 & 1 & -b_1 + b_3 \\ 0 & 1 & 1 & b_4 \end{array}\right]$$

$$\xrightarrow{R_2 \leftrightarrow R_4} \left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 0 & 1 & 1 & b_4 \\ 0 & 2 & 1 & -b_1 + b_3 \\ 0 & 3 & 1 & 2b_1 + b_2 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 - 3R_2]{R_3 \leftarrow R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 0 & 1 & 1 & b_4 \\ 0 & 0 & -1 & -b_1 + b_3 - 2b_4 \\ 0 & 0 & -2 & 2b_1 + b_2 - 3b_4 \end{array}\right]$$

$$\xrightarrow{R_4 \leftarrow R_4 - 2R_3} \left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 0 & 1 & 1 & b_4 \\ 0 & 0 & -1 & -b_1 + b_3 - 2b_4 \\ 0 & 0 & 0 & 4b_1 + b_2 - 2b_3 + b_4 \end{array}\right]$$

2. **Step 3.** The last row reads $0 = 4b_1 + b_2 - 2b_3 + b_4$.
3. **Self-check** with the columns: $(1, -2, 1, 0)$: $4 - 2 - 2 + 0 = 0$ ✓; $(1, 1, 3, 1)$: $4 + 1 - 6 + 1 = 0$ ✓; $(0, 1, 1, 1)$: $0 + 1 - 2 + 1 = 0$ ✓.
4. **(b)** $\mathbf{b} = (1, 1, 1, 1)$: $4 + 1 - 2 + 1 = 4 \neq 0$, so **no solution**.
5. $\mathbf{b} = (1, 0, 2, 0)$: $4 + 0 - 4 + 0 = 0$, so the system is **consistent**. (Bonus: back-substitute in the echelon form: $-x_3 = -1 + 2 - 0 = 1 \Rightarrow x_3 = -1$; $x_2 + x_3 = b_4 = 0 \Rightarrow x_2 = 1$; $x_1 + x_2 = b_1 = 1 \Rightarrow x_1 = 0$. Check: $A(0, 1, -1) = (1, 0, 2, 0)$ ✓.)

ANSWER

(a) $4b_1 + b_2 - 2b_3 + b_4 = 0$. (b) $(1, 1, 1, 1)$: no solution ($4 \neq 0$). $(1, 0, 2, 0)$: has a solution, $\mathbf{x} = (0, 1, -1)$.

Two zero rows, two constraints

PRACTICE · MODELED ON P.33 Find all constraint equations for $A\mathbf{x} = \mathbf{b}$ to be consistent, where $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 2 & 3 & 1 \\ 0 & 1 & 1 \end{bmatrix}$. Is $\mathbf{b} = (1, 2, 3, 1)$ in the column span? And $\mathbf{b} = (1, 2, 3, 0)$?

1. $$\left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 1 & 2 & 1 & b_2 \\ 2 & 3 & 1 & b_3 \\ 0 & 1 & 1 & b_4 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - 2R_1]{R_2 \leftarrow R_2 - R_1} \left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 0 & 1 & 1 & -b_1 + b_2 \\ 0 & 1 & 1 & -2b_1 + b_3 \\ 0 & 1 & 1 & b_4 \end{array}\right]$$
$$\xrightarrow[R_4 \leftarrow R_4 - R_2]{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 1 & 0 & b_1 \\ 0 & 1 & 1 & -b_1 + b_2 \\ 0 & 0 & 0 & -b_1 - b_2 + b_3 \\ 0 & 0 & 0 & b_1 - b_2 + b_4 \end{array}\right]$$
2. Two zero rows in the A-part, so **two** constraints: $-b_1 - b_2 + b_3 = 0$ and $b_1 - b_2 + b_4 = 0$. $(m - r = 4 - 2 = 2)$
3. $(1, 2, 3, 1)$: $-1 - 2 + 3 = 0$ and $1 - 2 + 1 = 0$. Both hold: consistent (indeed it is column 2 of A).
4. $(1, 2, 3, 0)$: first holds, second gives $1 - 2 + 0 = -1 \neq 0$. One failure is enough: inconsistent.

ANSWER
$b_3 = b_1 + b_2$ and $b_4 = b_2 - b_1$, both required. $(1, 2, 3, 1)$ is in the span; $(1, 2, 3, 0)$ is not.

"If any": when there is no constraint

PRACTICE · MODELED ON P.25 AND P.37 Find the constraint equation (if any) that $\mathbf{b}$ must satisfy for $A\mathbf{x} = \mathbf{b}$ to be consistent, $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 1 \\ 1 & 3 & 1 \end{bmatrix}$.

1. $$\left[\begin{array}{ccc|c} 1 & 2 & 1 & b_1 \\ 2 & 5 & 1 & b_2 \\ 1 & 3 & 1 & b_3 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 1 & b_1 \\ 0 & 1 & -1 & -2b_1 + b_2 \\ 0 & 1 & 0 & -b_1 + b_3 \end{array}\right]$$
$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & b_1 \\ 0 & 1 & -1 & -2b_1 + b_2 \\ 0 & 0 & 1 & b_1 - b_2 + b_3 \end{array}\right]$$
2. Every row of the A-part has a pivot. There is no row $0 = \ldots$, so nothing can go wrong.

ANSWER
No constraint equation: $A\mathbf{x} = \mathbf{b}$ is consistent for **every** $\mathbf{b} \in \mathbb{R}^3$ (here $\operatorname{rank} A = 3 = m$).

Classic ways to lose points on Question 3

- Doing the row operations on A only and forgetting the b-column. Every operation acts on the whole row.
- A sign slip when subtracting a combination like $2(b_3 - b_1)$. Expand carefully, then use the column self-check.
- For part (b), re-solving the whole system instead of substituting into your constraint. The question says "by using the constraint equation", so show the substitution and the number you get.
- Writing "no solution" for a $\mathbf{b}$ that satisfies the constraint. Zero means consistent.

In one minute

- **Consistent** = at least one solution; **inconsistent** = none (D1). Example 1 is consistent, Example 2 is inconsistent.
- Inconsistent $\iff$ the echelon form of $[A|\mathbf{b}]$ has a row $[0 \cdots 0 \,|\, c]$ with $c \neq 0$.
- Consistent $\iff$ $\mathbf{b} \in \mathrm{Span}(\text{columns of } A)$ $\iff$ the hyperplanes $\mathbf{A}_i \cdot \mathbf{x} = b_i$ meet.
- **Constraint equation:** carry $b_1, \ldots, b_m$ through the elimination; each zero row of the A-part gives (its b-entry)$= 0$.
- Number of constraints = number of zero rows of the echelon form of A (= $m - r$). None $\Rightarrow$ consistent for every $\mathbf{b}$.
- Constraints are not unique: multiples and combinations describe the same set (Remark p.27).
- Test a vector by substituting it into the constraint(s).
- A shifted plane (Example 6) gives "$= 28$", not "$= 0$": $10x - 8y + 7z = 28$.

CHAPTER 8

The rank of a matrix

One number, the rank, counts the pivots. Compare the rank of A with the rank of [A|b] and you know whether a system has a solution without solving it (the Rouché–Capelli theorem).

From Lecture 4, pages 34–38

Definition

Definition D2, p.34

The **rank** of a matrix A is the number of nonzero rows (equivalently, the number of pivot positions) in any echelon form of A, denoted by $\operatorname{rank}(A) = r$. The rank is usually denoted by r, and we always have

$$\operatorname{rank}(A) = r \leq \min(m, n).$$

Why $r \leq \min(m, n)$? Each pivot needs its own row **and** its own column. An $m \times n$ matrix has only m rows and n columns, so there can be at most m pivots and at most n pivots.

"Any echelon form": different row operations give different echelon forms, but they all have the same number of nonzero rows. So you may reduce in whatever order is convenient.

PRACTICE · MODELED ON P.34 Find the rank of $M = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 8 \\ -1 & -2 & 1 & -1 \end{bmatrix}$ and its pivot columns.

1

$$\begin{bmatrix} 1 & 2 & 0 & 3 \\ 2 & 4 & 1 & 8 \\ -1 & -2 & 1 & -1 \end{bmatrix} \xrightarrow[R_3 \leftarrow R_3 + R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

2 Echelon form reached: two nonzero rows, pivots in columns 1 and 3. (column 2 has no pivot: its entries below row 1 vanished together with column 1's)

ANSWER

$\operatorname{rank} M = 2 \leq \min(3, 4) = 3$. Pivot columns: 1 and 3.

Rank of A and rank of [A|b], from one reduction

When you reduce the augmented matrix $[A|\mathbf{b}]$ to echelon form $[U|\mathbf{c}]$, the part U is automatically an echelon form of A. So one reduction gives both ranks:

- $\operatorname{rank} A$ = number of nonzero rows of U (look **left** of the bar only);
- $\operatorname{rank}[A|\mathbf{b}]$ = number of nonzero rows of the whole $[U|\mathbf{c}]$.

Example (Lecture 4)	last row of the echelon form	$\operatorname{rank} A$	$\operatorname{rank}[A\|\mathbf{b}]$	consistent?
Example 1, $\mathbf{b} = (3, 0, 5, 6)$	$[0\ 0\ 0 \mid 0]$	3	3	yes
Example 2, $\mathbf{b}' = (2, 0, 1, 4)$	$[0\ 0\ 0 \mid 21]$	3	4	no
Example 6 (3×2)	$[0\ 0 \mid \frac{10b_1 - 8b_2 + 7b_3 - 28}{7}]$	2	2 or 3	iff entry $= 0$

→ **The two ranks differ by at most one**

Adding the column $\mathbf{b}$ can only turn a zero row of U into a nonzero row $[0 \cdots 0 \mid c]$. In an echelon form of the whole augmented matrix, the last column can hold **at most one** pivot (if two rows were $[0 \cdots 0 \mid 5]$ and $[0 \cdots 0 \mid 3]$, one more operation turns the second into a zero row). So either $\operatorname{rank}[A|\mathbf{b}] = \operatorname{rank} A$ (consistent) or $\operatorname{rank}[A|\mathbf{b}] = \operatorname{rank} A + 1$ (inconsistent). Nothing else.

! **The matrix you are given may not be in echelon form**

This is exactly what happened in the 2025 midterm Question 2(b): the leading entries of the given rows were stacked in the same column. You must finish the reduction before counting.

PRACTICE · MODELED ON THE 2025 MIDTERM Q2(B) Find $\operatorname{rank} A$ and $\operatorname{rank}[A|\mathbf{b}]$ for the system with augmented matrix $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -2 & 3 \end{array}\right]$.

1. Rows 2 and 3 both start in column 3, so this is **not** an echelon form yet. Counting "3 nonzero rows" now would be wrong.
2. $$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & -2 & 3 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 + R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 4 \end{array}\right]$$
3. Left of the bar: two nonzero rows. Whole matrix: three nonzero rows (the last row is $[0\ 0\ 0 \mid 4]$).

ANSWER

$\operatorname{rank} A = 2$, $\operatorname{rank}[A|\mathbf{b}] = 3$: the system is inconsistent (the last row says $0 = 4$).

Rank counts the zero rows, and so the constraints

Summary: Consistency and Constraint, p.35

Given a matrix $A \in M_{m\times n}(\mathbb{R})$, let $\operatorname{rank}(A) = r$. Then:

- The number of **zero rows** in any row echelon form of A is $m - r$.
- Therefore, a vector $\mathbf{b} \in \mathbb{R}^m$ must satisfy $m - r$ **constraint equations** in order for the system $A\mathbf{x} = \mathbf{b}$ to be consistent.

p.36

Now consider a system of m linear equations in n variables, with coefficient matrix A and rank r. We summarize the discussion as follows:

- The system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ satisfies $m - r$ **constraint equations**.
- These constraints arise from the zero rows of A in the row echelon form of the augmented matrix $[A \,|\, \mathbf{b}]$.
- The rank r represents the number of pivot rows, i.e., the number of **independent** equations in the system.

$$\#\text{constraint equations} = \#\text{zero rows in an echelon form of } A = m - r$$

COUNT CONSTRAINTS BEFORE YOU EVEN START (P.35)

Examples: Example 4 and Example 5 have $m = 4$, $r = 3$, so exactly one constraint each. A 4×3 matrix of rank 2 would give two. A matrix with $r = m$ gives none.

Consistency and rank: the Rouché–Capelli theorem

Proposition P1 (Rouché–Capelli Theorem), p.37

Let $A \in M_{m\times n}(\mathbb{R})$ be the coefficient matrix of a linear system $A\mathbf{x} = \mathbf{b}$ with m equations and n variables, and let $r = \operatorname{rank}(A)$. The linear system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if the rank of the augmented matrix equals the rank of the coefficient matrix:

$$A\mathbf{x} = \mathbf{b} \text{ is consistent} \iff \operatorname{rank}(A) = \operatorname{rank}([A \,|\, \mathbf{b}]).$$

In particular: If A has full row rank, i.e., $\operatorname{rank}(A) = m$, then $A\mathbf{x} = \mathbf{b}$ has at least one solution for every $\mathbf{b} \in \mathbb{R}^m$.

The slide "Proof of Proposition P1" (p.38) is left blank for the lecture. Here is a complete proof using only what we already know.

PROOF OF PROPOSITION P1 · P.38 (BLANK ON THE SLIDE) Prove: $A\mathbf{x} = \mathbf{b}$ is consistent $\iff$ $\operatorname{rank}(A) = \operatorname{rank}([A|\mathbf{b}])$.

1. Row-reduce $[A|\mathbf{b}]$ to an echelon form $[U|\mathbf{c}]$ of the **whole** augmented matrix. Row operations do not change the solution set, and U is an echelon form of A. (the operations look only at the columns of A to choose pivots)
2. By D2, $\operatorname{rank}(A)$ = number of nonzero rows of U, and $\operatorname{rank}([A|\mathbf{b}])$ = number of nonzero rows of $[U|\mathbf{c}]$.
3. A row of $[U|\mathbf{c}]$ is nonzero while its U-part is zero exactly when it has the form $[0 \cdots 0 \,|\, c]$ with $c \neq 0$. So $\operatorname{rank}([A|\mathbf{b}]) - \operatorname{rank}(A)$ = number of such rows.
4. ($\Rightarrow$) If the system is consistent, there is no row $[0 \cdots 0 \,|\, c]$ with $c \neq 0$ (p.18), so the two ranks are equal.
5. ($\Leftarrow$) If the ranks are equal, there is no such row, so by p.18 the system is consistent. ■
6. **"In particular".** If $\operatorname{rank}(A) = m$, then U has a pivot in every one of its m rows, so U has no zero row at all. Then a row $[0 \cdots 0 \,|\, c]$ cannot occur, whatever $\mathbf{b}$ is: the system is consistent for every $\mathbf{b} \in \mathbb{R}^m$. (Also: $m \leq \operatorname{rank}[A|\mathbf{b}] \leq m$ since it has only m rows.)

ANSWER
Consistent $\iff$ no row $[0 \cdots 0 \,|\, c \neq 0]$ $\iff$ $\operatorname{rank} A = \operatorname{rank}[A|\mathbf{b}]$. Full row rank ($r = m$) $\Rightarrow$ consistent for every $\mathbf{b}$.

★ **Where this appears in the midterm**
Question 2 asks for "the rank of the coefficient matrix A and the augmented matrix $[A|\mathbf{b}]$ ": this is P1 in action. The 2025 Question 4(b) used the "in particular" part: $\operatorname{rank}(A) = m$ means $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$, so the map $\mathbf{x} \mapsto A\mathbf{x}$ is onto.

! **Rank is about A, the count of unknowns is n**
Do not mix up m (rows, equations) and n (columns, unknowns). Full **row** rank $r = m$ is about existence for every $\mathbf{b}$. Uniqueness will need $r = n$ (next chapters).

In one minute

- $\operatorname{rank}(A)$ = number of nonzero rows = number of pivots in **any** echelon form of A; $r \leq \min(m, n)$.
- Reduce $[A|\mathbf{b}]$ once: count nonzero rows left of the bar for $\operatorname{rank} A$, in the whole matrix for $\operatorname{rank}[A|\mathbf{b}]$.
- Always finish the echelon form before counting (midterm Q2(b) trap).
- Zero rows of an echelon form of A: $m - r$; number of constraint equations: $m - r$.
- **P1 (Rouché–Capelli):** consistent $\iff$ $\operatorname{rank} A = \operatorname{rank}[A|\mathbf{b}]$. Otherwise $\operatorname{rank}[A|\mathbf{b}] = \operatorname{rank} A + 1$.
- $r = m$ (full row rank): at least one solution for every $\mathbf{b} \in \mathbb{R}^m$.

CHAPTER 9

Homogeneous and inhomogeneous systems

Every system Ax = b has a twin with zero on the right, Ax = 0. Solve the twin, find one solution of the original, add them, and you have every solution. You will also learn to spot a homogeneous system in one glance.

From Lecture 4, pages 39–47

Two kinds of systems

We now turn to the question of **how many solutions** a given consistent system of equations has (p.39). The first step is to compare a system with its "zero right-hand side" version.

Definition D3, p.39

A system $A\mathbf{x} = \mathbf{b}$ of linear equations is called **inhomogeneous** if $\mathbf{b} \neq \mathbf{0}$. The corresponding system

$$A\mathbf{x} = \mathbf{0}$$

is called the **associated homogeneous system**.

So a system is **homogeneous** when every right-hand side is zero. In an augmented matrix, that means **the column after the bar is all zeros**. This is part i.) of midterm Question 2.

Inhomogeneous means: b is not the zero vector

$\mathbf{b} = (0, 0, 1, 0)$ is **not** the zero vector, so a system with that right-hand side is **inhomogeneous**, even though most of its entries are zero. One nonzero entry anywhere in the last column is enough.

Note, p.47

A homogeneous system of linear equations, $A\mathbf{x} = \mathbf{0}$, is always consistent because the trivial solution $\mathbf{x} = \mathbf{0}$ always satisfies the equation.

Never 'no solution' for a homogeneous system

If your work on a homogeneous system ends with "inconsistent", you made an arithmetic mistake: row operations keep a zero column zero, so a row $[0 \cdots 0 \,|\, c \neq 0]$ cannot appear. For a homogeneous system, $\operatorname{rank} A = \operatorname{rank}[A|\mathbf{0}]$ always.

The distributive property

To relate the solutions of $A\mathbf{x} = \mathbf{b}$ to those of $A\mathbf{x} = \mathbf{0}$ we need one algebraic fact (p.40).

Proposition P2, p.40

Let $A \in M_{m\times n}(\mathbb{R})$ and $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}.$$

This is known as the **distributive property** of matrix multiplication.

PROOF OF PROPOSITION P2 · P.41 (COMPLETED) Prove $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$.

1. Recall that, by definition, the ith entry of the product $A\mathbf{x}$ is equal to the dot product $\mathbf{A}_i \cdot \mathbf{x}$, where $\mathbf{A}_i$ is the ith row vector of A. (this much is on the slide)
2. So the ith entry of $A(\mathbf{x} + \mathbf{y})$ is $\mathbf{A}_i \cdot (\mathbf{x} + \mathbf{y})$.
3. The dot product distributes over addition: $\mathbf{A}_i \cdot (\mathbf{x} + \mathbf{y}) = \mathbf{A}_i \cdot \mathbf{x} + \mathbf{A}_i \cdot \mathbf{y}$. ($\sum_j a_{ij}(x_j + y_j) = \sum_j a_{ij}x_j + \sum_j a_{ij}y_j$)
4. The right side is the ith entry of $A\mathbf{x}$ plus the ith entry of $A\mathbf{y}$, i.e. the ith entry of $A\mathbf{x} + A\mathbf{y}$.
5. This holds for every $i = 1, \ldots, m$, so the two vectors are equal. ■

ANSWER

$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$, entry by entry. In the same way $A(c\mathbf{x}) = cA\mathbf{x}$, so also $A(\mathbf{x} - \mathbf{y}) = A\mathbf{x} - A\mathbf{y}$.

The general form of all solutions

Theorem T1, p.42

Assume the system $A\mathbf{x} = \mathbf{b}$ is consistent, and let $\mathbf{u}_1$ be a particular solution. Then, all solutions $\mathbf{u}$ can be written as

$$\mathbf{u} = \mathbf{u}_1 + \mathbf{v}$$

for some solution $\mathbf{v}$ of the associated homogeneous system $A\mathbf{x} = \mathbf{0}$.

PROOF OF THEOREM T1 · P.43 (BLANK ON THE SLIDE) Let $A\mathbf{u}_1 = \mathbf{b}$. Show (a) every solution $\mathbf{u}$ has the form $\mathbf{u}_1 + \mathbf{v}$ with $A\mathbf{v} = \mathbf{0}$; and (b) conversely, every such $\mathbf{u}_1 + \mathbf{v}$ is a solution.

1. (a) Let $\mathbf{u}$ be any solution, so $A\mathbf{u} = \mathbf{b}$. Define $\mathbf{v} = \mathbf{u} - \mathbf{u}_1$; then $\mathbf{u} = \mathbf{u}_1 + \mathbf{v}$ automatically.
2. By P2, $A\mathbf{v} = A(\mathbf{u} - \mathbf{u}_1) = A\mathbf{u} - A\mathbf{u}_1 = \mathbf{b} - \mathbf{b} = \mathbf{0}$. So $\mathbf{v}$ solves the associated homogeneous system.
3. (b) Let $A\mathbf{v} = \mathbf{0}$. By P2, $A(\mathbf{u}_1 + \mathbf{v}) = A\mathbf{u}_1 + A\mathbf{v} = \mathbf{b} + \mathbf{0} = \mathbf{b}$. So $\mathbf{u}_1 + \mathbf{v}$ is a solution. ■

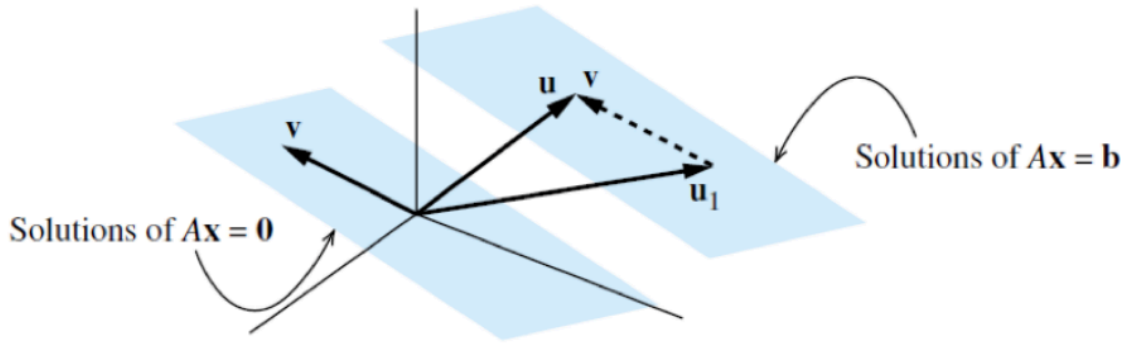
ANSWER

$\{\text{solutions of } A\mathbf{x} = \mathbf{b}\} = \{\mathbf{u}_1 + \mathbf{v} \mid A\mathbf{v} = \mathbf{0}\}$. Part (a) is T1 as stated; part (b) is the converse used in the Remark on p.44.

Remark, p.44

When the inhomogeneous system $A\mathbf{x} = \mathbf{b}$ is consistent, its solutions form a set obtained by translating the solution set of the associated homogeneous system $A\mathbf{x} = \mathbf{0}$ by a particular solution $\mathbf{u}_1$. In other words, the entire solution set of $A\mathbf{x} = \mathbf{b}$ is the affine subset

$$\{\mathbf{u}_1 + \mathbf{v} \mid A\mathbf{v} = \mathbf{0}\}.$$



Solutions of Ax = b are the solutions of Ax = 0 shifted by a particular solution u1: u = u1 + v (Lecture 4 p.44)

The picture: shifted hyperplanes

Since $\mathbf{u}_1$ lies on each of the hyperplanes $\mathbf{A}_i \cdot \mathbf{x} = b_i$, $i = 1, \ldots, m$, we can translate each hyperplane $\mathbf{A}_i \cdot \mathbf{x} = 0$, which passes through the origin, by the vector $\mathbf{u}_1$. Therefore, translating the intersection of the hyperplanes $\mathbf{A}_i \cdot \mathbf{x} = 0$ by $\mathbf{u}_1$ yields the intersection of the hyperplanes $\mathbf{A}_i \cdot \mathbf{x} = b_i$ (p.45).

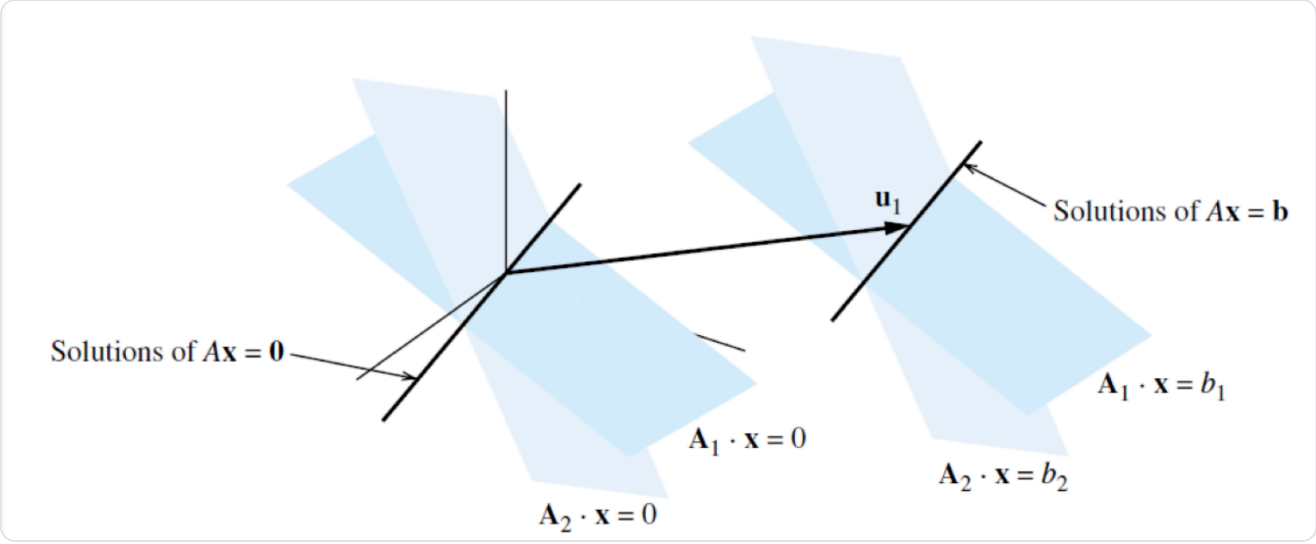


Figure A: two planes through the origin meet in a line (solutions of Ax = 0); the parallel planes A1·x = b1, A2·x = b2 meet in the parallel line through u1 (Lecture 4 p.46)

How many free variables?

Solutions to homogeneous systems, p.47

If the coefficient matrix A has rank r, then the system has exactly r pivot variables and $n - r$ free variables in its general solution. **In particular:** If $r = n$, then there are no free variables, and the trivial solution $\mathbf{x} = \mathbf{0}$ is the only solution to the system.

$$\underbrace{n}_{\text{unknowns}} = \underbrace{r}_{\text{pivot variables}} + \underbrace{n - r}_{\text{free variables}}$$

EVERY COLUMN OF A IS EITHER A PIVOT COLUMN OR A FREE COLUMN (P.47)

Worked example: homogeneous, then inhomogeneous

PRACTICE · MODELED ON P.42–47 Let $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 0 \\ 1 & 2 & 2 & -3 \end{bmatrix}$. (a) Solve $A\mathbf{x} = \mathbf{0}$. (b) Solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = (2, 7, 5)$ and write the answer in the form $\mathbf{u}_1 + \mathbf{v}$ of Theorem T1.

1 **(a)** Homogeneous: the last column is zero and stays zero.

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 3 & 0 \\ 2 & 4 & 1 & 0 & 0 \\ 1 & 2 & 2 & -3 & 0 \end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cccc|c} 1 & 2 & -1 & 3 & 0 \\ 0 & 0 & 3 & -6 & 0 \\ 0 & 0 & 3 & -6 & 0 \end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{cccc|c} 1 & 2 & -1 & 3 & 0 \\ 0 & 0 & 3 & -6 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{R_2 \leftarrow \frac{1}{3} R_2} \left[\begin{array}{cccc|c} 1 & 2 & -1 & 3 & 0 \\ 0 & 0 & 1 & -2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

2 $r = 2$, $n = 4$: pivot variables x_1, x_3; $n - r = 2$ free variables $x_2 = s$, $x_4 = t$.

3. Back-substitute: $x_3 = 2t$; $x_1 = -2s + x_3 - 3t = -2s - t$. So

$$\mathbf{v} = s\begin{bmatrix}-2\\1\\0\\0\end{bmatrix} + t\begin{bmatrix}-1\\0\\2\\1\end{bmatrix}, \quad s,t \in \mathbb{R}.$$

4. **(b)** Same operations with $\mathbf{b} = (2,7,5)$:

$$\left[\begin{array}{cccc|c}1&2&-1&3&2\\2&4&1&0&7\\1&2&2&-3&5\end{array}\right] \xrightarrow[R_3 \leftarrow R_3 - R_1]{R_2 \leftarrow R_2 - 2R_1} \left[\begin{array}{cccc|c}1&2&-1&3&2\\0&0&3&-6&3\\0&0&3&-6&3\end{array}\right]$$

$$\xrightarrow{R_3 \leftarrow R_3 - R_2} \left[\begin{array}{cccc|c}1&2&-1&3&2\\0&0&3&-6&3\\0&0&0&0&0\end{array}\right] \xrightarrow{R_2 \leftarrow \frac{1}{3}R_2} \left[\begin{array}{cccc|c}1&2&-1&3&2\\0&0&1&-2&1\\0&0&0&0&0\end{array}\right]$$

5. The last row is $[0\ 0\ 0\ 0 \mid 0]$: consistent ($\operatorname{rank} A = \operatorname{rank}[A|\mathbf{b}] = 2$).

6. Back-substitute: $x_3 = 1 + 2t$; $x_1 = 2 - 2s + x_3 - 3t = 3 - 2s - t$. Setting $s = t = 0$ gives a particular solution $\mathbf{u}_1 = (3,0,1,0)$. Check: $A\mathbf{u}_1 = (3+0-1+0,\ 6+0+1+0,\ 3+0+2+0) = (2,7,5)$ ✓.

ANSWER

(a) $\mathbf{v} = s(-2,1,0,0) + t(-1,0,2,1)$. (b) $\mathbf{x} = \underbrace{(3,0,1,0)}_{\mathbf{u}_1} + \underbrace{s(-2,1,0,0) + t(-1,0,2,1)}_{\mathbf{v},\ A\mathbf{v}=\mathbf{0}}$, $s,t \in \mathbb{R}$:

infinitely many solutions.

Any particular solution works

With $s = t = 1$ we get another solution $\mathbf{u}_2 = (0,1,3,1)$ (check: $A\mathbf{u}_2 = (2,7,5)$). Theorem T1 says $\mathbf{u}_2 - \mathbf{u}_1 = (-3,1,2,1)$ must solve $A\mathbf{x} = \mathbf{0}$, and it does: it is $\mathbf{v}$ with $s = t = 1$. Different particular solutions give the same solution set, just written differently.

Where the constraint is hiding

Here $m - r = 3 - 2 = 1$, so some $\mathbf{b}$ are not allowed. Carrying b_1, b_2, b_3 through the same operations, the last row gives $b_1 - b_2 + b_3 = 0$; our $\mathbf{b} = (2,7,5)$ satisfies it ($2 - 7 + 5 = 0$), which is why (b) was consistent.

In one minute

- **Homogeneous:** $A\mathbf{x} = \mathbf{0}$ (last column of the augmented matrix all zero). **Inhomogeneous:** $\mathbf{b} \neq \mathbf{0}$ (D3).
- A homogeneous system is always consistent: $\mathbf{x} = \mathbf{0}$ is a solution. Never answer "no solution".
- **P2:** $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$ (entry i is $\mathbf{A}_i \cdot (\mathbf{x} + \mathbf{y})$).
- **T1:** if $A\mathbf{u}_1 = \mathbf{b}$, every solution is $\mathbf{u}_1 + \mathbf{v}$ with $A\mathbf{v} = \mathbf{0}$ (proof: $A(\mathbf{u} - \mathbf{u}_1) = \mathbf{b} - \mathbf{b} = \mathbf{0}$).
- Solution set of $A\mathbf{x} = \mathbf{b}$ = the solution set of $A\mathbf{x} = \mathbf{0}$ shifted by $\mathbf{u}_1$ (affine subset, Figure A).
- Rank r: r pivot variables, $n - r$ free variables. $r = n \Rightarrow$ only the trivial solution.

CHAPTER 10

Uniqueness, and how to classify any system

None, exactly one, or infinitely many: three numbers decide it, rank A, rank [A|b] and n. This chapter gives you the one-glance table and a fixed four-line answer for midterm Question 2, plus the nonsingular-matrix theorem.

From Lecture 4, pages 48–54

When is the solution unique?

Definition, p.48

A system of linear equations $A\mathbf{x} = \mathbf{b}$ is said to have a **unique solution** if there is exactly one vector $\mathbf{x}$ that satisfies the equation.

Homogeneous case, p.48

The system $A\mathbf{x} = \mathbf{0}$ (homogeneous) has a unique solution if and only if the rank of A is equal to the number of unknowns, i.e., $r = n$. In this case, the only solution is the trivial solution $\mathbf{x} = \mathbf{0}$. **Infinitely Many Solutions:** If $r < n$, then the homogeneous system has infinitely many solutions due to the presence of free variables.

Why infinitely many and not, say, two? A free variable can be **any** real number, and each choice gives a different solution. One free variable is already enough for infinitely many.

Important Observations, p.49

- It is impossible for $r > n$, since the number of pivot positions cannot exceed the number of columns.
- Likewise, there cannot be more pivot positions than rows, thus: $r \leq m$.
- Therefore, if $n > m$ (more variables than equations), the system $A\mathbf{x} = \mathbf{0}$ must have infinitely many solutions.

The last point in one line: $r \leq m < n$, so $r < n$, so there is a free variable.

Back to inhomogeneous systems

From Theorem T1, if the inhomogeneous system $A\mathbf{x} = \mathbf{b}$ is consistent, then its solutions can be described as $\mathbf{x} = \mathbf{u} + \mathbf{v}$, where $\mathbf{u}$ is a particular solution to $A\mathbf{x} = \mathbf{b}$, and $\mathbf{v}$ is a solution to the associated homogeneous system $A\mathbf{x} = \mathbf{0}$ (p.50).

Proposition P3, p.50

Suppose the system $A\mathbf{x} = \mathbf{b}$ is consistent. Then:

- It has a **unique solution** if and only if the associated homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
- This occurs precisely when the rank of A is equal to the number of unknowns, i.e., $r = n$.

Why (not on the slides, but it is two lines): the solution set is $\{\mathbf{u} + \mathbf{v} \mid A\mathbf{v} = \mathbf{0}\}$. It has exactly one element if and only if the only $\mathbf{v}$ is $\mathbf{0}$, i.e. $A\mathbf{x} = \mathbf{0}$ has only the trivial solution, which by p.48 means $r = n$.

Consistency first, then uniqueness

$r = n$ does **not** by itself mean "unique solution" for $A\mathbf{x} = \mathbf{b}$. P3 starts with "Suppose the system is consistent". Example 2 had $r = n = 3$ and **no** solution. Always check $\operatorname{rank} A = \operatorname{rank}[A|\mathbf{b}]$ first.

The one-glance table

Put P1 (Rouché–Capelli), P3 and the free-variable count together. With $r = \operatorname{rank} A$ and n = number of unknowns (columns of A):

$\operatorname{rank} A$ vs $\operatorname{rank}[A\|\mathbf{b}]$	r vs n	Number of solutions	Why
$\operatorname{rank}[A\|\mathbf{b}] = r + 1$ (different)	anything	**none** (inconsistent)	P1: a row $[0 \cdots 0 \mid c \neq 0]$
equal	$r = n$	**exactly one** (unique)	P3: no free variables
equal	$r < n$	**infinitely many**	$n - r$ free variables (parameters)

Homogeneous $A\mathbf{x} = \mathbf{0}$: the first row of the table never happens. Unique (only $\mathbf{x} = \mathbf{0}$) if $r = n$; infinitely many if $r < n$; always infinitely many if $n > m$.

Shortcuts from the shape: $n > m$ (wide) $\Rightarrow$ never unique: none or infinitely many. $r = m$ (full row rank) $\Rightarrow$ never "none". $r = m = n \Rightarrow$ exactly one, for every $\mathbf{b}$.

A special case: unique solution for every b

p.51

When does the inhomogeneous system $A\mathbf{x} = \mathbf{b}$ have a **unique solution for every** $\mathbf{b} \in \mathbb{R}^m$?

- From Proposition P1, for the system to be consistent for all $\mathbf{b}$, we must have $\text{rank}(A) = m$.
- From Proposition P3, for the solution to be unique, we must have $\text{rank}(A) = n$.

Conclusion: Both conditions can hold simultaneously only when $\text{rank}(A) = m = n$. That is, A must be a square matrix of full rank.

Why 'must' in the first bullet

P1's "in particular" gives: $r = m \Rightarrow$ consistent for every $\mathbf{b}$. The converse also holds: if $r < m$, the echelon form of A has $m - r \geq 1$ zero rows, so $\mathbf{b}$ must satisfy at least one constraint equation, and any $\mathbf{b}$ breaking it gives no solution.

Nonsingular and singular matrices

Definition, p.52

An $n \times n$ matrix is called **nonsingular** if its rank $r = n$, and **singular** if its rank $r < n$.
Observation: An $n \times n$ matrix is nonsingular if and only if its row echelon form has a pivot in every row, and hence also in every column. This means its reduced echelon form must be the identity matrix:

$$R = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix} = I_n.$$

Remark: Although it may seem trivial, when $m = n$, having $r = n$ automatically implies $r = m$, and vice versa.

Proposition P4 (Characterizations of nonsingular matrices), p.53

Let A be an $n \times n$ matrix. The following statements are **equivalent**:

1. A is **nonsingular** (i.e., $\text{rank}(A) = n$).
2. The reduced echelon form of A must be **the identity matrix**.
3. The homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the **trivial solution**.
4. For every $\mathbf{b} \in \mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has a **unique solution**.

Conclusion: Any one of these conditions guarantees the others. Together, they provide several equivalent ways to characterize when a square matrix is nonsingular.

Example: the matrix $\begin{bmatrix}1 & 2 & 1\\ 2 & 5 & 1\\ 1 & 3 & 1\end{bmatrix}$ from the "if any" example in the constraint chapter reduced to three pivots, so it is nonsingular: $A\mathbf{x}=\mathbf{b}$ has exactly one solution for every $\mathbf{b}\in\mathbb{R}^3$, and $A\mathbf{x}=\mathbf{0}$ only $\mathbf{x}=\mathbf{0}$.

The recipe for midterm Question 2

The question gives several augmented matrices and asks, for each: i) homogeneous or inhomogeneous, ii) $\operatorname{rank}A$ and $\operatorname{rank}[A|\mathbf{b}]$, iii) pivot and free variables, iv) the number of solutions if it exists. Answer in that order, every time:

1. **Set up.** Columns left of the bar = A. Count n = number of those columns (unknowns $x_1,\dots,x_n$) and m = number of rows.
2. **Echelon?** Check the staircase. If two rows start in the same column (or a zero row sits above a nonzero row), reduce first.
3. **i) Type.** Last column all zeros $\Rightarrow$ homogeneous; otherwise inhomogeneous ($\mathbf{b}\neq\mathbf{0}$).
4. **ii) Ranks.** $\operatorname{rank}A$ = nonzero rows left of the bar; $\operatorname{rank}[A|\mathbf{b}]$ = nonzero rows of the whole matrix.
5. **iii) Variables.** Pivot variables = x_j for the pivot columns j; free variables = the other x_j.
6. **iv) Count.** Read the one-glance table. Give a reason: "consistent and rank = number of variables", "row $[0\ 0\ 0\,|\,1]$", "free variable x_2".

> ★ **What the official key writes when there is no solution**
> In the 2025 key, for the inconsistent system the pivot and free variables are written as "—" ("system inconsistent, no solution"): with no solutions there is no general solution to describe. Safest answer: write "—, the system is inconsistent", and you may add "(the echelon form of A has pivots in columns ...)". That shows you know both.

EXAM-STYLE · MODELED ON THE 2025 MIDTERM Q2 For each system, given by its augmented matrix, determine i) homogeneous or inhomogeneous, ii) $\operatorname{rank}A$ and $\operatorname{rank}[A|\mathbf{b}]$, iii) pivot and free variables, iv) the number of solutions.

$$\text{(a)}\left[\begin{array}{cccc|c}1 & -2 & 0 & 1 & 0\\ 0 & 0 & 1 & -1 & 0\\ 0 & 0 & 0 & 0 & 0\end{array}\right]\qquad \text{(b)}\left[\begin{array}{ccc|c}1 & 2 & 1 & 4\\ 0 & 1 & -1 & 1\\ 0 & 2 & -2 & 3\end{array}\right]\qquad \text{(c)}\left[\begin{array}{ccc|c}1 & 0 & 2 & 1\\ 0 & 1 & -1 & 2\\ 0 & 0 & 3 & 3\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\end{array}\right]$$

1 **(a)** $n=4$ unknowns, already in echelon form. i) Last column zero: **homogeneous**. ii) $\operatorname{rank}A=\operatorname{rank}[A|\mathbf{b}]=2$. iii) Pivots in columns 1 and 3: pivot variables x_1,x_3; free variables x_2,x_4. iv) Consistent (homogeneous) and $r=2<4=n$: **infinitely many**. (general solution: $x_2=s,\ x_4=t,\ x_3=t,\ x_1=2s-t$)

2 **(b)** $n = 3$. Rows 2 and 3 both start in column 2: not echelon yet.

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 2 & -2 & 3 \end{array}\right] \xrightarrow{R_3 \leftarrow R_3 - 2R_2} \left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

i) $\mathbf{b} = (4, 1, 3) \neq \mathbf{0}$: **inhomogeneous**. ii) $\operatorname{rank} A = 2$, $\operatorname{rank}[A|\mathbf{b}] = 3$. iii) —, inconsistent (the echelon form of A has pivots in columns 1, 2). iv) Row $[0\,0\,0\,|\,1]$ means $0 = 1$: **no solution**.

3 **(c)** $n = 3$ (five rows, but only three unknowns!), echelon form. i) **Inhomogeneous**. ii) $\operatorname{rank} A = \operatorname{rank}[A|\mathbf{b}] = 3$ (the two zero rows are $0 = 0$). iii) Pivot variables x_1, x_2, x_3; no free variables. iv) Consistent and $r = 3 = n$: **unique**. ($3x_3 = 3 \Rightarrow x_3 = 1$, $x_2 = 2 + x_3 = 3$, $x_1 = 1 - 2x_3 = -1$)

ANSWER

(a) homogeneous, ranks 2 and 2, pivot x_1, x_3, free x_2, x_4, infinitely many. (b) inhomogeneous, ranks 2 and 3, no solution. (c) inhomogeneous, ranks 3 and 3, pivot x_1, x_2, x_3, no free variable, unique solution $(-1, 3, 1)$.

Where the points go on Question 2

- **n is the number of columns of A, not the number of rows.** System (c) has five rows but three unknowns; zero rows do not add unknowns.
- Counting ranks on a matrix that is not yet in echelon form (system (b)).
- Counting the $\mathbf{b}$-column as a variable or a "pivot variable". Variables live left of the bar only.
- Answering "unique" from $r = n$ without checking consistency first.
- Calling a system with a zero last column "inconsistent". Homogeneous systems always have $\mathbf{x} = \mathbf{0}$.

Exercises slide, p.54

The slide lists: "1. Find constraint equation (if any) that $\mathbf{b}$ must satisfy in order for $A\mathbf{x} = \mathbf{b}$ to be consistent." (no matrix is given on the slide; practise with the recipe in the constraint chapter and the problem bank) and "2. Solve all the questions provided in Section 1.5 and Exercise 1.5", from *Shifrin & Adams (2010), Linear Algebra: A Geometric Approach*.

In one minute

- **Unique solution** = exactly one $\mathbf{x}$. Homogeneous: unique ($\mathbf{x} = \mathbf{0}$) $\iff$ $r = n$; $r < n \Rightarrow$ infinitely many.
- $r \leq m$ and $r \leq n$. More unknowns than equations ($n > m$) $\Rightarrow$ $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions.
- **P3:** if consistent, unique $\iff$ $A\mathbf{x} = \mathbf{0}$ only trivial $\iff$ $r = n$.
- **Table:** ranks differ $\Rightarrow$ none; equal and $r = n \Rightarrow$ one; equal and $r < n \Rightarrow$ infinitely many.

- Unique solution for every $\mathbf{b} \iff \operatorname{rank} A = m = n$ (square, full rank).
- **Nonsingular** $n \times n$: $r = n \iff$ RREF $= I_n \iff A\mathbf{x} = \mathbf{0}$ only trivial $\iff$ unique solution for every $\mathbf{b}$ (P4). Singular: $r < n$.
- Question 2 order: type, ranks, pivot/free, count. Reduce to echelon first; n = columns of A.

CHAPTER 11

Matrices, sums and scalar multiples

What an m×n matrix is, how to name its entries, rows and columns, the special square shapes (diagonal, upper and lower triangular), and the two easy operations: adding matrices and multiplying them by a number.

From Lecture 5, pages 3–16

What is an m×n matrix?

A matrix is a table of numbers. Its **size** is always written "rows × columns", in that order: first how many rows (going down), then how many columns (going across).

Matrix notation, Lecture 5 p.3

An $m \times n$ matrix A is a rectangular array consisting of mn real numbers, organized into m rows and n columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

and denoted by $A = [a_{ij}]$. The entry a_{ij} is located in the i-th row and j-th column.

Row first, column second

Both the size and the subscripts follow the same rule: **row first, then column**. So a_{23} means "row 2, column 3". A 2×3 matrix has 2 rows and 3 columns, which makes $2 \cdot 3 = 6$ entries.

READING ENTRIES · LECTURE 5 P.11 MATRICES Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $C = \begin{bmatrix} 13 & 14 \\ 15 & 16 \\ 17 & 18 \end{bmatrix}$. Give the sizes, a_{12}, a_{21} and c_{32}.

1. A has 2 rows and 3 columns, so A is 2×3. C has 3 rows and 2 columns, so C is 3×2.
2. a_{12}: row 1, column 2 of A, which is 2. a_{21}: row 2, column 1, which is 4. ($a_{12} \neq a_{21}$: the order of the subscripts matters)
3. c_{32}: row 3, column 2 of C, which is 18.

ANSWER

A is 2×3, C is 3×2; $a_{12} = 2$, $a_{21} = 4$, $c_{32} = 18$.

Equality of matrices

1.1 Matrix equality, p.4

Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ of the same dimensions $(m \times n)$ are equal if and only if $a_{ij} = b_{ij}$ for all $i = 1, \ldots, m$ and $j = 1, \ldots, n$.

So equality needs two things: the **same size**, and the **same number in every position**. A 2×3 matrix is never equal to a 3×2 matrix, even if they contain the same six numbers.

Rows and columns as vectors

An $m \times n$ matrix can be read as a stack of rows or as a row of columns (p.5):

- **Row vectors:** A has m rows $A_1 = (a_{11}, \ldots, a_{1n}),\ \ldots,\ A_m = (a_{m1}, \ldots, a_{mn})$. Each row has n entries, so $A_i \in \mathbb{R}^n$.
- **Column vectors:** A has n columns $\mathbf{a}_1 = \begin{bmatrix} a_{11} \\ \vdots \\ a_{m1} \end{bmatrix},\ \ldots,\ \mathbf{a}_n = \begin{bmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{bmatrix}$. Each column has m entries, so $\mathbf{a}_j \in \mathbb{R}^m$.

Rows have n entries, columns have m entries

It feels backwards at first. A row runs *across* all n columns, so it has n entries and lies in $\mathbb{R}^n$. A column runs *down* all m rows, so it lies in $\mathbb{R}^m$. For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$: rows are in $\mathbb{R}^3$, columns are in $\mathbb{R}^2$.

Zero matrix, matrix sets, square matrices

Zero matrix O (p.6)	The $m \times n$ matrix in which all entries are zero. (There is one zero matrix for each size.)
$\mathbb{M}_{m\times n}(\mathbb{R})$ **(p.6)**	The set of all $m \times n$ matrices with real entries: $\mathbb{M}_{m\times n}(\mathbb{R}) = \{A \mid A \text{ is an } m \times n \text{ matrix with real entries}\}$.
Square matrix (p.7)	Same number of rows and columns, $m = n$. We write $\mathbb{M}_n(\mathbb{R})$ instead of $\mathbb{M}_{n\times n}(\mathbb{R})$.
Diagonal entries (p.7)	For $A = [a_{ij}] \in \mathbb{M}_n(\mathbb{R})$, the entries $a_{11}, a_{22}, \ldots, a_{nn}$ (the a_{ii}). They form the **main diagonal**, from top-left to bottom-right.

Special types of square matrices

Each type is described by *where the zeros must be*. Compare the row number i with the column number j: below the diagonal $i > j$, above it $i < j$, on it $i = j$.

Definition, p.8

Let $A = [a_{ij}] \in \mathbb{M}_n(\mathbb{R})$.

1. **Diagonal matrix:** all non-diagonal entries are zero, i.e. $a_{ij} = 0$ whenever $i \neq j$.
2. **Upper triangular:** all entries below the main diagonal are zero, i.e. $a_{ij} = 0$ whenever $i > j$.
3. **Lower triangular:** all entries above the main diagonal are zero, i.e. $a_{ij} = 0$ whenever $i < j$.

Diagonal	Upper triangular	Lower triangular
$\begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$	$\begin{bmatrix} 2 & 7 & 1 \\ 0 & -1 & 4 \\ 0 & 0 & 5 \end{bmatrix}$	$\begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 0 \\ 6 & 8 & 5 \end{bmatrix}$
zeros where $i \neq j$	zeros where $i > j$ (below)	zeros where $i < j$ (above)

Name it by what survives

"Upper" triangular: the non-zero part is the *upper* triangle, so the zeros are below. A diagonal matrix is both upper and lower triangular. The diagonal entries themselves may be zero too; the definitions only force zeros off the diagonal.

One more important diagonal matrix, the **identity matrix** I_n (ones on the diagonal, zeros elsewhere), comes in the next chapter together with multiplication (p.25).

Scalar multiplication

3. Scalar multiplication, p.9

Let $A = [a_{ij}] \in \mathbb{M}_{m\times n}(\mathbb{R})$ and $c \in \mathbb{R}$. The scalar multiple cA is obtained by multiplying **each entry** of A by c:

$$(cA)_{ij} = c \cdot a_{ij}, \quad \text{for all } i = 1, \ldots, m \text{ and } j = 1, \ldots, n.$$

The size does not change: cA is again $m \times n$.

Matrix addition

4. Matrix addition, p.10

Let $A = [a_{ij}],\ B = [b_{ij}] \in \mathbb{M}_{m\times n}(\mathbb{R})$. The sum $A + B$ is defined entry by entry:

$$(A + B)_{ij} = a_{ij} + b_{ij}, \quad \text{for all } i = 1, \ldots, m \text{ and } j = 1, \ldots, n.$$

Same size or nothing

You can only add matrices of the **same size**. If the sizes differ, the sum is **undefined**. Don't pad with zeros, and don't write "$= 0$". The correct answer is simply "$A + C$ is not defined".

EXAMPLE 1 · LECTURE 5 P.11–12 Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $B = \begin{bmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}$, $C = \begin{bmatrix} 13 & 14 \\ 15 & 16 \\ 17 & 18 \end{bmatrix}$.

Compute $A + B$, $4B$, and $A + C$, $B + C$ where possible.

1. **Sum.** A and B are both 2×3, so they can be added:

$$A + B = \begin{bmatrix} 1+7 & 2+8 & 3+9 \\ 4+10 & 5+11 & 6+12 \end{bmatrix} = \begin{bmatrix} 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}.$$

2. **Scalar multiple**, $c = 4$: multiply every entry of B by 4:

$$4B = 4\begin{bmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix} = \begin{bmatrix} 28 & 32 & 36 \\ 40 & 44 & 48 \end{bmatrix}.$$

3. **Incompatible sum.** A and B are 2×3, but C is 3×2. Addition requires the same dimensions. (p.12)

ANSWER

$A + B = \begin{bmatrix} 8 & 10 & 12 \\ 14 & 16 & 18 \end{bmatrix}$, $4B = \begin{bmatrix} 28 & 32 & 36 \\ 40 & 44 & 48 \end{bmatrix}$; $A + C$ and $B + C$ are undefined.

PRACTICE (MODELED ON P.11–12) With A, B as above, compute $2A - B$.

1. $2A = \begin{bmatrix} 2 & 4 & 6 \\ 8 & 10 & 12 \end{bmatrix}$. (scalar multiple first)
2. $2A - B = 2A + (-1)B = \begin{bmatrix} 2-7 & 4-8 & 6-9 \\ 8-10 & 10-11 & 12-12 \end{bmatrix}$.

ANSWER

$2A - B = \begin{bmatrix} -5 & -4 & -3 \\ -2 & -1 & 0 \end{bmatrix}$.

Properties of addition and scalar multiplication

Proposition 1.1, p.13–14

Let $A, B, C \in \mathbb{M}_{m\times n}(\mathbb{R})$ and $c, d \in \mathbb{R}$. Then:

1. $A + B = B + A$ (commutativity of matrix addition).
2. $(A + B) + C = A + (B + C)$ (associativity of matrix addition).
3. $O + A = A$ (existence of an additive identity; O is the zero matrix).
4. There exists a matrix $-A$ such that $A + (-A) = O$ (existence of an additive inverse).
5. $c(dA) = (cd)A$ (scalar multiplication is associative).
6. $c(A + B) = cA + cB$ (distributivity over matrix addition).
7. $(c + d)A = cA + dA$ (distributivity over scalar addition).
8. $1A = A$.

Proof (the slide on p.15 is left blank, so here it is)

One idea proves all eight

Two matrices of the same size are equal when every (i, j)-entry agrees (p.4). Every operation here works **entry by entry**. So each property reduces to the same rule for **real numbers**, applied in one position (i, j).

Fix any $i \in \{1, \dots, m\}$ and $j \in \{1, \dots, n\}$. All matrices involved are $m \times n$.

1. $(A + B)_{ij} = a_{ij} + b_{ij} = b_{ij} + a_{ij} = (B + A)_{ij}$. (real addition is commutative)
2. $((A + B) + C)_{ij} = (a_{ij} + b_{ij}) + c_{ij} = a_{ij} + (b_{ij} + c_{ij}) = (A + (B + C))_{ij}$. (real addition is associative)
3. $(O + A)_{ij} = 0 + a_{ij} = a_{ij}$.
4. Define $-A = [-a_{ij}]$, which is the same as $(-1)A$. Then $(A + (-A))_{ij} = a_{ij} + (-a_{ij}) = 0 = O_{ij}$.
5. $(c(dA))_{ij} = c\,(d\,a_{ij}) = (cd)\,a_{ij} = ((cd)A)_{ij}$.
6. $(c(A + B))_{ij} = c\,(a_{ij} + b_{ij}) = c\,a_{ij} + c\,b_{ij} = (cA + cB)_{ij}$.
7. $((c + d)A)_{ij} = (c + d)\,a_{ij} = c\,a_{ij} + d\,a_{ij} = (cA + dA)_{ij}$.
8. $(1A)_{ij} = 1 \cdot a_{ij} = a_{ij}$.

Both sides of each identity have the same size and agree in every position (i, j), so they are equal by the definition of matrix equality. ■

How to write such a proof in the exam

Three lines: (1) "Both sides are $m \times n$." (2) "For every i, j: (entry of left side) $= \cdots =$ (entry of right side)", with the real-number rule you used. (3) "Hence the matrices are equal."

A matrix is also a vector in $\mathbb{R}^{mn}$

An $m \times n$ matrix holds mn numbers. List them in a fixed order and you get a vector in $\mathbb{R}^{mn}$ (p.16). For example

$$A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \\ -5 & 4 \end{bmatrix} \in \mathbb{M}_{3\times 2}(\mathbb{R}) \quad \longleftrightarrow \quad (1, -1, 2, 3, -5, 4) \in \mathbb{R}^6,$$

listing the entries row-wise (column-wise also works, if you stick to one convention).

> **Why this matters**
> Matrix addition and scalar multiplication are entry by entry, exactly like vector addition and scalar multiplication in $\mathbb{R}^{mn}$. That is why the eight rules of Proposition 1.1 look exactly like the rules for vectors.

In one minute

- $m \times n$ = m rows, n columns; a_{ij} = row i, column j. Always row first.
- Rows are in $\mathbb{R}^n$, columns are in $\mathbb{R}^m$.
- Equal matrices: same size **and** same entry in every position.
- Square $\mathbb{M}_n(\mathbb{R})$: diagonal ($a_{ij} = 0$ for $i \neq j$), upper triangular ($a_{ij} = 0$ for $i > j$), lower triangular ($a_{ij} = 0$ for $i < j$).
- $(cA)_{ij} = c\,a_{ij}$; $(A + B)_{ij} = a_{ij} + b_{ij}$, defined only for equal sizes.
- Proposition 1.1 (8 rules) is proved entry by entry from the rules of real numbers.
- An $m \times n$ matrix can be encoded as a vector in $\mathbb{R}^{mn}$.

CHAPTER 12

Matrix multiplication

The one operation that is not entry by entry. Learn the row-times-column rule, the size check that tells you in two seconds whether a product exists, the identity matrix, the rules that do hold, and the classic traps (AB is not BA, and friends).

From Lecture 5, pages 17–32

When is AB defined? The size check

Multiplication glues two matrices together along a shared length. Write the sizes next to each other. The two **inner** numbers must match; the two **outer** numbers give the size of the answer.

$$\underset{m\times\boxed{n}}{A}\cdot\underset{\boxed{n}\times p}{B}=\underset{m\times p}{AB}\qquad \mathbb{M}_{m\times n}(\mathbb{R})\times\mathbb{M}_{n\times p}(\mathbb{R})\longrightarrow\mathbb{M}_{m\times p}(\mathbb{R})$$

SIZE CHECK (P.17): INNER SIZES EQUAL, OUTER SIZES GIVE THE RESULT

Note, p.19

The matrix product AB is defined only when the number of **columns of** A equals the number of **rows of** B.

A	B	AB?	BA?
2×3	3×2	2×2	3×3
2×3	3×1	2×1	undefined ($1\neq 2$)
2×2	2×2	2×2	2×2
3×2	3×1	undefined ($2\neq 3$)	undefined ($1\neq 3$)

The row-times-column rule

Definition: matrix multiplication, p.18

Let $A\in\mathbb{M}_{m\times n}(\mathbb{R})$ and $B\in\mathbb{M}_{n\times p}(\mathbb{R})$. The matrix product AB is the $m\times p$ matrix whose (i,j)-entry is

$$(AB)_{ij}=a_{i1}b_{1j}+a_{i2}b_{2j}+\cdots+a_{in}b_{nj}=\sum_{k=1}^{n}a_{ik}b_{kj}.$$

That is, $(AB)_{ij}$ is the dot product of the i-th row of A and the j-th column of B, both viewed as vectors in $\mathbb{R}^n$.

→ Where does entry (i, j) come from?

Row i of the answer uses **row i of A**. Column j of the answer uses **column j of B**. Lay the row on top of the column, multiply the pairs, and add. The size check is exactly what makes the row of A and the column of B the same length n.

ONE ENTRY IN COLOUR · EXAMPLE 2, LECTURE 5 P.19–21 Let $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 1 \end{bmatrix}$. Find the $(2,1)$-entry of AB.

1. Size check: $(2 \times \boxed{3})(\boxed{3} \times 2)$. Inner sizes match, so AB exists and is 2×2.
2. Entry $(2,1)$: take **row 2 of A** and **column 1 of B**:

$$\begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 1 \end{bmatrix} = \begin{bmatrix} \cdot & \cdot \\ \boxed{(AB)_{21}} & \cdot \end{bmatrix}$$

3. Multiply matching pairs and add:

$$(AB)_{21} = 1 \cdot 1 + 3 \cdot (-1) + 4 \cdot 3 = 1 - 3 + 12.$$

ANSWER

$(AB)_{21} = 10.$

★ Finger trick

Left index finger runs **across** a row of A, right index finger runs **down** a column of B, both at the same speed. Multiply what's under your fingers and keep a running sum.

Example 2: compare with the matrix–vector product (p.20–21)

A vector $x \in \mathbb{R}^n$ is an $n \times 1$ matrix, so Ax from Chapter 1 is just a special case of AB. The slides leave the answers blank ("....."); here they are.

EXAMPLE 2 · LECTURE 5 P.20–21 Let $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & 4 \end{bmatrix}$, $x = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 2 \\ -1 & 0 \\ 3 & 1 \end{bmatrix}$.

Compute Ax, AB, BA, and decide whether Bx is defined.

1. Ax: $(2 \times 3)(3 \times 1) = 2 \times 1$.

$$Ax = \begin{bmatrix} 2 \cdot 1 + (-1) \cdot 0 + 0 \cdot (-2) \\ 1 \cdot 1 + 3 \cdot 0 + 4 \cdot (-2) \end{bmatrix} = \begin{bmatrix} 2 \\ -7 \end{bmatrix}.$$

(slide p.20 blank)

2. AB: $(2 \times 3)(3 \times 2) = 2 \times 2$.

$$AB = \begin{bmatrix} 2\cdot 1 + (-1)(-1) + 0\cdot 3 & 2\cdot 2 + (-1)\cdot 0 + 0\cdot 1 \\ 1\cdot 1 + 3\cdot(-1) + 4\cdot 3 & 1\cdot 2 + 3\cdot 0 + 4\cdot 1 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 10 & 6 \end{bmatrix}.$$

(slide p.21 blank; entry $(2,1) = 10$ is the coloured one above)

3. BA: $(3 \times 2)(2 \times 3) = 3 \times 3$.

$$BA = \begin{bmatrix} 1\cdot 2 + 2\cdot 1 & 1\cdot(-1) + 2\cdot 3 & 1\cdot 0 + 2\cdot 4 \\ -1\cdot 2 + 0\cdot 1 & (-1)(-1) + 0\cdot 3 & -1\cdot 0 + 0\cdot 4 \\ 3\cdot 2 + 1\cdot 1 & 3\cdot(-1) + 1\cdot 3 & 3\cdot 0 + 1\cdot 4 \end{bmatrix} = \begin{bmatrix} 4 & 5 & 8 \\ -2 & 1 & 0 \\ 7 & 0 & 4 \end{bmatrix}.$$

(slide p.21 blank)

4. AB is 2×2 and BA is 3×3. They are not even the same size, so certainly $AB \neq BA$: matrix multiplication is generally **not commutative**.

5. Bx: $(3 \times \boxed{2})(\boxed{3} \times 1)$. Inner sizes $2 \neq 3$, so Bx is **not defined**.

ANSWER

$Ax = \begin{bmatrix} 2 \\ -7 \end{bmatrix}$, $AB = \begin{bmatrix} 3 & 4 \\ 10 & 6 \end{bmatrix}$, $BA = \begin{bmatrix} 4 & 5 & 8 \\ -2 & 1 & 0 \\ 7 & 0 & 4 \end{bmatrix}$, Bx undefined.

Slide p.21 wording

The slide says "the scalar multiplication Bx is not defined: B is a 3×2 matrix and x is an 1×3 matrix". It means the *matrix product* Bx, and x is a 3×1 column. The mismatch is: B has 2 columns, x has 3 rows.

Example 3: AB defined, BA not (p.22–23)

EXAMPLE 3 · LECTURE 5 P.22–23 Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{M}_{2\times 3}(\mathbb{R})$, $B = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \in \mathbb{M}_{3\times 1}(\mathbb{R})$.

Compute AB and BA if possible.

1. AB: $(2 \times 3)(3 \times 1) = 2 \times 1$, defined.

$$AB = \begin{bmatrix} 1\cdot 1 + 2\cdot 0 + 3\cdot(-1) \\ 4\cdot 1 + 5\cdot 0 + 6\cdot(-1) \end{bmatrix} = \begin{bmatrix} -2 \\ -2 \end{bmatrix}.$$

(slide p.22 blank)

2. BA: $(3 \times \boxed{1})(\boxed{2} \times 3)$. Inner dimensions 1 and 2 do not match, so BA is undefined.

ANSWER

$AB = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$; BA is undefined.

The slide calls this "not always reversible". It means only that you cannot swap the order of the factors; it has nothing to do with inverse matrices.

Example 4: even square matrices don't commute (p.24)

EXAMPLE 4 · LECTURE 5 P.24 Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix}$. Compute AB and BA.

1. $AB = \begin{bmatrix} 1\cdot 3 + 2\cdot 1 & 1\cdot 0 + 2\cdot 4 \\ 0\cdot 3 + 1\cdot 1 & 0\cdot 0 + 1\cdot 4 \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ 1 & 4 \end{bmatrix}$. (slide blank)
2. $BA = \begin{bmatrix} 3\cdot 1 + 0\cdot 0 & 3\cdot 2 + 0\cdot 1 \\ 1\cdot 1 + 4\cdot 0 & 1\cdot 2 + 4\cdot 1 \end{bmatrix} = \begin{bmatrix} 3 & 6 \\ 1 & 6 \end{bmatrix}$. (slide blank)
3. Same size 2×2, but different entries.

ANSWER

$$AB = \begin{bmatrix} 5 & 8 \\ 1 & 4 \end{bmatrix} \neq BA = \begin{bmatrix} 3 & 6 \\ 1 & 6 \end{bmatrix}.$$

Non-commutativity (p.21, p.24, p.26)

Matrix multiplication is, in general, **not commutative**: $AB \neq BA$ in most cases. Three ways it fails:

1. AB defined, BA not (Example 3);
2. both defined, different sizes (Example 2);
3. both defined, same size, different entries (Example 4).

To **disprove** "$AB = BA$ for all A, B", one counterexample such as Example 4 is a complete answer.

The identity matrix

Definition, p.25

The identity matrix I_n is the $n \times n$ square matrix with ones on the main diagonal and zeros elsewhere:

$$(I_n)_{ij} = \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j, \end{cases} \qquad I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}.$$

It plays the role of the number 1: $AI_n = A = I_m A$ for every $m \times n$ matrix A. Note the two different sizes: on the right of A you need I_n (n = columns of A), on the left I_m (m = rows of A), or the size check fails.

PRACTICE (PROP. 1.2 (1), P.26) Check $AI_3 = A = I_2A$ for $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$.

1. AI_3: $(2 \times 3)(3 \times 3) = 2 \times 3$. Row i of A dotted with column j of I_3 (a single 1 in position j) picks out a_{ij}. So $AI_3 = A$.
2. I_2A: $(2 \times 2)(2 \times 3) = 2 \times 3$. Row i of I_2 (a single 1 in position i) dotted with column j of A picks out a_{ij}. So $I_2A = A$.
3. I_2A would not work with I_3: $(3 \times 3)(2 \times 3)$ fails the size check.

ANSWER

$AI_3 = I_2A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$.

Properties that do hold

Proposition 1.2, p.26–27

Let $A, A' \in \mathbb{M}_{m\times n}(\mathbb{R})$, $B, B' \in \mathbb{M}_{n\times p}(\mathbb{R})$, $C \in \mathbb{M}_{p\times q}(\mathbb{R})$ and $c \in \mathbb{R}$. Then:

1. **Identity:** $AI_n = A = I_mA$. Therefore I_n is called the $n \times n$ identity matrix.
2. **Distributivity:** $(A + A')B = AB + A'B$ and $A(B + B') = AB + AB'$.
3. **Compatibility with scalars:** $(cA)B = c(AB) = A(cB)$.
4. **Associativity:** $(AB)C = A(BC)$.

Keep the order in distributivity

$A(B + B') = AB + AB'$: A stays on the **left** in both terms. Writing $A(B + B') = AB + B'A$ is wrong, since order matters.

Proof of associativity (p.28–30)

Claim. For $A \in \mathbb{M}_{m\times n}(\mathbb{R})$, $B \in \mathbb{M}_{n\times p}(\mathbb{R})$, $C \in \mathbb{M}_{p\times q}(\mathbb{R})$: $(AB)C = A(BC)$. Both sides are $m \times q$.

Strategy: show both sides have the same columns.

1. Let $c^{(j)}$ be the j-th column of C. The j-th column of a product is the left factor times the j-th column of the right factor, so

$$[(AB)C]^{(j)} = (AB)c^{(j)}, \qquad [A(BC)]^{(j)} = A\left(Bc^{(j)}\right).$$

   So we need $(AB)c^{(j)} = A(Bc^{(j)})$ for each column $c^{(j)}$ of C.

2. Let $b_1, \ldots, b_p$ be the columns of B and $c^{(j)} = \begin{bmatrix} c_1 \\ \vdots \\ c_p \end{bmatrix} \in \mathbb{R}^p$. Then $Bc^{(j)} = \sum_{k=1}^{p} c_k b_k$, a linear combination of the columns of B.
3. By linearity of the matrix–vector product: $A(Bc^{(j)}) = A\left(\sum_{k=1}^{p} c_k b_k\right) = \sum_{k=1}^{p} c_k\, Ab_k$.

4. Ab_k is the k-th column of AB, so $\sum_{k=1}^{p} c_k\,(AB)^{(k)} = (AB)c^{(j)}$. (again "matrix times vector = combination of its columns")
5. **Conclusion:** this holds for every column $c^{(j)}$ of C, so $(AB)C = A(BC)$. ■

Two facts used in the proof

(i) $Mv = v_1 m_1 + \cdots + v_p m_p$ where the m_k are the columns of M (matrix times vector is a combination of the columns). (ii) The j-th column of MN is M times the j-th column of N. Both come straight from the row-times-column rule. On p.29 the numbers $c_1, \ldots, c_p$ are really the entries $c_{1j}, \ldots, c_{pj}$ of column j; the slide drops the j.

EXAMPLE 5 · LECTURE 5 P.31–32 Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, $C = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Compute $(AB)C$ and $A(BC)$.

1. $AB = \begin{bmatrix} 1\cdot 2 + 2\cdot 1 & 1\cdot 0 + 2\cdot 3 \\ 0\cdot 2 + 1\cdot 1 & 0\cdot 0 + 1\cdot 3 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ 1 & 3 \end{bmatrix}$. (slide p.31 blank)
2. $(AB)C = \begin{bmatrix} 4 & 6 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 + 12 \\ 1 + 6 \end{bmatrix} = \begin{bmatrix} 16 \\ 7 \end{bmatrix}$.
3. $BC = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 + 6 \end{bmatrix} = \begin{bmatrix} 2 \\ 7 \end{bmatrix}$. (slide p.32 blank)
4. $A(BC) = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 7 \end{bmatrix} = \begin{bmatrix} 2 + 14 \\ 7 \end{bmatrix} = \begin{bmatrix} 16 \\ 7 \end{bmatrix}$.

ANSWER

$(AB)C = A(BC) = \begin{bmatrix} 16 \\ 7 \end{bmatrix}$. This confirms associativity in this example.

The classic traps

The slides prove that $AB \neq BA$ in general (p.24). Everything below follows from that fact and the distributive law (p.26). They are favourite "prove or disprove" items, so learn one counterexample for each.

Added for practice

The four traps below are not written on the Lecture 5 slides. They are direct consequences of p.24 and p.26. Every counterexample is checked with sympy. Here A^2 means AA, which needs A square.

1. AB and BA are different

Never swap factors. Example 4 above: $AB = \begin{bmatrix} 5 & 8 \\ 1 & 4 \end{bmatrix}$, $BA = \begin{bmatrix} 3 & 6 \\ 1 & 6 \end{bmatrix}$.

2. $(A+B)^2$ is not $A^2 + 2AB + B^2$

Expand with distributivity, keeping the order:

$$(A+B)^2 = (A+B)(A+B) = A^2 + AB + BA + B^2.$$

This equals $A^2 + 2AB + B^2$ **only if** $AB = BA$. With Example 4's matrices: $(A+B)^2 = \begin{bmatrix} 18 & 18 \\ 9 & 27 \end{bmatrix}$, but $A^2 + 2AB + B^2 = \begin{bmatrix} 20 & 20 \\ 9 & 25 \end{bmatrix}$. Similarly $(A+B)(A-B) = A^2 - AB + BA - B^2$, not $A^2 - B^2$ in general.

3. AB = O does not force A = O or B = O

For numbers, $ab = 0 \Rightarrow a = 0$ or $b = 0$. Not for matrices:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix},\ B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}: \quad AB = \begin{bmatrix} 1-1 & -1+1 \\ 1-1 & -1+1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

although neither A nor B is the zero matrix. Even $A^2 = O$ with $A \neq O$ happens: $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}^2 = O$.

4. Cancellation fails: AB = AC does not give B = C

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix},\ B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix},\ C = \begin{bmatrix} 1 & 2 \\ 5 & 6 \end{bmatrix}: \quad AB = AC = \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}, \text{ but } B \neq C.$$

Reason: $AB = AC \iff A(B-C) = O$, and by trap 3 this can happen with $B - C \neq O$.

In one minute

- Size check: $(m \times n)(n \times p) = (m \times p)$. Columns of A must equal rows of B.
- $(AB)_{ij} = \sum_k a_{ik} b_{kj}$ = row i of A · column j of B.
- Slide answers: $Ax = (2, -7)^T$; $AB = \begin{bmatrix} 3 & 4 \\ 10 & 6 \end{bmatrix}$, BA is 3×3; Ex.3 $AB = (-2, -2)^T$; Ex.4 $AB = \begin{bmatrix} 5 & 8 \\ 1 & 4 \end{bmatrix} \neq BA = \begin{bmatrix} 3 & 6 \\ 1 & 6 \end{bmatrix}$; Ex.5 $(16, 7)^T$.
- $AB \neq BA$ in general: one may be undefined, sizes may differ, or entries differ.
- I_n: ones on the diagonal; $AI_n = A = I_m A$.
- True: distributivity, $(cA)B = c(AB) = A(cB)$, $(AB)C = A(BC)$ (proved column by column).
- False in general: $(A+B)^2 = A^2 + 2AB + B^2$; $AB = O \Rightarrow A = O$ or $B = O$; $AB = AC \Rightarrow B = C$.

CHAPTER 13

Matrices as functions

Every m×n matrix A gives a function that sends a vector x with n entries to the vector Ax with m entries. Multiplying matrices is then just doing one function after another. That one idea explains associativity and solves the slide exercises, and it leads straight to midterm Q4(b): when is this function onto?

From Lecture 5, pages 33–38, and the MAT2037 Midterm 2025 (Q4)

The idea

In Chapter 1, matrices were bookkeeping: a compact way to write and solve linear systems. Here the same matrix becomes a **machine**. You feed in a vector, it gives back a vector (p.33).

Matrix as a function, p.34

Let $A \in \mathbb{M}_{m\times n}(\mathbb{R})$. Multiplying A by a vector $x \in \mathbb{R}^n$ defines a function

$$\mu_A : \mathbb{R}^n \to \mathbb{R}^m, \qquad \mu_A(x) = Ax.$$

μ_A maps vectors from the **domain** $\mathbb{R}^n$ to the **codomain** $\mathbb{R}^m$. This lets us study matrices as **linear transformations**: they stretch, rotate, reflect, or project vectors in space.

Domain = number of columns

For A of size $m \times n$, the input lives in $\mathbb{R}^n$ (n = number of **columns**, so that Ax passes the size check) and the output lives in $\mathbb{R}^m$ (m = number of **rows**). So a 2×3 matrix gives $\mu_A : \mathbb{R}^3 \to \mathbb{R}^2$, not $\mathbb{R}^2 \to \mathbb{R}^3$.

APPLYING THE FUNCTION OF A · MATRICES OF EXAMPLE 2, P.20 Let $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & 4 \end{bmatrix}$. State the domain and codomain of μ_A, and compute $\mu_A(1, 0, -2)$ and $\mu_A(e_2)$, where $e_2 = (0, 1, 0)$.

1. A is 2×3, so $\mu_A : \mathbb{R}^3 \to \mathbb{R}^2$.
2. $\mu_A \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 2\cdot 1 - 1\cdot 0 + 0\cdot(-2) \\ 1\cdot 1 + 3\cdot 0 + 4\cdot(-2) \end{bmatrix} = \begin{bmatrix} 2 \\ -7 \end{bmatrix}$. (this is Ax of p.20)
3. $\mu_A(e_2) = \begin{bmatrix} 2\cdot 0 - 1\cdot 1 + 0\cdot 0 \\ 1\cdot 0 + 3\cdot 1 + 4\cdot 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$, exactly the 2nd column of A.

ANSWER

$\mu_A : \mathbb{R}^3 \to \mathbb{R}^2$; $\mu_A(1, 0, -2) = (2, -7)$; $\mu_A(e_2) = (-1, 3) =$ column 2 of A.

The columns are the images of the unit vectors

Matrix times vector is a combination of the columns: $Ax = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ (used on p.29). With $x = e_j$ (a 1 in place j, zeros elsewhere) only $\mathbf{a}_j$ survives. So **the columns of A are exactly where μ_A sends** $e_1, \ldots, e_n$. In particular, if $\mu_A = \mu_B$ (same output for every input), then A and B have the same columns, so $A = B$.

Why "linear"?

Take B and B' in Proposition 1.2 to be column vectors $x, y \in \mathbb{R}^n$. Distributivity and compatibility with scalars (p.26–27) become

$$\mu_A(x+y) = \mu_A(x) + \mu_A(y), \qquad \mu_A(cx) = c\,\mu_A(x) \qquad (x, y \in \mathbb{R}^n,\ c \in \mathbb{R})$$

SUMS AND SCALAR MULTIPLES ARE PRESERVED: A LINEAR TRANSFORMATION

This is the "linearity of matrix multiplication" quoted in the associativity proof on p.29.

Multiplication is composition

From the associativity proof, $(AB)c = A(Bc)$ for every vector $c \in \mathbb{R}^p$, when A is $m \times n$ and B is $n \times p$ (p.35). Written with μ:

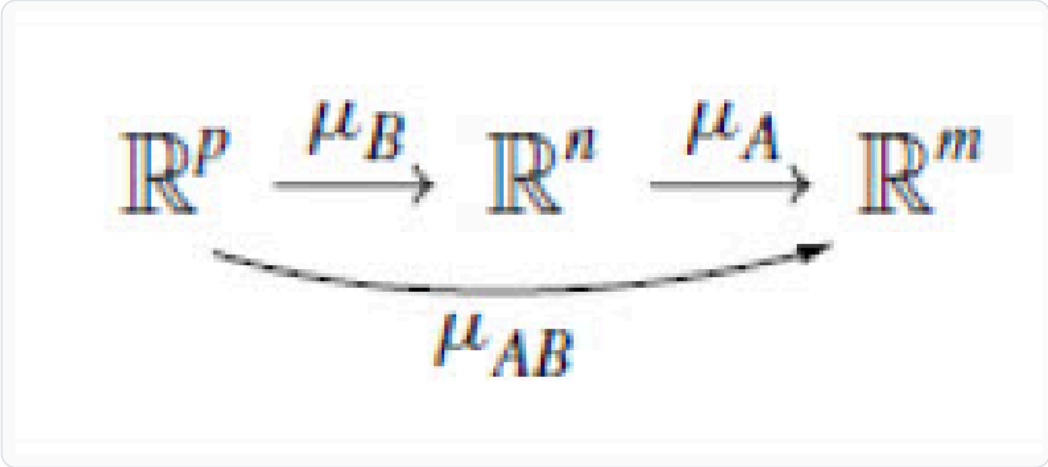
6.1 Matrix multiplication as composition of functions, p.35–37

$$\mu_{AB}(c) = \mu_A(\mu_B(c)) = (\mu_A \circ \mu_B)(c), \qquad \text{i.e.} \qquad \mu_{AB} = \mu_A \circ \mu_B,$$

where $\circ$ denotes composition of functions. Schematically:

$$\mathbb{R}^p \xrightarrow{\mu_B} \mathbb{R}^n \xrightarrow{\mu_A} \mathbb{R}^m, \qquad \mu_{AB} : \mathbb{R}^p \to \mathbb{R}^m \text{ goes straight across.}$$

This formula is the true motivation behind the definition of matrix multiplication: one may define AB as the composition $\mu_A \circ \mu_B$, and then derive the computational formula.



The commutative diagram for the product AB as a composition (Lecture 5 p.37)

Right to left

$\mu_A \circ \mu_B$ means "first μ_B, then μ_A", just like $AB\,c = A(Bc)$: the matrix next to the vector acts first. And the sizes must chain: μ_B must land in $\mathbb{R}^n$, the domain of μ_A. This is the size check $(m \times n)(n \times p)$ again.

COMPOSITION IN NUMBERS · MATRICES OF EXAMPLE 5, P.31–32 With $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}$, $c = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, compute $\mu_A(\mu_B(c))$ and $\mu_{AB}(c)$.

1. First μ_B: $\mu_B(c) = Bc = \begin{bmatrix} 2 \\ 7 \end{bmatrix}$.
2. Then μ_A: $\mu_A \begin{bmatrix} 2 \\ 7 \end{bmatrix} = \begin{bmatrix} 2+14 \\ 7 \end{bmatrix} = \begin{bmatrix} 16 \\ 7 \end{bmatrix}$.
3. In one go: $AB = \begin{bmatrix} 4 & 6 \\ 1 & 3 \end{bmatrix}$, $\mu_{AB}(c) = \begin{bmatrix} 4+12 \\ 1+6 \end{bmatrix} = \begin{bmatrix} 16 \\ 7 \end{bmatrix}$.

ANSWER

$\mu_A(\mu_B(c)) = \mu_{AB}(c) = (16, 7)$.

The conceptual reason for associativity (p.36)

Composition of functions is always associative: $(f \circ g) \circ h = f \circ (g \circ h)$. Hence

$$\mu_{(AB)C} = \mu_{AB} \circ \mu_C = (\mu_A \circ \mu_B) \circ \mu_C = \mu_A \circ (\mu_B \circ \mu_C) = \mu_A \circ \mu_{BC} = \mu_{A(BC)},$$

and so $(AB)C = A(BC)$. (The last step, "same function $\Rightarrow$ same matrix", is the column fact from the idea box above.)

Exercises on p.38, solved

EXERCISE 1 · LECTURE 5 P.38 How is μ_{I_n} defined, where I_n denotes the identity matrix?

1. I_n is $n \times n$, so $\mu_{I_n} : \mathbb{R}^n \to \mathbb{R}^n$.
2. $\mu_{I_n}(x) = I_n x = x$ for every $x \in \mathbb{R}^n$. (row i of I_n has a single 1 in place i, so it picks out x_i)

ANSWER

$\mu_{I_n} : \mathbb{R}^n \to \mathbb{R}^n$, $\mu_{I_n}(x) = x$: the identity function of $\mathbb{R}^n$.

EXERCISE 2 · LECTURE 5 P.38 Use matrix multiplication as function composition to prove Properties (1)–(3) of Proposition 1.2.

1. **Tool 1:** $\mu_{XY} = \mu_X \circ \mu_Y$ (p.35). **Tool 2:** if $\mu_X(v) = \mu_Y(v)$ for all v, then $X = Y$ (apply to e_j: same columns). **Tool 3** (Chapter 1, matrix times vector): $(X + X')v = Xv + X'v$, $X(u + v) = Xu + Xv$, $(cX)v = c(Xv) = X(cv)$.

2. **(1) Identity.** By Exercise 1, $\mu_{I_n} = \mathrm{id}_{\mathbb{R}^n}$. So $\mu_{AI_n} = \mu_A \circ \mu_{I_n} = \mu_A \circ \mathrm{id} = \mu_A$, hence $AI_n = A$. Likewise $\mu_{I_m A} = \mathrm{id}_{\mathbb{R}^m} \circ \mu_A = \mu_A$, hence $I_m A = A$.
3. **(2) Distributivity.** For every $v \in \mathbb{R}^p$: $\mu_{(A+A')B}(v) = (A+A')(Bv) = A(Bv) + A'(Bv) = (AB)v + (A'B)v = (AB + A'B)v$. So $(A+A')B = AB + A'B$.
   Also $\mu_{A(B+B')}(v) = A\big((B+B')v\big) = A(Bv + B'v) = A(Bv) + A(B'v) = (AB)v + (AB')v = (AB + AB')v$. So $A(B+B') = AB + AB'$.
4. **(3) Scalars.** For every $v \in \mathbb{R}^p$: $\mu_{(cA)B}(v) = (cA)(Bv) = c\,(A(Bv)) = c\,(AB)v = (c(AB))v$, and $\mu_{A(cB)}(v) = A\big((cB)v\big) = A\big(c(Bv)\big) = c\,(A(Bv)) = (c(AB))v$. So $(cA)B = c(AB) = A(cB)$.

ANSWER
In each case the two matrices define the same function, so by Tool 2 they are equal. ■

> **Exercise 3 (p.38)**
> "Solve all the questions in Section 2.1 and Exercise 2.1" refers to the textbook Shifrin & Adams (2010), *Linear Algebra: A Geometric Approach*, which is not part of this guide's sources. The practice problems in this guide cover the same skills.

From the midterm: when is μ_A onto?

> **Beyond Lecture 5**
> The slides stop at composition. The 2025 midterm (Q4(b)) asks about μ_A being **surjective**, using rank from the earlier chapters on linear systems. This section connects the two and goes no further.

Onto (surjective) — $\mu_A : \mathbb{R}^n \to \mathbb{R}^m$ is onto if for **every** $b \in \mathbb{R}^m$ there is at least one $x \in \mathbb{R}^n$ with $Ax = b$. In words: $Ax = b$ is consistent for every right-hand side b.

One-to-one (injective) — Different inputs give different outputs: $Ax = Ay \Rightarrow x = y$. In words: $Ax = b$ never has more than one solution.

MIDTERM 2025 Q4(B) Let A be an $m \times n$ matrix with $n \neq m$, and assume that $\mathrm{rank}(A) = m$. Consider $\mu_A : \mathbb{R}^n \to \mathbb{R}^m$, $\mu_A(x) = Ax$. Prove or disprove that μ_A is surjective.

1. **True.** Rank is the number of pivots, and there is at most one pivot per row and per column, so $\mathrm{rank}\, A \leq \min(m, n)$. From $\mathrm{rank}\, A = m$ and $n \neq m$ we get $m < n$.
2. Let $b \in \mathbb{R}^m$ be arbitrary and row reduce $[A \mid b]$. The coefficient part reduces to an echelon form of A with m pivots, i.e. **a pivot in every row**. So no row of the form $(0 \cdots 0 \mid \text{nonzero})$ can appear.
3. Hence $\mathrm{rank}[A \mid b] = \mathrm{rank}\, A = m$ and $Ax = b$ is consistent. (and since $n > m$ there are $n - m$ free variables, so there are infinitely many solutions)
4. So for every $b \in \mathbb{R}^m$ there is at least one $x \in \mathbb{R}^n$ with $\mu_A(x) = Ax = b$.

ANSWER

μ_A is surjective (onto). ■

The key sentence

The key sentence is "**a pivot in every row, so no $(0 \cdots 0 \mid c)$ row with $c \neq 0$ can appear, so $Ax = b$ is consistent for every b**". The infinitely many solutions are not needed for onto. They are exactly the reason this μ_A is *not* one-to-one. The assumption $n \neq m$ is not needed either: if $m = n$, rank $= m$ still gives a pivot in every row.

Flip the hypothesis and the answer flips. Midterm Q3 used the 4×3 matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & 2 & 3 \end{bmatrix}$. It has rank $3 = n < m = 4$, and $Ax = b$ is consistent only when $5b_1 - b_2 - 2b_3 - 2b_4 = 0$. For $b = (1, 0, 2, 3)$ this gives $5 - 0 - 4 - 6 = -5 \neq 0$, so no x works and $\mu_A : \mathbb{R}^3 \to \mathbb{R}^4$ is **not** onto. It is one-to-one, though: every column has a pivot, so there are no free variables.

A is $m \times n$	Condition	Why	Midterm example
μ_A onto	$\operatorname{rank} A = m$ (pivot in every **row**)	no $(0 \cdots 0 \mid c \neq 0)$ row: consistent for all b	Q4(b): onto, not 1-1 ($m < n$)
μ_A one-to-one	$\operatorname{rank} A = n$ (pivot in every **column**)	no free variables: at most one solution (as in Q2(c))	Q3 matrix: 1-1, not onto (4×3, rank 3)
always	$\operatorname{rank} A \leq \min(m, n)$	at most one pivot per row and per column	if $m < n$, never 1-1; if $m > n$, never onto

In one minute

- A is $m \times n \Rightarrow \mu_A : \mathbb{R}^n \to \mathbb{R}^m$, $\mu_A(x) = Ax$. Domain = columns, codomain = rows.
- $\mu_A(e_j) =$ column j of A; so $\mu_A = \mu_B \Rightarrow A = B$.
- Linear: $\mu_A(x + y) = \mu_A(x) + \mu_A(y)$, $\mu_A(cx) = c\,\mu_A(x)$.
- $\mu_{AB} = \mu_A \circ \mu_B$: $\mathbb{R}^p \xrightarrow{\mu_B} \mathbb{R}^n \xrightarrow{\mu_A} \mathbb{R}^m$; B acts first.
- Associativity of matrix multiplication = associativity of composition.
- $\mu_{I_n}(x) = x$, the identity function (Exercise 1).
- Onto $\iff \operatorname{rank} A = m$; one-to-one $\iff \operatorname{rank} A = n$; $\operatorname{rank} A \leq \min(m, n)$. Midterm Q4(b): rank $= m \Rightarrow$ onto, **true**.

CHAPTER 14

Right and left inverses

For numbers, 5 has the inverse 1/5. Matrices can be rectangular and AB is usually not BA, so there are two one-sided kinds of inverse. After this chapter you can tell them apart, use them to solve Ax = b, and say exactly what each one guarantees.

From Lecture 6 (Inverse Matrices), pages 3–17

Why two kinds of inverse?

For a number $a \neq 0$ we have $a \cdot \frac{1}{a} = 1 = \frac{1}{a} \cdot a$. One number does both jobs.

Matrices are different in two ways. First, AB and BA are usually different matrices (they may not even have the same size). Second, A can be rectangular, $m \times n$. So we ask separately: is there a matrix that gives the identity when it stands to the **right** of A? And one that works on the **left**?

Definition (D1), Lecture 6 p.3

Let A be an $m \times n$ matrix.

- An $n \times m$ matrix B is called a **right inverse** of A if
$$AB = I_m,$$
where I_m denotes the $m \times m$ identity matrix.
- An $n \times m$ matrix C is called a **left inverse** of A if
$$CA = I_n,$$
where I_n is the $n \times n$ identity matrix.

Get the sizes right first

Both one-sided inverses of an $m \times n$ matrix are $n \times m$ (rows and columns swapped). Then AB is $(m \times n)(n \times m) = m \times m$, so it is compared with I_m; and CA is $(n \times m)(m \times n) = n \times n$, compared with I_n. The identity's size is "the number of rows of the product".

Symmetry (p.4)

The definition reads in both directions. If B is a right inverse of A, meaning $AB = I_m$, then A serves as a **left inverse of** B (the same equation, read from B's point of view). Conversely, if C is a left inverse of A, i.e. $CA = I_n$, then A is a **right inverse of** C.

EXAMPLE 1 · LECTURE 6 P.5–6 (BLANKS FILLED) Let $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$.

Compute AB and BA. What kind of inverse is B?

1. A is 2×3 and B is 3×2, so AB is 2×2 and BA is 3×3.
2. $AB = \begin{bmatrix} 1 \cdot 1 + 2 \cdot 0 + 0 \cdot 0 & 1 \cdot (-2) + 2 \cdot 1 + 0 \cdot 0 \\ 0 \cdot 1 + 1 \cdot 0 + 1 \cdot 0 & 0 \cdot (-2) + 1 \cdot 1 + 1 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2.$
3. So B is a **right** inverse of A. Thus A is a **left** inverse of B, as well. (symmetry, p.4)
4. However, $BA = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$, which is not the 3×3 identity matrix I_3.
5. Therefore B is **not a left inverse** of A. Furthermore, A is not a **right** inverse of B, neither.

ANSWER

$AB = I_2$ but $BA \neq I_3$: B is a right inverse (not a left inverse) of A; A is a left inverse (not a right inverse) of B.

One product is not enough

$AB = I$ does **not** give $BA = I$ for rectangular matrices. Example 1 is the standard counterexample: always compute the product in the order the definition asks for.

Why right inverses matter: solving $A\mathbf{x} = \mathbf{b}$

Let $A \in M_{m \times n}(\mathbb{R})$ and let B be a right inverse, so $AB = I_m$. Take any $\mathbf{b} \in \mathbb{R}^m$ and set $\mathbf{x} = B\mathbf{b}$. Then (p.7)

$$A\mathbf{x} = A(B\mathbf{b}) = (AB)\mathbf{b} = I_m\mathbf{b} = \mathbf{b}.$$

A RIGHT INVERSE HANDS YOU A SOLUTION: $\mathbf{x} = B\mathbf{b}$

The middle step is associativity of matrix multiplication. Nothing else is needed.

EXAMPLE 2 · LECTURE 6 P.8 With A, B from Example 1, check whether $A\mathbf{x} = \mathbf{b}$ is consistent for $\mathbf{b} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$. If it is, find at least one solution.

1. B is a right inverse of A (Example 1), so the system is consistent for *every* $\mathbf{b} \in \mathbb{R}^2$, and $\mathbf{x} = B\mathbf{b}$ is a solution.
2. $\mathbf{x} = B\mathbf{b} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 - 4 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} -5 \\ 2 \\ 0 \end{bmatrix}.$
3. Check: $A\mathbf{x} = \begin{bmatrix} 1 \cdot (-5) + 2 \cdot 2 + 0 \\ 0 + 2 + 0 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \mathbf{b}$. ✓

ANSWER
Consistent; one solution is $\mathbf{x} = (-5,\ 2,\ 0)^\top$. (There are more: A has 3 unknowns and rank 2, so one free variable.)

Solvability via right inverses, p.9–11
If a matrix A has a **right inverse**, say B, then for every $\mathbf{b} \in \mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ always has **at least one solution**, which is $B\mathbf{b}$, and so is **consistent** with either unique or infinitely many solutions.

The slides then ask two true/false questions (p.10–11). Both are **True**:

- **Q1)** If B is a right inverse of A, then the system $A\mathbf{x} = \mathbf{b}$ is solvable for any given $\mathbf{b} \in \mathbb{R}^m$ with the solution $\mathbf{x} = B\mathbf{b}$. **[T]** — this is exactly the computation $A(B\mathbf{b}) = (AB)\mathbf{b} = \mathbf{b}$.
- **Q2)** If A has a right inverse, a solution for $A\mathbf{x} = \mathbf{b}$ always exists for every right-hand side $\mathbf{b} \in \mathbb{R}^m$. **[T]** — same reason; "exists" is all it claims (not uniqueness).

Left inverses: uniqueness, not existence

EXAMPLE 3 · LECTURE 6 P.12 (Q3) Let $A \in M_{m\times n}(\mathbb{R})$. True or false: if A has a **left inverse**, then for every vector $\mathbf{b} \in \mathbb{R}^m$, the linear system $A\mathbf{x} = \mathbf{b}$ is consistent. If true, prove it; if false, give a counterexample.

1. **False.** Take the 3×2 matrix $M = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ (the B of Example 1).
2. M has a left inverse: $C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$ satisfies $CM = I_2$ (Example 1 says exactly this).
3. Take $\mathbf{b} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$. The third equation of $M\mathbf{x} = \mathbf{b}$ reads $0x_1 + 0x_2 = 1$, which is impossible.

ANSWER
False: M has a left inverse, yet $M\mathbf{x} = (0, 0, 1)^\top$ has no solution.

So what *does* a left inverse give? The slides answer on p.13. Suppose $A \in M_{m\times n}(\mathbb{R})$ has a left inverse C (where $CA = I_n$).

Left inverse: uniqueness vs. existence, p.13
1. Consistency is NOT guaranteed: having a left inverse does **not** mean the system $A\mathbf{x} = \mathbf{b}$ is always consistent. The system may have **no solution** for some $\mathbf{b} \in \mathbb{R}^m$.
2. If consistent $\Longrightarrow$ solution is unique: if a solution exists (i.e. if the system is consistent), it must be **unique** and equals $\mathbf{x} = C\mathbf{b}$:

$$A\mathbf{x} = \mathbf{b} \implies C(A\mathbf{x}) = C\mathbf{b} \implies (CA)\mathbf{x} = C\mathbf{b} \implies \mathbf{x} = C\mathbf{b}.$$

Read the arrows carefully. They start from "suppose $\mathbf{x}$ is a solution". So they prove: *any* solution must equal $C\mathbf{b}$, hence there is at most one. They do not prove that $C\mathbf{b}$ is a solution.

That is why the candidate must be checked (p.14): $A(C\mathbf{b}) = \mathbf{b} \iff (AC)\mathbf{b} = \mathbf{b}$. This may fail for some $\mathbf{b} \in \mathbb{R}^m$ because C is only a left inverse ($AC \neq I_m$ in general). For $C\mathbf{b}$ to be a valid solution for **every** $\mathbf{b} \in \mathbb{R}^m$, we must have $AC = I_m$, which means C must be **both a left and a right inverse** of A (i.e. A is invertible).

SEEING P.13–14 ON EXAMPLE 3'S MATRIX $M = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, $CM = I_2$. Test the candidate $C\mathbf{b}$ for $\mathbf{b} = (1,1,0)^\top$ and for $\mathbf{b} = (0,0,1)^\top$.

1. $\mathbf{b} = (1,1,0)^\top$: $C\mathbf{b} = (1+2,\ 1)^\top = (3,1)^\top$, and $M(3,1)^\top = (3-2,\ 1,\ 0)^\top = (1,1,0)^\top = \mathbf{b}$. ✓ Consistent, and by p.13 this is the **only** solution.
2. $\mathbf{b} = (0,0,1)^\top$: $C\mathbf{b} = (0,\ 1)^\top$, but $M(0,1)^\top = (-2,1,0)^\top \neq \mathbf{b}$. The candidate fails, so the system has no solution. (if it had one, it would have to be $C\mathbf{b}$)

ANSWER
With a left inverse: compute $C\mathbf{b}$, then **check** it. Works $\Rightarrow$ the unique solution. Fails $\Rightarrow$ no solution.

	Right inverse B ($AB = I_m$)	Left inverse C ($CA = I_n$)
Guarantees	**existence**: $A\mathbf{x} = \mathbf{b}$ consistent for every $\mathbf{b}$	**uniqueness**: at most one solution
Formula	$\mathbf{x} = B\mathbf{b}$ is always a solution	if a solution exists, it is $\mathbf{x} = C\mathbf{b}$ (must check)
Rank (Lemma L1)	$\operatorname{rank}(A) = m$ (rows)	$\operatorname{rank}(A) = n$ (columns)

The rank picture

Recall from Chapter 1 (p.15) that for an $m \times n$ matrix A the system $A\mathbf{x} = \mathbf{b}$ is tied to the rank:

- The system is **consistent for all** $\mathbf{b} \in \mathbb{R}^m$ if and only if the echelon form of A has **no zero rows**, that is, $\operatorname{rank}(A) = m$, the number of rows of A. (*Existence part.*)
- The system has a **unique solution** (no free variables) if and only if $\operatorname{rank}(A) = n$, the number of columns of A. (*Uniqueness part.*)

Unique means 'at most one'
When $\operatorname{rank}(A) = n < m$, some $\mathbf{b}$ still give an inconsistent system (Example 3). Read the uniqueness part as: no free variables, so *whenever* the system is consistent, the solution is unique.

Lemma (L1), p.16

Let A be an $m \times n$ matrix.

- If A has a **right inverse**, then $\operatorname{rank}(A) = m$.
- If A has a **left inverse**, then $\operatorname{rank}(A) = n$.

Therefore, if A has both a left inverse and a right inverse, then A must be a **square** matrix ($m = n$) with full rank: $\operatorname{rank}(A) = n$.

Why it fits: a right inverse makes every system consistent (existence), and existence for all $\mathbf{b}$ is exactly $\operatorname{rank}(A) = m$. A left inverse makes solutions unique, and uniqueness is exactly $\operatorname{rank}(A) = n$.

Quick size test

The rank is never bigger than the number of rows or the number of columns. So a tall matrix ($m > n$) can never reach rank m: **no right inverse**. A wide matrix ($m < n$) can never reach rank n: **no left inverse**. Example 1 fits: the wide 2×3 matrix A has a right inverse only; the tall 3×2 matrix B has a left inverse only.

When left and right inverses agree

Suppose A is a square matrix of size $n \times n$, and assume it has both a right inverse B and a left inverse C, so that $AB = I_n = CA$. Using the associativity of matrix multiplication, we compute (the blank on p.17):

$$C = CI_n = C(AB) = (CA)B = I_nB = B.$$

LEFT INVERSE = RIGHT INVERSE (P.17)

Thus, if a square matrix A has **both a left inverse and a right inverse**, they must be **equal**. In this case, we refer to this common matrix as the (two-sided) inverse of A. This is the bridge to the next chapter.

How to remember the trick

Put the identity next to C, replace it by AB, and regroup so that CA meets. Each step uses one fact: I is neutral, $AB = I$, associativity, $CA = I$, I is neutral.

In one minute

- A is $m \times n$. Right inverse: $n \times m$ matrix B with $AB = I_m$. Left inverse: $n \times m$ matrix C with $CA = I_n$.
- Symmetry: $AB = I$ means B is a right inverse of A *and* A is a left inverse of B.
- $AB = I$ does not imply $BA = I$ (Example 1: $AB = I_2$, $BA \neq I_3$).
- Right inverse $\Rightarrow A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$, with $\mathbf{x} = B\mathbf{b}$ (existence).

- Left inverse $\Rightarrow$ at most one solution, and it must be $C\mathbf{b}$; consistency is **not** guaranteed, so check $A(C\mathbf{b}) = \mathbf{b}$.
- Lemma L1: right inverse $\Rightarrow \operatorname{rank} A = m$; left inverse $\Rightarrow \operatorname{rank} A = n$; both $\Rightarrow$ square with full rank.
- Square A with $AB = I_n = CA$: $C = CI_n = C(AB) = (CA)B = I_nB = B$.

CHAPTER 15

Invertible matrices

A square matrix is invertible when one matrix B undoes it from both sides. You will learn the definition word for word, why invertible and nonsingular mean the same thing (Theorems T1 and T2), and the short proofs the exam asks for, such as $BA = I \Rightarrow B = A^{-1}$ and $(AB)^{-1} = B^{-1}A^{-1}$.

From Lecture 6 (Inverse Matrices), pages 18–31 and 47

Two-sided inverses

The last chapter ended with this: if a square matrix has a left inverse and a right inverse, they are the same matrix. That common matrix gets a name.

Definition (D2), Lecture 6 p.18

A square matrix $A \in M_n(\mathbb{R})$ is called **invertible** if there exists a matrix $B \in M_n(\mathbb{R})$ such that

$$AB = I_n = BA.$$

In this case, B is called the (two-sided) **inverse** of A, and we write $B = A^{-1}$.

The definition is symmetric in A and B. So if $B = A^{-1}$, then it follows immediately that $A = B^{-1}$ (p.18). In other words, $\left(A^{-1}\right)^{-1} = A$.

The inverse is unique (p.19)

The inverse of a matrix is **unique**. Indeed, if B and C are both inverses of A, then (the dotted lines on p.19):

$$B = BI_n = B(AC) = (BA)C = I_nC = C.$$

Therefore $B = C$, meaning a square matrix can have **at most one** (two-sided) inverse. That is why we may write "*the* inverse" and use the symbol A^{-1}.

Same trick as p.17

Uniqueness uses exactly the identity-sandwich from the last chapter: $B = BI = B(AC) = (BA)C = IC = C$. Learn it once, use it three times (left = right, uniqueness, and the exam proof below).

EXAMPLE 4 · LECTURE 6 P.20 (BLANKS FILLED) Let $A = \begin{bmatrix} 3 & 1 \\ -1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -1 \\ 1 & 3 \end{bmatrix}$. Compute AB and BA.

1. $AB = \begin{bmatrix} 3\cdot 0 + 1\cdot 1 & 3\cdot(-1) + 1\cdot 3 \\ -1\cdot 0 + 0\cdot 1 & (-1)(-1) + 0\cdot 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
2. $BA = \begin{bmatrix} 0\cdot 3 + (-1)(-1) & 0\cdot 1 + (-1)\cdot 0 \\ 1\cdot 3 + 3\cdot(-1) & 1\cdot 1 + 3\cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
3. Both products are I_2. (for a square matrix the definition asks for both)

ANSWER
$AB = BA = I_2$, and hence $B = A^{-1}$ and $A = B^{-1}$.

EXAMPLE 5 · LECTURE 6 P.21 (THE 'VERIFY!' DONE) Let $A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 1 & 1 \\ 0 & 1 & 2 \end{bmatrix}$, $B = \begin{bmatrix} \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & -\frac{1}{2} \\ -\frac{1}{4} & -\frac{1}{2} & \frac{3}{4} \end{bmatrix}$. The slide gives $AB = I_3$. Verify $BA = I_3$.

1. Row 1 of B times the columns of A: $\frac{1}{4}\cdot 2 - \frac{1}{2}(-1) + \frac{1}{4}\cdot 0 = 1$; $\frac{1}{4} - \frac{1}{2} + \frac{1}{4} = 0$; $0 - \frac{1}{2} + \frac{1}{2} = 0$.
2. Row 2: $\frac{1}{2}\cdot 2 + 1\cdot(-1) + 0 = 0$; $\frac{1}{2} + 1 - \frac{1}{2} = 1$; $0 + 1 - 1 = 0$.
3. Row 3: $-\frac{1}{4}\cdot 2 - \frac{1}{2}(-1) + 0 = 0$; $-\frac{1}{4} - \frac{1}{2} + \frac{3}{4} = 0$; $0 - \frac{1}{2} + \frac{3}{2} = 1$.

ANSWER
$BA = I_3$. Together with $AB = I_3$: B is the inverse of A, $A^{-1} = B$.

Unit vectors e_j and the identity

Definition, p.22

Let $e_j \in \mathbb{R}^n$ denote the vector whose entries are **zero**, except for a **1** in the jth position:

$$e_j = (0, 0, \ldots, 0, \underbrace{1}_{j\text{th position}}, 0, \ldots, 0)^\top \in \mathbb{R}^n.$$

Each vector e_j appears as the jth column of the identity matrix I_n. That is, $I_n = [\, e_1\ e_2\ \ldots\ e_n \,]$, where each e_j forms a column.

For example, in $\mathbb{R}^4$ (p.23), $e_3 = (0, 0, 1, 0)^\top$ is the third column of I_4, the unit vector in the direction of the third coordinate axis, and $e_1 = (1, 0, 0, 0)^\top$.

Why we need them: $AB = I_n$ is really n separate systems. Column j of AB is A times column j of B, and column j of I_n is e_j. So

$$AB = I_n \iff A\mathbf{b}_j = e_j \text{ for every } j = 1, \ldots, n,$$

$\mathbf{B}_J$ = THE JTH COLUMN OF B

Invertible $\iff$ nonsingular

"Nonsingular" comes from an earlier lecture. The slides recall it:

Proposition P4∗ (see Lecture Notes 4), p.24

Let A be an $n \times n$ matrix. The following statements are **equivalent**:

1. A is **nonsingular** (i.e., $\operatorname{rank}(A) = n$).
2. The reduced row echelon form RREF of A must be the $n \times n$ **identity matrix** I_n.
3. The homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the **trivial solution**.
4. For every $\mathbf{b} \in \mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has a **unique solution**.

Invertible $\Rightarrow$ nonsingular (p.25)

Let $A \in M_n(\mathbb{R})$ be invertible. Then for **every** $\mathbf{b} \in \mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution $\mathbf{x} = A^{-1}\mathbf{b}$:

- *it is a solution:* $A(A^{-1}\mathbf{b}) = (AA^{-1})\mathbf{b} = \mathbf{b}$ (A^{-1} is a right inverse);
- *it is the only one:* $A\mathbf{x} = \mathbf{b} \Rightarrow A^{-1}A\mathbf{x} = A^{-1}\mathbf{b} \Rightarrow \mathbf{x} = A^{-1}\mathbf{b}$ (A^{-1} is a left inverse).

That is statement 4 of P4∗, so A must be **nonsingular**.

Nonsingular $\Rightarrow$ invertible (p.25–29)

"But what about the converse? If A is nonsingular, must A be invertible?" The slides build the inverse column by column.

1. Suppose $A \in M_n(\mathbb{R})$ is nonsingular. Then for every $\mathbf{c} \in \mathbb{R}^n$, the system $A\mathbf{x} = \mathbf{c}$ has a unique solution, by P4∗.
2. So for each e_j there is a unique vector $\mathbf{b}_j \in \mathbb{R}^n$ with $A\mathbf{b}_j = e_j$. Let $B = [\,\mathbf{b}_1\ \mathbf{b}_2\ \cdots\ \mathbf{b}_n\,]$ be the matrix with these columns (p.26).
3. Since $A\mathbf{b}_j = e_j$ for each j: $AB = A[\,\mathbf{b}_1\ \cdots\ \mathbf{b}_n\,] = [\,e_1\ \cdots\ e_n\,] = I_n$ (p.27).
4. All n systems have the same coefficient matrix, so they are solved at once: Gaussian elimination takes $[\,A \mid I_n\,] = [\,A \mid e_1\ \cdots\ e_n\,]$ to the reduced echelon form $[\,I_n \mid \mathbf{b}_1\ \cdots\ \mathbf{b}_n\,] = [\,I_n \mid B\,]$ (p.28).
5. Now we must verify $BA = I_n$ as well. Reverse the row operations: they transform $[\,B \mid I_n\,] \longrightarrow [\,I_n \mid A\,]$. This shows $BA = I_n$, completing the argument. Hence $B = A^{-1}$ (p.29).

About step 5

The slide gives step 5 as a sketch (row operations can always be undone). For the exam you will not be asked to re-prove it; you cite the result, Theorem T1, below.

Theorem T1, p.30

Let $A \in M_n(\mathbb{R})$. Then A is **nonsingular** if and only if it is **invertible**.

Remark: Gaussian elimination provides a practical method to determine when a matrix is not invertible. If, during the reduction of A to echelon form, a row of all zeros appears, then A is singular and thus not invertible.

Therefore, A is invertible **if and only if** $\operatorname{rank}(A) = n$ **if and only if** its reduced row echelon form RREF must be the identity matrix.

Theorem T2 (Characterizations of invertible matrices), p.31

By Proposition P4* and Theorem T1, the following statements are **equivalent**.

1. A is **invertible**.
2. A is **nonsingular**, i.e. $\operatorname{rank}(A) = n$.
3. The reduced row echelon form RREF of A must be the $n \times n$ **identity matrix** I_n.
4. The homogeneous system $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
5. For every $\mathbf{b} \in \mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution.

How to use T2

T2 is a menu. To **prove** A is invertible, pick the easiest statement to show; statement 4 ($A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$) is usually the one-liner. To show A is **not** invertible, exhibit a nonzero $\mathbf{x}$ with $A\mathbf{x} = \mathbf{0}$, or a zero row in the echelon form.

USING T2 (MODELED ON P.25 AND P.30) Is $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 1 & 3 & 4 \end{bmatrix}$ invertible?

1. Look for a nonzero solution of $A\mathbf{x} = \mathbf{0}$. Row 3 + row 1 = $(2, 5, 7)$ = row 2, and the columns satisfy $\text{col}_1 + \text{col}_2 - \text{col}_3 = \mathbf{0}$.
2. So $\mathbf{x} = (1, 1, -1)^\top \neq \mathbf{0}$ gives $A\mathbf{x} = (1+2-3,\ 2+5-7,\ 1+3-4)^\top = \mathbf{0}$.
3. Statement 4 of T2 fails, so statement 1 fails.

ANSWER

A is singular, hence not invertible. (Row reduction shows the same: a zero row appears; see Problem 3 in the next chapter.)

Proof patterns for the exam

The 2025 midterm asked (Q4(a)): *Let A and B be $n \times n$ matrices such that $BA = I_n$. Prove that $B = A^{-1}$.* The point: the definition D2 needs **both** $AB = I_n$ and $BA = I_n$, and we are only given one. T1/T2 fill the gap.

PROOF 1 · BA = I_n ⇒ B = A^{-1} (EXAM PATTERN) Let $A, B \in M_n(\mathbb{R})$ with $BA = I_n$. Prove that A is invertible and $B = A^{-1}$.

1. **Step 1: A is invertible.** Suppose $A\mathbf{x} = \mathbf{0}$ for some $\mathbf{x} \in \mathbb{R}^n$. Then
$$\mathbf{x} = I_n\mathbf{x} = (BA)\mathbf{x} = B(A\mathbf{x}) = B\mathbf{0} = \mathbf{0}.$$
So $A\mathbf{x} = \mathbf{0}$ has only the trivial solution. (statement 4 of T2)
2. By Theorem T2 (4 ⇒ 1), A is invertible: A^{-1} exists with $AA^{-1} = I_n = A^{-1}A$.
3. **Step 2: $B = A^{-1}$.** Multiply $BA = I_n$ on the right by A^{-1}:
$$(BA)A^{-1} = I_nA^{-1} \;\Rightarrow\; B(AA^{-1}) = A^{-1} \;\Rightarrow\; BI_n = A^{-1} \;\Rightarrow\; B = A^{-1}.$$
(associativity, then $AA^{-1} = I_n$)

ANSWER
A is invertible and $B = A^{-1}$; in particular $AB = I_n$ too.

Do not assume what you must prove
Writing "multiply $BA = I$ by A^{-1}" as the **first** line loses the points: A^{-1} is not known to exist yet. Step 1 (via $A\mathbf{x} = \mathbf{0}$) is what earns them. Also never write "$BA = AB$": matrices do not commute in general.

A second way to do Step 1
$BA = I_n$ says B is a left inverse of A. By Lemma L1, $\operatorname{rank}(A) = n$, so A is nonsingular, hence invertible by T1. Then finish exactly as in Step 2. Either route is fine; the $A\mathbf{x} = \mathbf{0}$ route is the one in the official midterm solution.

PROOF 2 · AB = I_n ⇒ BA = I_n (SQUARE MATRICES) Let $A, B \in M_n(\mathbb{R})$ with $AB = I_n$. Prove $BA = I_n$.

1. Read $AB = I_n$ from B's point of view: A is a left inverse of B. (symmetry, p.4)
2. Apply Proof 1 with the roles swapped (A plays "B", B plays "A"): B is invertible and $A = B^{-1}$.
3. Then $BA = BB^{-1} = I_n$.

ANSWER
For **square** matrices one of $AB = I_n$, $BA = I_n$ implies the other. (For rectangular ones it fails: Example 1 of the last chapter.)

Inverse of a product

Proposition P1, p.47

Let A and B be invertible matrices in $M_n(\mathbb{R})$. Then the product AB is also invertible, and its inverse is given by

$$(AB)^{-1} = B^{-1}A^{-1}.$$

PROOF OF P1 (BLANK ON P.47) Prove that $(AB)^{-1} = B^{-1}A^{-1}$.

1. By definition D2 we must show that $B^{-1}A^{-1}$ works from **both** sides.
2. Right: $(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AI_nA^{-1} = AA^{-1} = I_n$. (associativity)
3. Left: $(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}I_nB = B^{-1}B = I_n$.
4. So AB is invertible, and since the inverse is unique (p.19), it equals $B^{-1}A^{-1}$. □

ANSWER

$(AB)^{-1} = B^{-1}A^{-1}$: the order reverses.

Order reverses

$(AB)^{-1} = A^{-1}B^{-1}$ is **wrong** in general. Think "socks and shoes": you put on socks, then shoes; to undo, take off shoes first. In the product $(AB)(A^{-1}B^{-1})$, the B and A^{-1} cannot meet, so nothing cancels.

Solving $A\mathbf{x} = \mathbf{b}$ with A^{-1}

If A is invertible, the unique solution of $A\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = A^{-1}\mathbf{b}$ (p.25). Example: with $A, B = A^{-1}$ from Example 4 and $\mathbf{b} = (1, 2)^\top$, $\mathbf{x} = \begin{bmatrix} 0 & -1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -2 \\ 7 \end{bmatrix}$. Check: $3(-2) + 7 = 1$, $-(-2) + 0 = 2$. ✓

In one minute

- **D2:** $A \in M_n(\mathbb{R})$ is invertible if some $B \in M_n(\mathbb{R})$ has $AB = I_n = BA$; write $B = A^{-1}$. Then also $A = B^{-1}$.
- The inverse is unique: $B = BI = B(AC) = (BA)C = IC = C$.
- $AB = I_n \iff A\mathbf{b}_j = e_j$ for every column $\mathbf{b}_j$ of B.
- **T1:** nonsingular $\iff$ invertible. **T2:** invertible $\iff \operatorname{rank} A = n \iff$ RREF$= I_n \iff A\mathbf{x} = \mathbf{0}$ only trivially $\iff A\mathbf{x} = \mathbf{b}$ uniquely solvable for all $\mathbf{b}$.
- Zero row during elimination $\Rightarrow$ singular $\Rightarrow$ not invertible.
- **Exam proof** $BA = I_n \Rightarrow B = A^{-1}$: $A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = (BA)\mathbf{x} = B\mathbf{0} = \mathbf{0}$, so A invertible (T2); then $B = BAA^{-1} = A^{-1}$.
- **P1:** $(AB)^{-1} = B^{-1}A^{-1}$; check both products give I_n.

CHAPTER 16

Computing A^{-1} by Gauss–Jordan elimination

One procedure does everything: write [A | I], row-reduce until the left block is I, and read A^{-1} off the right. If a zero row shows up on the left instead, A is not invertible. At the end, the same idea finds right inverses of wide matrices.

From Lecture 6 (Inverse Matrices), pages 32–51

The method

Last chapter we saw why it works: the columns of A^{-1} are the solutions $\mathbf{b}_j$ of $A\mathbf{x} = e_j$, and all these systems share the matrix A. So we solve them together, with all the right-hand sides $e_1, \ldots, e_n$ (that is, I_n) written next to A.

Method for finding the inverse in $M_n(\mathbb{R})$, Lecture 6 p.32

Let $A \in M_n(\mathbb{R})$. To find the inverse of A, provided it exists, one can apply the following procedure.

Method: Perform Gaussian elimination on the augmented matrix $[\,A \mid I_n\,]$, where I_n is the $n \times n$ identity matrix. Apply row operations to transform the matrix into the form $[\,I_n \mid A^{-1}\,]$.

If this row reduction is successful (i.e., A is row-equivalent to I_n), then A is invertible, and the resulting right-hand side is the inverse A^{-1}.

$$[\,A \mid I_n\,] \xrightarrow{\text{row operations}} [\,I_n \mid A^{-1}\,]$$

GAUSS–JORDAN FOR THE INVERSE

The recipe, step by step

1. **Write** $[\,A \mid I_n\,]$: A on the left, the identity on the right.
2. **Go down, column by column** (forward elimination). In column j, get a nonzero pivot in row j (swap rows $R_j \leftrightarrow R_k$ if the entry is 0; a pivot 1 or −1 saves fractions). Use $R_i \to R_i - cR_j$ to make every entry **below** the pivot zero.
3. **Watch the left block.** If a row of the left block becomes all zeros, **stop**: A is singular, not invertible (Remark, p.30).
4. **Scale** each pivot to 1: $R_j \to \frac{1}{p}R_j$.
5. **Go up** (backward elimination). From the last column to the first, use $R_i \to R_i - cR_j$ to make every entry **above** each pivot zero.
6. **Read** A^{-1} from the right block, then **verify** $AA^{-1} = I_n$ (the exam also asks for $A^{-1}A = I_n$).

The three allowed row operations: swap two rows ($R_i \leftrightarrow R_j$); multiply a row by a **nonzero** number ($R_i \to cR_i$); add a multiple of another row ($R_i \to R_i + cR_j$).

> **Whole rows, both blocks**
> Every row operation acts on the **entire** row, left block and right block. Changing only the left part is the most common way to get a wrong inverse. And use row operations only, never column operations.

Example 6 from the slides, every step named

EXAMPLE 6 · LECTURE 6 P.33–34 Find A^{-1} for $A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 1 & 0 & 1 \end{bmatrix}$.

1. Form the augmented matrix:

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

2. $R_3 \to R_3 - R_1$ (clear the 1 below the first pivot)

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & -2 & -2 & -1 & 0 & 1 \end{array}\right]$$

3. $R_3 \to R_3 + 2R_2$ (clear the -2 below the second pivot)

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 6 & -1 & 2 & 1 \end{array}\right]$$

4. $R_3 \to \frac{1}{6}R_3$ (make the third pivot 1)

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{6} & \frac{1}{3} & \frac{1}{6} \end{array}\right]$$

5. $R_2 \to R_2 - 4R_3,\ R_1 \to R_1 - 3R_3$ (clear column 3 above its pivot)

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & \frac{3}{2} & -1 & -\frac{1}{2} \\ 0 & 1 & 0 & \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & -\frac{1}{6} & \frac{1}{3} & \frac{1}{6} \end{array}\right]$$

6 $R_1 \to R_1 - 2R_2$ (clear column 2 above its pivot; the left block is now I_3)

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{6} & -\frac{1}{3} & \frac{5}{6} \\ 0 & 1 & 0 & \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & -\frac{1}{6} & \frac{1}{3} & \frac{1}{6} \end{array}\right]$$

ANSWER

$A^{-1} = \begin{bmatrix} \frac{1}{6} & -\frac{1}{3} & \frac{5}{6} \\ \frac{2}{3} & -\frac{1}{3} & -\frac{2}{3} \\ -\frac{1}{6} & \frac{1}{3} & \frac{1}{6} \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 1 & -2 & 5 \\ 4 & -2 & -4 \\ -1 & 2 & 1 \end{bmatrix}$ (the slide's result, p.34).

Verify (p.34 asks for it): $AA^{-1} = \frac{1}{6}\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & -2 & 5 \\ 4 & -2 & -4 \\ -1 & 2 & 1 \end{bmatrix} = \frac{1}{6}\begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} = I_3$. For instance, entry (1,1) is $1 \cdot 1 + 2 \cdot 4 + 3 \cdot (-1) = 6$, and entry (1,2) is $1 \cdot (-2) + 2 \cdot (-2) + 3 \cdot 2 = 0$.

Pull out the common denominator

Writing $A^{-1} = \frac{1}{6}(\text{integer matrix})$ makes the check $AA^{-1} = I$ much quicker: multiply integer matrices, then divide by 6.

Examples 7 and 8 (left blank on the slides)

EXAMPLE 7 · LECTURE 6 P.35–36 Find the inverse (if it exists) of $A = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 1 & 3 \\ 4 & 0 & 1 \end{bmatrix} \in M_3(\mathbb{R})$.

1

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ -1 & 1 & 3 & 0 & 1 & 0 \\ 4 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

2 $R_1 \leftrightarrow R_2$ (a pivot ± 1 in row 1 avoids fractions)

$$\left[\begin{array}{ccc|ccc} -1 & 1 & 3 & 0 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 0 \\ 4 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

3 $R_1 \to -R_1$ (make the first pivot 1)

$$\left[\begin{array}{ccc|ccc} 1 & -1 & -3 & 0 & -1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 0 \\ 4 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$$

4 $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - 4R_1$ (clear column 1 below the pivot)

$$\left[\begin{array}{ccc|ccc} 1 & -1 & -3 & 0 & -1 & 0 \\ 0 & 3 & 6 & 1 & 2 & 0 \\ 0 & 4 & 13 & 0 & 4 & 1 \end{array}\right]$$

5 $R_2 \to \frac{1}{3}R_2$ (make the second pivot 1)

$$\left[\begin{array}{ccc|ccc} 1 & -1 & -3 & 0 & -1 & 0 \\ 0 & 1 & 2 & \frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 4 & 13 & 0 & 4 & 1 \end{array}\right]$$

6 $R_3 \to R_3 - 4R_2$ (clear column 2 below the pivot)

$$\left[\begin{array}{ccc|ccc} 1 & -1 & -3 & 0 & -1 & 0 \\ 0 & 1 & 2 & \frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 5 & -\frac{4}{3} & \frac{4}{3} & 1 \end{array}\right]$$

7 $R_3 \to \frac{1}{5}R_3$ (make the third pivot 1)

$$\left[\begin{array}{ccc|ccc} 1 & -1 & -3 & 0 & -1 & 0 \\ 0 & 1 & 2 & \frac{1}{3} & \frac{2}{3} & 0 \\ 0 & 0 & 1 & -\frac{4}{15} & \frac{4}{15} & \frac{1}{5} \end{array}\right]$$

8 $R_2 \to R_2 - 2R_3,\ R_1 \to R_1 + 3R_3$ (clear column 3 above its pivot)

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & -\frac{4}{5} & -\frac{1}{5} & \frac{3}{5} \\ 0 & 1 & 0 & \frac{13}{15} & \frac{2}{15} & -\frac{2}{5} \\ 0 & 0 & 1 & -\frac{4}{15} & \frac{4}{15} & \frac{1}{5} \end{array}\right]$$

9 $R_1 \to R_1 + R_2$ (clear column 2 above its pivot; left block is I_3)

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{1}{15} & -\frac{1}{15} & \frac{1}{5} \\ 0 & 1 & 0 & \frac{13}{15} & \frac{2}{15} & -\frac{2}{5} \\ 0 & 0 & 1 & -\frac{4}{15} & \frac{4}{15} & \frac{1}{5} \end{array}\right]$$

ANSWER

$A^{-1} = \frac{1}{15}\begin{bmatrix} 1 & -1 & 3 \\ 13 & 2 & -6 \\ -4 & 4 & 3 \end{bmatrix}$. Check row 1 of A with column 1: $2 \cdot 1 + 1 \cdot 13 + 0 = 15$ ✓, so $AA^{-1} = I_3$.

EXAMPLE 8 · LECTURE 6 P.37 Find the inverse (if it exists) of $A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 2 \\ 0 & 1 & 1 \end{bmatrix} \in M_3(\mathbb{R})$.

1

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 3 & 0 & 2 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

2 $R_2 \to R_2 - 3R_1$ (clear column 1 below the pivot; row 3 already has 0)

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & -6 & 5 & -3 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$$

3 $R_2 \leftrightarrow R_3$ (row 3 already has a 1 in column 2: use it as the pivot)

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & -6 & 5 & -3 & 1 & 0 \end{array}\right]$$

4 $R_3 \to R_3 + 6R_2$ (clear column 2 below the pivot)

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 11 & -3 & 1 & 6 \end{array}\right]$$

5 $R_3 \to \frac{1}{11}R_3$ (make the third pivot 1)

$$\left[\begin{array}{ccc|ccc} 1 & 2 & -1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & -\frac{3}{11} & \frac{1}{11} & \frac{6}{11} \end{array}\right]$$

6 $R_2 \to R_2 - R_3,\ R_1 \to R_1 + R_3$ (clear column 3 above its pivot)

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 0 & \frac{8}{11} & \frac{1}{11} & \frac{6}{11} \\ 0 & 1 & 0 & \frac{3}{11} & -\frac{1}{11} & \frac{5}{11} \\ 0 & 0 & 1 & -\frac{3}{11} & \frac{1}{11} & \frac{6}{11} \end{array}\right]$$

7 $R_1 \to R_1 - 2R_2$ (clear column 2 above its pivot; left block is I_3)

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{2}{11} & \frac{3}{11} & -\frac{4}{11} \\ 0 & 1 & 0 & \frac{3}{11} & -\frac{1}{11} & \frac{5}{11} \\ 0 & 0 & 1 & -\frac{3}{11} & \frac{1}{11} & \frac{6}{11} \end{array}\right]$$

ANSWER

$A^{-1} = \frac{1}{11}\begin{bmatrix} 2 & 3 & -4 \\ 3 & -1 & 5 \\ -3 & 1 & 6 \end{bmatrix}$; $AA^{-1} = I_3$ ✓.

★ **Swapping is allowed and often smart**

In Example 8 the row $(0, 1, 1)$ already has a 1 where we need the second pivot, so swapping $R_2 \leftrightarrow R_3$ avoids dividing by −6.

When A is not invertible

EXAMPLE 10 · LECTURE 6 P.39–40 Determine whether $A = \begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix} \in M_2(\mathbb{R})$ is invertible.

1 $$\left[\begin{array}{cc|cc} 2 & 4 & 1 & 0 \\ 1 & 2 & 0 & 1 \end{array}\right]$$

2 $R_1 \leftrightarrow R_2$ (pivot 1 on top)

$$\left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 0 \end{array}\right]$$

3 $R_2 \to R_2 - 2R_1$ (clear below the pivot: the left block of row 2 becomes $0 \;\; 0$)

$$\left[\begin{array}{cc|cc} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & -2 \end{array}\right]$$

ANSWER

The left block has a zero row, so it cannot be turned into I_2. Since the left side does not reduce to the identity matrix I_2, the matrix A is **not invertible**, so is singular.

→ **How to recognise 'not invertible' during the process**

A row whose **left part** is all zeros, like $[0 \;\; 0 \mid 1 \;\; -2]$. The right part does **not** matter (it is never all zero). A zero row on the left means $\operatorname{rank}(A) < n$, so by Theorem T2 A is singular and has no inverse. Write exactly that sentence as your conclusion.

! **Don't keep going**

Once the zero row appears, stop and conclude. Dividing that row by 0 to "force" a pivot is not a row operation.

The 2×2 case: a formula

The slides (p.41–43) run the same method on a general $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, with $a \neq 0$ initially.

1 $$\left[\begin{array}{cc|cc} a & b & 1 & 0 \\ c & d & 0 & 1 \end{array}\right]$$

Divide the first row by a:

$$\left[\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ c & d & 0 & 1 \end{array}\right]$$

2 Subtract $c \times (\text{row } 1)$ from row 2:

$$\left[\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & d-\frac{bc}{a} & -\frac{c}{a} & 1 \end{array}\right]$$

3 Assuming $ad - bc \neq 0$, divide row 2 by $d - \frac{bc}{a} = \frac{ad-bc}{a}$:

$$\left[\begin{array}{cc|cc} 1 & \frac{b}{a} & \frac{1}{a} & 0 \\ 0 & 1 & -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right]$$

4 Eliminate the upper-right element, $\text{Row } 1 \to \text{Row } 1 - \frac{b}{a} \times \text{Row } 2$:

$$\left[\begin{array}{cc|cc} 1 & 0 & \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ 0 & 1 & -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{array}\right]$$

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad ad - bc \neq 0 \implies A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

INVERSE OF A 2×2 MATRIX (P.43)

Check (p.44): $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \cdot \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{1}{ad-bc}\begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix} = I_2$, and similarly in the other order. Thus the formula holds as the inverse of A.

Remark on nonsingularity, p.45

Although the formula was derived under the assumption that $a \neq 0$, it is valid for any 2×2 matrix A. Because, if $a = 0$, we can instead check whether $c \neq 0$. If both a and c are zero, then A is noninvertible; otherwise, if at least one of them is nonzero, the formula can still be applied. From the row reduction process, A is invertible if and only if $ad - bc \neq 0$, because if $ad - bc = 0$, the echelon form of A contains a zero row, indicating A is singular. Therefore,

$$A \text{ is invertible} \iff ad - bc \neq 0.$$

Memory trick

Swap the diagonal ($a \leftrightarrow d$), negate the off-diagonal ($b, c \to -b, -c$), divide by $ad - bc$.

EXAMPLE 9 · LECTURE 6 P.38 AND P.46 (BOTH WAYS) Find the inverse (if it exists) of $A = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} \in M_2(\mathbb{R})$.

1 **By row reduction.**

$$\left[\begin{array}{cc|cc} 4 & 3 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{array}\right]$$

2 $R_1 \leftrightarrow R_2$ (put the row with the smaller pivot 2 on top)

$$\left[\begin{array}{cc|cc} 2 & 1 & 0 & 1 \\ 4 & 3 & 1 & 0 \end{array}\right]$$

3 $R_2 \to R_2 - 2R_1$ (clear below the first pivot)

$$\left[\begin{array}{cc|cc} 2 & 1 & 0 & 1 \\ 0 & 1 & 1 & -2 \end{array}\right]$$

4 $R_1 \to R_1 - R_2$ (clear above the second pivot)

$$\left[\begin{array}{cc|cc} 2 & 0 & -1 & 3 \\ 0 & 1 & 1 & -2 \end{array}\right]$$

5 $R_1 \to \frac{1}{2}R_1$ (make the first pivot 1; left block is I_2)

$$\left[\begin{array}{cc|cc} 1 & 0 & -\frac{1}{2} & \frac{3}{2} \\ 0 & 1 & 1 & -2 \end{array}\right]$$

6 **By the formula** (p.46): $ad - bc = 4 \cdot 1 - 3 \cdot 2 = -2 \neq 0$, so $A^{-1} = \frac{1}{-2}\begin{bmatrix} 1 & -3 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} & \frac{3}{2} \\ 1 & -2 \end{bmatrix}$. Same answer.

ANSWER

$A^{-1} = \begin{bmatrix} -\frac{1}{2} & \frac{3}{2} \\ 1 & -2 \end{bmatrix}$. Check: $\begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}\begin{bmatrix} -\frac{1}{2} & \frac{3}{2} \\ 1 & -2 \end{bmatrix} = \begin{bmatrix} -2+3 & 6-6 \\ -1+1 & 3-2 \end{bmatrix} = I_2$ ✓.

For Example 10 (p.46): $ad - bc = 2 \cdot 2 - 4 \cdot 1 = 0$, so the matrix is singular and not invertible, confirming the row reduction.

Sign slips in the formula

The minus signs go on b and c (off-diagonal), not on a and d. And $\frac{1}{ad-bc}$ keeps its sign: in Example 9 it is $-\frac{1}{2}$, which flips every entry.

A 4×4 example, exam style

The 2025 midterm (Q5) asked: "Use Gaussian elimination to find A^{-1} of a 4×4 matrix, if it exists. If A is invertible, verify your answer by checking $AA^{-1} = I_4$ and $A^{-1}A = I_4$." Same method, just more rows. Here is a practice matrix of the same size.

4×4 PRACTICE (MODELED ON LECTURE 6 P.32 AND MIDTERM Q5) Find A^{-1} for $A = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$ and verify.

1

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

2 $R_2 \to R_2 - R_1$ (clear column 1; rows 3 and 4 already have 0 there)

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

3 $R_3 \to R_3 - R_2$ (clear column 2 below its pivot)

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \end{array}\right]$$

4 $R_3 \leftrightarrow R_4$ (row 4 has a 1 in column 3, a better pivot than the 2)

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 1 & 1 & -1 & 1 & 0 \end{array}\right]$$

5 $R_4 \to R_4 - 2R_3$ (clear column 3 below its pivot)

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -1 & 1 & -1 & 1 & -2 \end{array}\right]$$

6 $R_4 \to -R_4$ (make the last pivot 1)

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 2 \end{array}\right]$$

7 $R_3 \to R_3 - R_4$ (clear column 4 above its pivot; rows 1 and 2 already have 0)

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 2 \end{array}\right]$$

8 $R_1 \to R_1 - R_3$ (clear column 3 above its pivot; row 2 already has 0; left block is I_4)

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & 1 & 0 & 0 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 & -1 & 1 & -1 & 2 \end{array}\right]$$

ANSWER

$$A^{-1} = \begin{bmatrix} 0 & 1 & -1 & 1 \\ -1 & 1 & 0 & 0 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 2 \end{bmatrix}$$

Verification (write it out; it is worth points). Row 3 of A, $(0,1,2,1)$, times the columns of A^{-1}: $0 \cdot 0 + 1 \cdot (-1) + 2 \cdot 1 + 1 \cdot (-1) = 0$, $0 + 1 - 2 + 1 = 0$, $0 + 0 + 2 - 1 = 1$, $0 + 0 - 2 + 2 = 0$: the third row of I_4. Doing all rows gives $AA^{-1} = I_4$. In the other order, row 4 of A^{-1}, $(-1,1,-1,2)$, times column 3 of A, $(1,1,2,1)^\top$: $-1 + 1 - 2 + 2 = 0$, and times column 4, $(0,0,1,1)^\top$: $-1 + 2 = 1$. All entries give $A^{-1}A = I_4$.

Use the zeros

If the matrix is already nearly triangular (like the midterm's, which had zeros in the lower-left corner), skip the columns that are already clean and go straight to the rows that need work. Fewer operations, fewer slips.

Right inverses via Gaussian elimination

The same idea works for a rectangular $m \times n$ matrix, "as long as $\operatorname{rank}(A) = m$. When $m < n$, the existence of free variables provides infinitely many right inverses" (p.48). We want $B \in M_{n \times m}$ with $AB = I_m$, i.e. $A\mathbf{b}_j = e_j$ for each column.

SECTION 4 EXAMPLE · LECTURE 6 P.48–51 Find right inverses of $A = \begin{bmatrix} 2 & -3 & 1 \\ 1 & -2 & 0 \end{bmatrix}$.

1 Augment with I_2:

$$\left[\begin{array}{ccc|cc} 2 & -3 & 1 & 1 & 0 \\ 1 & -2 & 0 & 0 & 1 \end{array}\right]$$

2 $R_1 \leftrightarrow R_2$ (the slide's first step: pivot 1 on top)

$$\left[\begin{array}{ccc|cc} 1 & -2 & 0 & 0 & 1 \\ 2 & -3 & 1 & 1 & 0 \end{array}\right]$$

3 $R_2 \to R_2 - 2R_1$ (clear below the first pivot)

$$\left[\begin{array}{ccc|cc} 1 & -2 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & -2 \end{array}\right]$$

4 $R_1 \to R_1 + 2R_2$ (clear above the second pivot (reduced echelon form))

$$\left[\begin{array}{ccc|cc} 1 & 0 & 2 & 2 & -3 \\ 0 & 1 & 1 & 1 & -2 \end{array}\right]$$

5 The left block has pivots in columns 1, 2; x_3 is free. **First column** B_1 (solves $A\mathbf{x} = e_1$, right column $(2, 1)$): $x_1 + 2x_3 = 2,\ x_2 + x_3 = 1$. With $x_3 = s$:

$$B_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + s \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}, \quad s \in \mathbb{R}.$$

6 **Second column** B_2 (solves $A\mathbf{x} = e_2$, right column $(-3, -2)$): with $x_3 = t$,

$$B_2 = \begin{bmatrix} -3 \\ -2 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ -1 \\ 1 \end{bmatrix}, \quad t \in \mathbb{R}.$$

7 $s = t = 0$ gives the right inverse $B = [B_1 \mid B_2] = \begin{bmatrix} 2 & -3 \\ 1 & -2 \\ 0 & 0 \end{bmatrix}$ (p.50); $s = 1,\ t = -1$ gives another, $B' = \begin{bmatrix} 0 & -1 \\ 0 & -1 \\ 1 & -1 \end{bmatrix}$ (p.51).

8 Check $AB = I_2$: $\begin{bmatrix} 4-3+0 & -6+6+0 \\ 2-2+0 & -3+4+0 \end{bmatrix} = I_2$ ✓. Check $AB' = I_2$: $\begin{bmatrix} 0+0+1 & -2+3-1 \\ 0+0+0 & -1+2+0 \end{bmatrix} = I_2$ ✓.

ANSWER

A has infinitely many right inverses $\begin{bmatrix} 2-2s & -3-2t \\ 1-s & -2-t \\ s & t \end{bmatrix}$, $s, t \in \mathbb{R}$.

Why the same direction vector twice

Both columns have the free-variable part $s(-2, -1, 1)^\top$ or $t(-2, -1, 1)^\top$: that vector solves $A\mathbf{x} = \mathbf{0}$, and adding it to any solution gives another solution.

In one minute

- Recipe: $[\,A \mid I_n\,] \to [\,I_n \mid A^{-1}\,]$ using row operations on whole rows.
- Order: go down (clear below pivots), scale pivots to 1, go up (clear above pivots). Swap rows when a pivot is 0.
- Zero row in the **left block** $\Rightarrow \operatorname{rank} A < n \Rightarrow A$ singular, not invertible (T2). Stop there.
- Always verify $AA^{-1} = I_n$ (and $A^{-1}A = I_n$ if asked). Factor out a common denominator first.
- 2×2: $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$; invertible $\iff ad - bc \neq 0$.
- Wide A with $\operatorname{rank} A = m$: row-reduce $[\,A \mid I_m\,]$; free variables give infinitely many right inverses.

CHAPTER 17

The exam playbook: five question types

The 2025 midterm had five questions, and each one is a type that can come back with new numbers. For each type you get what it looks like, a fixed recipe, a fresh worked example, the answer layout to copy onto your paper, and the usual mistakes. The last part shows how to write a complete proof at this level.

Built on the 2025 midterm and Lectures 3–6

→ **The one habit that wins points**

Every answer has two parts: the **computation** and the **reason**. In the 2025 midterm all the official numbers were right; what was thin were the reasons ("rank 2 because consistent", "so A is invertible" with no theorem). Graders read the reasons. One short sentence such as "by Rouché–Capelli" or "by Theorem T2" turns a guess into an argument.

The five types at a glance

Type	You see…	You do…	Lecture
1 · Span	"Is $\mathbf{b} \in \mathrm{Span}\{\dots\}$? Find the scalars"	Vectors as columns, reduce $[\cdots \mid \mathbf{b}]$, back-substitute, check	L4 p.8–13
2 · Classify	Several augmented matrices, "homogeneous? rank? pivots? how many solutions?"	Echelon form, count rows, compare ranks, compare with n	L3 p.33–41, L4 p.34–50
3 · Constraint	Tall A (more rows than columns), $\mathbf{b} = (b_1, \dots, b_m)$ as letters	Reduce with letters; each zero row ⇒ its right side $= 0$	L4 p.20–35
4 · Proof	"Prove", "prove or disprove", inverses, μ_A, rank	Translate into a statement about $A\mathbf{x} = \mathbf{0}$ or $A\mathbf{x} = \mathbf{b}$, then cite a theorem	L4, L5 p.34, L6
5 · Inverse	"Find A^{-1} by Gaussian elimination, if it exists; verify"	$[A \mid I_n] \to [I_n \mid A^{-1}]$, then both products	L6 p.30–45

Type 1: is b in the span? (midterm Q1)

What it looks like. Three or four vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and a vector $\mathbf{b}$. "(a) Is $\mathbf{b}$ in the span? (b) If so, write $\mathbf{b}$ as a linear combination." Sometimes it is worded as "Can $\mathbf{b}$ be written as a linear combination of...".

Why it is a linear system. $\mathbf{b} \in \text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ means there are numbers a, c, d with $a\mathbf{u} + c\mathbf{v} + d\mathbf{w} = \mathbf{b}$. Read entry by entry, that is a system whose coefficient columns are $\mathbf{u}, \mathbf{v}, \mathbf{w}$: the "column view" of $A\mathbf{x} = \mathbf{b}$ (Lecture 4 p.6–7).

$$\mathbf{b} \in \text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} \iff \left[\begin{array}{ccc|c} \mathbf{u} & \mathbf{v} & \mathbf{w} & \mathbf{b} \end{array}\right] \text{ is consistent}$$

RECIPE FOR TYPE 1

1. Write the augmented matrix with the vectors as **columns**.
2. Row reduce to echelon form.
3. A row $(0 \ \cdots \ 0 \mid c)$ with $c \neq 0$? Then the answer is **no**, $\mathbf{b} \notin \text{Span}$. Otherwise **yes**.
4. If yes: back-substitute to find the scalars. Free variables mean many choices; any one works, so give one.
5. Check by multiplying out the combination.

PRACTICE FOR TYPE 1 (NEW NUMBERS) Let $\mathbf{u} = (1, 2, -1)$, $\mathbf{v} = (0, 1, 3)$, $\mathbf{w} = (2, 1, 0)$, $\mathbf{b} = (4, 4, -5)$. Is $\mathbf{b} \in \text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$? If yes, find a, c, d.

1. $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 2 & 1 & 1 & 4 \\ -1 & 3 & 0 & -5 \end{array}\right]$. (columns $\mathbf{u}, \mathbf{v}, \mathbf{w} \mid \mathbf{b}$)
2. $R_2 \to R_2 - 2R_1$, $R_3 \to R_3 + R_1$: $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -3 & -4 \\ 0 & 3 & 2 & -1 \end{array}\right]$.
3. $R_3 \to R_3 - 3R_2$: $\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & 11 & 11 \end{array}\right]$. No contradiction row ⇒ consistent ⇒ **yes**.
4. $11d = 11 \Rightarrow d = 1$; $c - 3d = -4 \Rightarrow c = -1$; $a + 2d = 4 \Rightarrow a = 2$.
5. Check: $2\mathbf{u} - \mathbf{v} + \mathbf{w} = (2 - 0 + 2,\ 4 - 1 + 1,\ -2 - 3 + 0) = (4, 4, -5)$. ✓

ANSWER
$\mathbf{b} \in \text{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$, $\mathbf{b} = 2\mathbf{u} - \mathbf{v} + \mathbf{w}$.

Answer template (Type 1)

1. "$\mathbf{b} \in \text{Span} \Leftrightarrow a\mathbf{u} + c\mathbf{v} + d\mathbf{w} = \mathbf{b}$ has a solution." Write the system or the augmented matrix.
2. Each row operation written above its arrow, e.g. $R_2 \to R_2 - 2R_1$.
3. "Echelon form has no row $(0\ 0\ 0 \mid c \neq 0)$, so the system is consistent and $\mathbf{b} \in \text{Span}$." (Or: "row $(0\ 0\ 0 \mid 5)$ means $0 = 5$, so $\mathbf{b} \notin \text{Span}$.")
4. Back substitution, then the boxed combination and the check.

Common mistakes

- Vectors as rows instead of columns.
- Stopping at "yes" in (a) with no reason.
- Forgetting that "no" is a possible answer: if a contradiction row appears, (b) has no scalars (Lecture 4 Example 2).
- Skipping the check, which takes 20 seconds and catches every slip.

Type 2: classify augmented matrices (midterm Q2)

What it looks like. Two to four augmented matrices, each with the same list of questions: homogeneous or not, $\text{rank}\, A$ and $\text{rank}[A \mid b]$, pivot and free variables, number of solutions. Usually most are already in echelon form, and one is a trap that is not.

The four facts you use. Rank = number of nonzero rows in an **echelon** form (Lecture 4, D2). Consistent $\Leftrightarrow \text{rank}\, A = \text{rank}[A \mid b]$ (Rouché–Capelli, P1). If consistent: unique $\Leftrightarrow r = n$, infinitely many $\Leftrightarrow r < n$ with $n - r$ free variables (Lecture 4 p.47–50). Homogeneous means $\mathbf{b} = \mathbf{0}$ (D3), and such a system is always consistent.

$$\begin{aligned} \text{rank}\, A < \text{rank}[A \mid b] &\Rightarrow \text{no solution} \\ \text{rank}\, A = \text{rank}[A \mid b] = n &\Rightarrow \text{exactly one} \\ \text{rank}\, A = \text{rank}[A \mid b] < n &\Rightarrow \text{infinitely many } (n - r \text{ free}) \end{aligned}$$

RECIPE FOR TYPE 2 (N = NUMBER OF UNKNOWNS = COLUMNS OF A)

1. **Is it in echelon form?** If not, row reduce first (only the left part decides the pivots).
2. Last column all zeros? Homogeneous; otherwise inhomogeneous.
3. Count nonzero rows of A and of $[A \mid b]$.
4. Pivot variables = pivot columns of A; free = the other columns of A.
5. Use the table above for the number of solutions.

PRACTICE FOR TYPE 2 (NEW NUMBERS) Classify (i) $\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right]$, (ii) $\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & 1 & 3 & 0 \\ 0 & 2 & 6 & 0 \end{array}\right]$, (iii) $\left[\begin{array}{cc|c} 2 & 1 & 1 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{array}\right]$.

1. (i) Inhomogeneous ($\mathbf{b} = (3, 2, 0) \neq \mathbf{0}$). Echelon form; ranks $2 = 2 \Rightarrow$ consistent. Pivots in columns 1, 3 $\Rightarrow$ pivot x_1, x_3, free x_2. $r = 2 < n = 3 \Rightarrow$ **infinitely many**: $x_3 = 2$, $x_1 = x_2 - 1$.
2. (ii) Homogeneous. **Not** in echelon form (the 2 below the pivot of row 2). $R_3 \to R_3 - 2R_2$ gives $(0\ 0\ 0 \mid 0)$. Ranks $2 = 2$. Pivot x_1, x_2, free x_3. **Infinitely many**: $\mathbf{x} = t(5, -3, 1)$.
3. (iii) Inhomogeneous. Echelon form; $\operatorname{rank} A = 2$ but $\operatorname{rank}[A \mid b] = 3$ (row 3 says $0 = 4$) $\Rightarrow$ **no solution**. Pivot columns of A: x_1, x_2; no free columns.

ANSWER
(i) infinitely many, 1 free; (ii) infinitely many, 1 free; (iii) inconsistent.

Answer template (Type 2), one block per matrix

- **Type:** homogeneous / inhomogeneous, because $\mathbf{b} = \ldots$
- **Echelon form:** (already / after $R_i \to \ldots$) ...
- **Ranks:** $\operatorname{rank} A = _$ (nonzero rows of A), $\operatorname{rank}[A \mid b] = _$.
- **Pivot / free:** pivot $x_.$; free $x_.$.
- **Solutions:** none / one / infinitely many, because (ranks differ) / (ranks equal $= n$) / (ranks equal $< n$).

Common mistakes

- Counting rows of a matrix that is not in echelon form.
- Giving the wrong reason for a right number ("rank 2 because consistent"). The rank comes from counting rows.
- "More equations than unknowns $\Rightarrow$ no solution" or "a zero row $\Rightarrow$ infinitely many". Only the ranks decide.
- Listing the $\mathbf{b}$-column as a pivot variable. Variables are columns of A only.
- For an inconsistent system, leaving pivot/free blank with no explanation. Name the pivot columns of A **and** say "no solution, so there is nothing to parametrise".

Type 3: the constraint equation (midterm Q3)

What it looks like. A matrix with more rows than columns (e.g. 4×3) and $\mathbf{b} = (b_1, \ldots, b_m)$ written with letters. "Find the constraint equation(s) that $\mathbf{b}$ must satisfy for consistency." Then: "Is the system consistent for this particular $\mathbf{b}$?"

Why it works. Row operations never mix the b's into the left block, so each row of the echelon form that is zero on the left reads $0 = (\text{combination of } b\text{'s})$. Consistent ⇔ all of these combinations are 0. There are $m - r$ of them (Lecture 4 p.35).

$$\#\{\text{constraint equations}\} = m - \operatorname{rank} A$$

HOW MANY TO EXPECT

1. Compute $m - r$ in your head so you know how many zero rows to expect (usually 1).
2. Row reduce $[A \mid \mathbf{b}]$ keeping $b_1, \ldots, b_m$ as letters. Combine like terms after every step.
3. For each zero row on the left: (right side) $= 0$. Clear fractions.
4. **Self-check:** every column of A must satisfy the constraint (because $A\mathbf{e}_j$ is the j-th column, that system is consistent).
5. Test the given $\mathbf{b}$: plug in. 0 means consistent, anything else means no solution.

PRACTICE FOR TYPE 3 (NEW NUMBERS) $A = \begin{bmatrix} 1 & 2 & 0 \\ 1 & 3 & 1 \\ 0 & 1 & 2 \\ 2 & 4 & 1 \end{bmatrix}$. Find the constraint on $\mathbf{b}$ and test $\mathbf{b} = (1, 2, 0, 1)$ and $\mathbf{b} = (1, 0, 0, 0)$.

1. $[A \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 1 & 2 & 0 & b_1 \\ 1 & 3 & 1 & b_2 \\ 0 & 1 & 2 & b_3 \\ 2 & 4 & 1 & b_4 \end{array}\right]$. $R_2 \to R_2 - R_1$, $R_4 \to R_4 - 2R_1$: $\left[\begin{array}{ccc|c} 1 & 2 & 0 & b_1 \\ 0 & 1 & 1 & b_2 - b_1 \\ 0 & 1 & 2 & b_3 \\ 0 & 0 & 1 & b_4 - 2b_1 \end{array}\right]$.
2. $R_3 \to R_3 - R_2$: row 3 becomes $(0\ 0\ 1 \mid b_3 - b_2 + b_1)$.
3. $R_4 \to R_4 - R_3$: row 4 becomes $(0\ 0\ 0 \mid b_4 - 2b_1 - b_3 + b_2 - b_1) = (0\ 0\ 0 \mid -3b_1 + b_2 - b_3 + b_4)$.
4. Constraint: $-3b_1 + b_2 - b_3 + b_4 = 0$. Self-check on the columns: $-3 + 1 - 0 + 2 = 0$, $-6 + 3 - 1 + 4 = 0$, $0 + 1 - 2 + 1 = 0$. ✓
5. $\mathbf{b} = (1, 2, 0, 1)$: $-3 + 2 - 0 + 1 = 0 \Rightarrow$ consistent (indeed $\mathbf{x} = (-3, 2, -1)$). $\mathbf{b} = (1, 0, 0, 0)$: $-3 \neq 0 \Rightarrow$ no solution.

ANSWER

$-3b_1 + b_2 - b_3 + b_4 = 0$; consistent for $(1, 2, 0, 1)$, inconsistent for $(1, 0, 0, 0)$.

Answer template (Type 3)

1. $[A \mid \mathbf{b}]$ with letters, each row operation labelled.
2. "Row k is $(0 \cdots 0 \mid \ \ldots)$; the system is consistent iff this is 0."
3. Boxed constraint with integer coefficients.
4. (b) "Substituting $\mathbf{b} = \ldots$: $\cdots = _ \neq 0$, so the system has no solution" (or "$= 0$, so it is consistent").

Common mistakes

- Sign errors when dividing a row with letters, e.g. $\frac{b_2-b_1}{-2} = \frac{b_1-b_2}{2}$.
- Not simplifying after each step, so expressions grow and slips hide.
- Thinking a different-looking answer is wrong. Any nonzero multiple of the constraint is the same constraint, and the column check settles it.
- Solving the whole system in (b) when the question says "using the constraint".

Type 4: short proofs (midterm Q4)

What it looks like. One or two statements about inverses, rank, consistency or the map μ_A. "Prove that..." or "Prove or disprove...". Typical ingredients: $BA = I_n$, $AB = I_m$, $\operatorname{rank} A = m$ or $= n$, "surjective", "unique solution for every $\mathbf{b}$".

The fixed recipe.

1. **Write what must be shown**, using the definition: "$B = A^{-1}$" means $AB = BA = I_n$ (Lecture 6, D2); "μ_A surjective" means $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b} \in \mathbb{R}^m$.
2. **Translate it into a linear system**: a question about $A\mathbf{x} = \mathbf{0}$ (only trivial solution?) or about $A\mathbf{x} = \mathbf{b}$ (consistent for all $\mathbf{b}$? unique?).
3. **Decide it with a theorem**: Theorem T2 for square matrices, Rouché–Capelli or rank $= m$ for existence, rank $= n$ for uniqueness.
4. **For "prove or disprove"**: first test the claim on a small example. If it fails, give **one concrete counterexample** with numbers and show the failure. If it holds, prove it for every case.

PRACTICE FOR TYPE 4(A): THE MIRROR OF THE MIDTERM Let A, B be $n \times n$ with $AB = I_n$. Prove that $B = A^{-1}$.

1. Let $B\mathbf{x} = \mathbf{0}$. Then $\mathbf{x} = I_n\mathbf{x} = (AB)\mathbf{x} = A(B\mathbf{x}) = A\mathbf{0} = \mathbf{0}$. (associativity, Lecture 5 Prop. 1.2)
2. So $B\mathbf{x} = \mathbf{0}$ has only the trivial solution. B is square, so B is invertible by Theorem T2 (Lecture 6).
3. $A = AI_n = A(BB^{-1}) = (AB)B^{-1} = B^{-1}$. So $A = B^{-1}$, which means $AB = BA = I_n$.
4. Hence B satisfies Definition D2 as the inverse of A: $B = A^{-1}$. ■

ANSWER
$AB = I_n$ (square) ⇒ B invertible ⇒ $A = B^{-1}$ ⇒ $B = A^{-1}$.

PRACTICE FOR TYPE 4(B): A 'DISPROVE' Let A be $m \times n$ with $n \neq m$ and $\operatorname{rank} A = n$. Prove or disprove: $\mu_A : \mathbb{R}^n \to \mathbb{R}^m$ is surjective.

1. **False.** $n = \operatorname{rank} A \leq \min(m, n) \leq m$ and $n \neq m$, so $n < m$: the echelon form of A has $m - n \geq 1$ zero rows, and some $\mathbf{b}$ breaks the constraint.

2 Counterexample (Lecture 6, Example 1): $A = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$, $m = 3$, $n = 2$, $\operatorname{rank} A = 2 = n$.

3 Take $\mathbf{b} = (0, 0, 1)$. Row 3 of $[A \mid \mathbf{b}]$ is $(0\,0 \mid 1)$, i.e. $0 = 1$. So there is no $\mathbf{x}$ with $A\mathbf{x} = \mathbf{b}$, and $\mathbf{b}$ is not in the image of μ_A. ■

ANSWER
Disproved: rank $= n < m$ gives uniqueness, not existence; $\mathbf{b} = (0, 0, 1)$ is never reached.

Notice this A even has a left inverse ($\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} A = I_2$). That settles the true/false question on Lecture 6 p.12: "if A has a left inverse, $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$" is **false**.

Answer template (Type 4)

1. **Claim** (true/false) and **what it means** (the definition).
2. **Given**: list the hypotheses, sizes included (square? $m \times n$?).
3. **Chain**: one equality or implication per line, each with its reason in brackets (associativity, Theorem T2, Rouché–Capelli...).
4. **Conclusion**: repeat the claim, then ■. For "disprove": the counterexample with numbers and the line where it fails.

Common mistakes

- Using what you have to prove ("since $B = A^{-1}$, $AB = I$...").
- Writing A^{-1} before showing it exists.
- Swapping the order in a product: $BA \neq AB$ in general (Lecture 5 Example 4).
- Confusing $\operatorname{rank} = m$ (existence, onto) with $\operatorname{rank} = n$ (uniqueness).
- "Disproving" with words only. Give a matrix and a vector.

Type 5: inverse by Gaussian elimination (midterm Q5)

What it looks like. A 3×3 or 4×4 matrix: "Use Gaussian elimination to find A^{-1}, if it exists. Verify." Often it is nearly triangular, so there is little work below the diagonal.

$$[A \mid I_n] \xrightarrow{\text{row operations}} [I_n \mid A^{-1}]$$

METHOD (LECTURE 6 P.32)

1. Write $[A \mid I_n]$.
2. Go **down**: make zeros below each pivot. If a zero row appears on the left, stop: A is not invertible (Lecture 6 p.30, Example 10).
3. Go **up**: starting from the last column, make zeros above each pivot. Scale the pivots to 1.
4. Read off the right block, then verify $AA^{-1} = I_n$ **and** $A^{-1}A = I_n$.

PRACTICE FOR TYPE 5 (NEW NUMBERS) Find A^{-1} for $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$.

1. $R_2 \to R_2 - R_1$: $\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right]$.
2. $R_3 \to R_3 - R_2$: $\left[\begin{array}{ccc|ccc} 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & -1 & 1 & 0 \\ 0 & 0 & 1 & 1 & -1 & 1 \end{array}\right]$. Three pivots ⇒ invertible.
3. $R_2 \to R_2 - R_3$: row 2 becomes $(0\ 1\ 0 \mid -2\ 2\ -1)$.
4. $R_1 \to R_1 - R_2$: row 1 becomes $(1\ 0\ 0 \mid 3\ -2\ 1)$.
5. Verify: row 2 of A times A^{-1}: $(1,2,1) \to (3-4+1,\ -2+4-1,\ 1-2+1) = (0,1,0)$; and similarly for the other rows, and for $A^{-1}A$.

ANSWER

$A^{-1} = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 2 & -1 \\ 1 & -1 & 1 \end{bmatrix}$, and $AA^{-1} = A^{-1}A = I_3$.

Answer template (Type 5)

1. $[A \mid I_n]$, then each new matrix in full, operations labelled.
2. One sentence: "n pivots, so A is invertible" (or "zero row, so A is not invertible").
3. Boxed A^{-1}.
4. "Check: $AA^{-1} = I_n$" and "$A^{-1}A = I_n$", with at least the row-by-row products shown.

Common mistakes

- Applying an operation to the left block only.
- Showing just one of the two products when the question asks for both.
- Copying slips in long rows. Rewrite the whole matrix each time, it is faster than hunting for errors.
- For a 2×2 check: $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ (Lecture 6 p.41–45). Swap the diagonal and negate the off-diagonal; do not transpose.

Writing proofs at this level

A proof here is short: usually 4–6 lines. It is **complete** when a reader can check every line without guessing. That needs three things: the goal stated precisely, every hypothesis used visibly (especially sizes such as "square"), and a named reason for every step that is not plain algebra.

What a complete answer to a Q4-type claim looks like

Incomplete (loses points) **Complete**

"$A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = BA\mathbf{x} = \mathbf{0}$, so A is invertible, so $B = A^{-1}$."

Missing: why "only trivial solution" gives "invertible"; that A is square; how $B = A^{-1}$ follows.

"Let $A\mathbf{x} = \mathbf{0}$. Then $\mathbf{x} = I_n\mathbf{x} = (BA)\mathbf{x} = B(A\mathbf{x}) = \mathbf{0}$. Since A is $n \times n$, Theorem T2 gives that A is invertible. Then $B = B(AA^{-1}) = (BA)A^{-1} = A^{-1}$. ■"

Results you may cite (Lectures 3–6)

Name	Statement	Where
Theorem 1 (L3)	Elementary row operations do not change the solution set.	L3 p.29
Definition D2 (L4)	$\operatorname{rank} A$ = number of nonzero rows in an echelon form; $\operatorname{rank} A \leq \min(m, n)$.	L4 p.34
Constraints	$\mathbf{b}$ must satisfy $m - r$ constraint equations.	L4 p.35
Proposition P1 (L4), Rouché–Capelli	Consistent ⇔ $\operatorname{rank} A = \operatorname{rank}[A \mid \mathbf{b}]$. If $\operatorname{rank} A = m$: solvable for every $\mathbf{b}$.	L4 p.37
Homogeneous systems	Always consistent; r pivot and $n - r$ free variables; $n > m \Rightarrow$ infinitely many solutions.	L4 p.47, p.49
Proposition P3 (L4)	A consistent system has a unique solution ⇔ $A\mathbf{x} = \mathbf{0}$ has only $\mathbf{0}$ ⇔ $r = n$.	L4 p.50
Proposition P4 (L4)	Square A: nonsingular ($r = n$) ⇔ RREF $= I_n$ ⇔ $A\mathbf{x} = \mathbf{0}$ only trivially ⇔ unique solution for every $\mathbf{b}$.	L4 p.53
Proposition 1.2 (L5)	$AI_n = A = I_mA$; distributivity; $(cA)B = c(AB)$; $(AB)C = A(BC)$.	L5 p.26–27
μ_A (L5)	$\mu_A : \mathbb{R}^n \to \mathbb{R}^m$, $\mu_A(\mathbf{x}) = A\mathbf{x}$; $\mu_{AB} = \mu_A \circ \mu_B$.	L5 p.34–37
D1 (L6)	Right inverse: $AB = I_m$; left inverse: $CA = I_n$.	L6 p.3
Rank conditions (L6)	Solvable for all $\mathbf{b}$ ⇔ $\operatorname{rank} A = m$; unique for all $\mathbf{b}$ ⇔ $\operatorname{rank} A = n$. Lemma L1: right inverse ⇒ rank m; left inverse ⇒ rank n.	L6 p.15–16
D2 (L6)	A invertible if there is B with $AB = I_n = BA$; then $B = A^{-1}$ (and it is unique).	L6 p.18–19
Theorem T1 (L6)	Square A: nonsingular ⇔ invertible.	L6 p.30
Theorem T2 (L6)	Square A: invertible ⇔ $\operatorname{rank} A = n$ ⇔ RREF $= I_n$ ⇔ $A\mathbf{x} = \mathbf{0}$ only trivially ⇔ unique solution for every $\mathbf{b}$.	L6 p.31

Two theorems share a name

Lecture 4 and Lecture 6 both have a "Theorem T1" and a "Proposition P1", with different contents (L4 T1: solutions $= \mathbf{u}_1 + \mathbf{v}$; L6 T1: nonsingular ⇔ invertible; L4 P1: Rouché–Capelli; L6 P1: $(AB)^{-1} = B^{-1}A^{-1}$). On the exam, write the statement in a few words as well, e.g. "by Rouché–Capelli" or "by Theorem T2 (invertible ⇔ only trivial solution)". The word **surjective** is not defined on the slides; it is the usual meaning for functions: every $\mathbf{b}$ in the codomain $\mathbb{R}^m$ is hit.

Course-sourced examples for "prove or disprove"

- **One-sided inverses are not inverses** (L6 Example 1): $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$, $AB = I_2$, $BA \neq I_3$.
- **A left inverse does not give existence** (L6 p.12–13): the 3×2 matrix B above has the left inverse A, but $B\mathbf{x} = (0, 0, 1)$ has no solution.
- **A singular square matrix** (L6 Example 10): $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$ gets a zero row, so it has no inverse; $ad - bc = 0$.
- $AB \neq BA$ (L5 Example 4): $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix}$.

In one minute

- Type 1: span ⇔ $[\mathbf{u}\ \mathbf{v}\ \mathbf{w} \mid \mathbf{b}]$ consistent; back-substitute; check.
- Type 2: echelon form **first**; rank = nonzero rows; compare ranks (existence) and r with n (uniqueness).
- Type 3: reduce with letters; $m - r$ zero rows ⇒ $m - r$ constraints; check them on the columns of A.
- Type 4: definition, then linear system, then theorem (T2, Rouché–Capelli, rank m / rank n); counterexamples with numbers.
- Type 5: $[A \mid I] \to [I \mid A^{-1}]$; zero row ⇒ no inverse; verify both products.
- Always write the reason next to the result. That is where the 2025 points were lost.

CHAPTER 18

The 2025 midterm, fully solved

The real MAT 2037 midterm of 05.11.2025: five questions, 20 points each. For each one you get how to recognise it, a clean solution that follows the official typed solution, the gaps in that solution filled in, and the traps that cost points.

From the 2025 midterm (05.11.2025, with typed solutions), pages 1–7

The exam at a glance

Q	Points	Type	Method	Budget
1	20	Is $\mathbf{b}$ in a span? Find the coefficients	$[\mathbf{u}\ \mathbf{v}\ \mathbf{w} \mid \mathbf{b}]$, echelon form, back substitution	12 min
2	20	Classify three augmented matrices	Echelon form, then ranks, then pivots, then number of solutions	18 min
3	20	Constraint equation on $b_1, \ldots, b_4$	Reduce $[A \mid \mathbf{b}]$ with letters; zero row ⇒ its right side $= 0$	15 min
4	20	Two short proofs	Theorem T2 (Lecture 6), Rouché–Capelli (Lecture 4)	20 min
5	20	Inverse of a 4×4 matrix	$[A \mid I_4] \to [I_4 \mid A^{-1}]$, then check both products	15 min

About the source and the time budget

The PDF has the printed exam and a typed solution for every question. **All numbers in the official solutions are correct** (we re-did each one with sympy, matrix by matrix). What it has are a few gaps in the reasoning: a wrong reason in Q2(a), a missing row reduction in Q2(b), proofs in Q4 that do not cite the theorems they use, and a verification in Q5 that is only claimed, not shown. Each gap is fixed below in a grey note. The exam sheet gives the start time (15:00) but not the length. The budget column assumes about 90 minutes and leaves 10 minutes for checking.

Question 1 (20 points): is b in the span?

Let $\mathbf{u} = (3, -2, 0)^T$, $\mathbf{v} = (0, 2, -3)^T$, $\mathbf{w} = (2, 1, -2)^T$, $\mathbf{b} = (1, 0, 2)^T$. *(a) Determine whether* $\mathbf{b} \in \mathrm{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$. *(b) If it does, find scalars* $a, c, d \in \mathbb{R}$ *such that* $\mathbf{b} = a\mathbf{u} + c\mathbf{v} + d\mathbf{w}$.

How to recognise it

The words "span" or "linear combination" mean **solve a linear system**. The vectors become the **columns** of the coefficient matrix and $\mathbf{b}$ is the right-hand side. This is exactly Example 1 of Lecture 4 (p.8–13), just with 3 equations instead of 4. **Time budget:** about 12 minutes.

MIDTERM 2025 Q1 · 2025 MIDTERM P.1–2 Is $\mathbf{b} \in \mathrm{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$? If yes, find a, c, d.

1. Write $a\mathbf{u} + c\mathbf{v} + d\mathbf{w} = \mathbf{b}$ entry by entry: $3a + 0c + 2d = 1$, $-2a + 2c + d = 0$, $0a - 3c - 2d = 2$.
2. Augmented matrix: $\left[\begin{array}{ccc|c} 3 & 0 & 2 & 1 \\ -2 & 2 & 1 & 0 \\ 0 & -3 & -2 & 2 \end{array}\right]$. (columns = $\mathbf{u}, \mathbf{v}, \mathbf{w}$, last column = $\mathbf{b}$)
3. $R_2 \to R_2 + \frac{2}{3}R_1$: $\left[\begin{array}{ccc|c} 3 & 0 & 2 & 1 \\ 0 & 2 & \frac{7}{3} & \frac{2}{3} \\ 0 & -3 & -2 & 2 \end{array}\right]$. $(-2 + \frac{2}{3}\cdot 3 = 0;\ 1 + \frac{2}{3}\cdot 2 = \frac{7}{3};\ 0 + \frac{2}{3}\cdot 1 = \frac{2}{3})$
4. $R_3 \to R_3 + \frac{3}{2}R_2$: $\left[\begin{array}{ccc|c} 3 & 0 & 2 & 1 \\ 0 & 2 & \frac{7}{3} & \frac{2}{3} \\ 0 & 0 & \frac{3}{2} & 3 \end{array}\right]$. $(-2 + \frac{3}{2}\cdot\frac{7}{3} = \frac{3}{2};\ 2 + \frac{3}{2}\cdot\frac{2}{3} = 3)$
5. Echelon form, and no row $(0\ 0\ 0 \mid c)$ with $c \neq 0 \Rightarrow$ the system is **consistent** $\Rightarrow \mathbf{b}$ is in the span. That answers (a).
6. Back substitution: row 3: $\frac{3}{2}d = 3 \Rightarrow d = 2$. Row 2: $2c + \frac{7}{3}\cdot 2 = \frac{2}{3} \Rightarrow 2c = -4 \Rightarrow c = -2$. Row 1: $3a + 2\cdot 2 = 1 \Rightarrow a = -1$.
7. Check: $-\mathbf{u} - 2\mathbf{v} + 2\mathbf{w} = (-3 + 0 + 4,\ 2 - 4 + 2,\ 0 + 6 - 4) = (1, 0, 2) = \mathbf{b}$. ✓

ANSWER

(a) Yes, $\mathbf{b} \in \mathrm{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$. (b) $a = -1,\ c = -2,\ d = 2$, i.e. $\mathbf{b} = -\mathbf{u} - 2\mathbf{v} + 2\mathbf{w}$.

Note

Because there are three pivots (rank 3 = number of unknowns), these are the **only** scalars that work. The question does not ask for this, but one sentence saying so shows you understand why the answer is "the" answer.

Where students lose points

- Putting $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in as **rows**. That is a different system, and the numbers come out wrong.
- Answering (a) with "yes" and no reason. The reason is "the echelon form has no row $(0\ 0\ 0 \mid c \neq 0)$", or "the system is consistent".
- Fraction slips in $R_2 + \frac{2}{3}R_1$. If you dislike fractions, use $R_2 \to 3R_2 + 2R_1$ (a legal combination of operations (ii) and (iii)). It gives $(0\ 6\ 7 \mid 2)$ and the same answer.
- Not doing the 20-second check at the end. It catches every arithmetic slip.

Question 2 (20 points): classify three systems

For each system, given by its augmented matrix, determine: i) homogeneous or inhomogeneous, ii) the rank of A and of $[A \mid b]$, iii) the pivot and free variables, iv) the number of solutions, if it exists.

How to recognise it

Several augmented matrices and a list of four things to find. It is pure Lecture 3–4 vocabulary: echelon form, rank (Lecture 4 D2), Rouché–Capelli (P1), homogeneous (D3), pivot and free variables (Lecture 3 D3). Run the same four-line recipe on every matrix (see the playbook chapter). The first thing to check every time is **whether the matrix is really in echelon form**. **Time budget:** about 18 minutes (6 per matrix).

(a)

MIDTERM 2025 Q2(A) · 2025 MIDTERM P.2

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

1. Last column is $\mathbf{0} \Rightarrow$ **homogeneous**.
2. Already in echelon form. A has two nonzero rows $\Rightarrow \operatorname{rank} A = 2$. Adding a zero column does not change that $\Rightarrow \operatorname{rank}[A \mid b] = 2$. (rank = number of nonzero rows in an echelon form)
3. Leading entries in columns 1 and 3 $\Rightarrow$ pivot variables x_1, x_3; free variable x_2.
4. Consistent (equal ranks; a homogeneous system always has $\mathbf{x} = \mathbf{0}$) and $r = 2 < n = 3 \Rightarrow$ **infinitely many** solutions: $x_3 = 0$, $x_1 = -2x_2$, i.e. $\mathbf{x} = t(-2, 1, 0)$.

ANSWER

Homogeneous; $\operatorname{rank} A = \operatorname{rank}[A \mid b] = 2$; pivot x_1, x_3, free x_2; infinitely many solutions.

The official solution gives the wrong reason

It writes "$\operatorname{rank}(A) = \operatorname{rank}(A \mid b) = 2$ (because the system is consistent)". Consistency only tells you the two ranks are **equal** (Rouché–Capelli). It does not tell you they equal **2**. The 2 comes from counting the nonzero rows of the echelon form. Write "rank 2: two nonzero rows in echelon form; ranks equal, so consistent". The final answers are right.

(b)

MIDTERM 2025 Q2(B) · 2025 MIDTERM P.3

$$\left[\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 3 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array}\right]$$

1. $\mathbf{b} = (3, 0, 1, 1) \neq \mathbf{0} \Rightarrow$ **inhomogeneous**.

2. **Not** in echelon form: the 3 and the 1 sit below the leading -1 of row 2 in column 3. Reduce: $R_3 \to R_3 + 3R_2$, $R_4 \to R_4 + R_2$: $\left[\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 3 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{array}\right]$, then $R_4 \to R_4 - R_3$: $\left[\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 3 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$.
3. Left block (A): 2 nonzero rows $\Rightarrow \operatorname{rank} A = 2$. Whole matrix: 3 nonzero rows $\Rightarrow \operatorname{rank}[A \mid b] = 3$.
4. Row $(0\,0\,0\,0 \mid 1)$ says $0 = 1 \Rightarrow$ **inconsistent**, no solution. (Rouché–Capelli: $2 \neq 3$)
5. Pivot/free: the pivot columns of A are 1 and 3, so x_1, x_3 would be pivot variables and x_2, x_4 free. But there is no solution to describe.

ANSWER
Inhomogeneous; $\operatorname{rank} A = 2$, $\operatorname{rank}[A \mid b] = 3$; no solution (pivot columns of A: 1 and 3; free: 2 and 4).

What the official solution skips

It says "inconsistent because we have a row like $(0\,0\,0\,0 \mid 1)$", but the given matrix has no such row. It only appears after the two row operations above, which you must show. For pivot/free it writes "— (system inconsistent)". That is defensible, but the question asks for pivot and free variables, so the safest paper answer gives both: "pivot columns of A: x_1, x_3; free: x_2, x_4; but the system has no solution."

(c)

MIDTERM 2025 Q2(C) · 2025 MIDTERM P.3

$$\left[\begin{array}{ccc|c} 1 & 2 & 2 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

1. $\mathbf{b} = (0, 0, 1, 0, 0) \neq \mathbf{0} \Rightarrow$ **inhomogeneous**. (one nonzero entry is enough)
2. Echelon form with 3 nonzero rows $\Rightarrow \operatorname{rank} A = \operatorname{rank}[A \mid b] = 3 \Rightarrow$ consistent.
3. Pivots in columns 1, 2, 3 $\Rightarrow$ pivot variables x_1, x_2, x_3; **no** free variables.
4. Consistent and $r = n = 3 \Rightarrow$ **unique** solution (Proposition P3, Lecture 4). The two zero rows say $0 = 0$ and change nothing.
5. Bonus check: $x_3 = -1$; $2x_2 - x_3 = 0 \Rightarrow x_2 = -\frac{1}{2}$; $x_1 + 2x_2 + 2x_3 = 0 \Rightarrow x_1 = 3$.

ANSWER
Inhomogeneous; rank $3 = 3$; pivot x_1, x_2, x_3, no free variables; exactly one solution, $(3, -\frac{1}{2}, -1)$.

Note

The official answer for (c) is correct, including the pivot/free claim and "unique". The system has 5 equations and 3 unknowns, but that does not make it inconsistent or give it many solutions: only the ranks matter.

Where students lose points

- Reading ranks off a matrix that is **not** in echelon form. In (b), "4 nonzero rows, so rank 4" is wrong.
- Calling (c) "homogeneous" because most of $\mathbf{b}$ is zero. One nonzero entry makes it inhomogeneous.
- "More equations than unknowns ⇒ no solution" (c) or "zero rows ⇒ infinitely many" (a, c). Decide with the ranks: compare $\operatorname{rank} A$ with $\operatorname{rank}[A \mid b]$ and with n.
- Giving a reason that is not the real reason (see the note on (a)). Graders read the reason.
- Counting the $\mathbf{b}$-column when you list pivot variables. Only columns of A are variables.

Question 3 (20 points): the constraint equation

Consider $A\mathbf{x} = \mathbf{b}$ *with* $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & 2 & 3 \end{bmatrix}$, $\mathbf{b} = (b_1, b_2, b_3, b_4)^T \in \mathbb{R}^4$. *(a) Determine a constraint equation that* $b_1, \ldots, b_4$ *must satisfy for the system to be consistent. (b) For* $\mathbf{b} = (1, 0, 2, 3)^T$, *use (a) to decide whether the system has a solution.*

How to recognise it

"Constraint equation", or "for which $\mathbf{b}$ is the system consistent", with a tall matrix (more rows than columns). Row reduce $[A \mid \mathbf{b}]$ keeping the b_i as letters. A is 4×3 of rank 3, so there is $m - r = 1$ zero row and exactly **one** constraint (Lecture 4 p.35). It is the same task as Examples 4–5 of Lecture 4 (p.22–27). **Time budget:** about 15 minutes.

MIDTERM 2025 Q3 · 2025 MIDTERM P.3–5 Find the constraint, then test $\mathbf{b} = (1, 0, 2, 3)$.

1. $[A \mid \mathbf{b}] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 1 & -1 & 1 & b_2 \\ 1 & 1 & -1 & b_3 \\ 1 & 2 & 3 & b_4 \end{array}\right]$.
2. $R_2 \to R_2 - R_1$, $R_3 \to R_3 - R_1$, $R_4 \to R_4 - R_1$: $\left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & -2 & 0 & b_2 - b_1 \\ 0 & 0 & -2 & b_3 - b_1 \\ 0 & 1 & 2 & b_4 - b_1 \end{array}\right]$.

3. $R_2 \to R_2/(-2)$: row 2 becomes $\left(\begin{array}{ccc|c} 0 & 1 & 0 & \frac{b_1-b_2}{2} \end{array}\right)$. $(\frac{b_2-b_1}{-2} = \frac{b_1-b_2}{2})$
4. $R_4 \to R_4 - R_2$: row 4 becomes $\left(\begin{array}{ccc|c} 0 & 0 & 2 & b_4 - \frac{3}{2}b_1 + \frac{1}{2}b_2 \end{array}\right)$. $(b_4 - b_1 - \frac{b_1-b_2}{2})$
5. $R_4 \to R_4 + R_3$: row 4 becomes $\left(\begin{array}{ccc|c} 0 & 0 & 0 & b_4 + b_3 - \frac{5}{2}b_1 + \frac{1}{2}b_2 \end{array}\right)$.
6. Row 4 now reads $0 = b_4 + b_3 - \frac{5}{2}b_1 + \frac{1}{2}b_2$. The system is consistent exactly when this is 0. Multiply by -2: $\boxed{5b_1 - b_2 - 2b_3 - 2b_4 = 0}$.
7. (b) $\mathbf{b} = (1, 0, 2, 3)$: $5(1) - 0 - 2(2) - 2(3) = 5 - 4 - 6 = -5 \neq 0 \Rightarrow$ the constraint fails $\Rightarrow$ **no solution**.

ANSWER
(a) $5b_1 - b_2 - 2b_3 - 2b_4 = 0$. (b) $-5 \neq 0$, so $A\mathbf{x} = \mathbf{b}$ has no solution.

Checked

We computed the constraint independently. Sympy's left null space of A is spanned by $(-5, 1, 2, 2)$, the same equation up to sign. We also tested it on 30 random $\mathbf{b}$: it holds exactly when $\operatorname{rank}[A \mid \mathbf{b}] = \operatorname{rank} A$. The official constraint is correct. A quick self-check you can do on paper: each **column** of A must satisfy the constraint (each column is $A\mathbf{e}_j$, so that system is consistent). Column 1: $5 - 1 - 2 - 2 = 0$ ✓. Column 2: $5 + 1 - 2 - 4 = 0$ ✓. Column 3: $5 - 1 + 2 - 6 = 0$ ✓.

Where students lose points

- Sign slips with the letters, especially in $R_2/(-2)$. Keep each entry as one fraction, and simplify only at the end.
- Stopping before the zero row: the constraint is the right side of the row whose left side is **all zeros**.
- In (b), solving the system from scratch instead of plugging into the constraint. The question says "by using the constraint equation", so plug in and write the number (-5).
- Different correct-looking answers: any nonzero multiple (e.g. $-5b_1 + b_2 + 2b_3 + 2b_4 = 0$) is the same constraint and gets full marks. Use the column check to be sure.

Question 4 (20 points): two proofs

(a) Let A and B be $n \times n$ matrices with $BA = I_n$. Prove that $B = A^{-1}$. (b) Let A be $m \times n$ with $n \neq m$ and $\operatorname{rank}(A) = m$. Consider $\mu_A : \mathbb{R}^n \to \mathbb{R}^m$, $\mu_A(\mathbf{x}) = A\mathbf{x}$. Prove or disprove that μ_A is surjective.

How to recognise it

"Prove", "prove or disprove", left/right inverses, μ_A, rank. Lecture 6 (inverses) and Lecture 4 (rank and consistency) give you the tools. The key move is to turn the claim into a statement about a **linear system** ($A\mathbf{x} = \mathbf{0}$ or $A\mathbf{x} = \mathbf{b}$) and then quote the theorem that decides it. **Time budget:** about 20 minutes (10 per part).

(a) $BA = I_n \Rightarrow B = A^{-1}$

MIDTERM 2025 Q4(A) · 2025 MIDTERM P.5 Prove $B = A^{-1}$.

1. **Step 1: $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.** Let $\mathbf{x} \in \mathbb{R}^n$ with $A\mathbf{x} = \mathbf{0}$. Then $\mathbf{x} = I_n\mathbf{x} = (BA)\mathbf{x} = B(A\mathbf{x}) = B\mathbf{0} = \mathbf{0}$. ($I_n\mathbf{x} = \mathbf{x}$ and associativity: Proposition 1.2, Lecture 5)
2. **Step 2: A is invertible.** A is square ($n \times n$), so by **Theorem T2** of Lecture 6, statement (4) "$A\mathbf{x} = \mathbf{0}$ has only the trivial solution" implies statement (1) "A is invertible". So there is A^{-1} with $AA^{-1} = A^{-1}A = I_n$.
3. **Step 3: $B = A^{-1}$.** $B = BI_n = B(AA^{-1}) = (BA)A^{-1} = I_nA^{-1} = A^{-1}$. ■

ANSWER

$A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = \mathbf{0}$, so A is invertible (Theorem T2), and $B = (BA)A^{-1} = A^{-1}$.

Is the official proof complete?

Its logic is right and the steps are the same as above. But it jumps from "the only solution of $A\mathbf{x} = \mathbf{0}$ is $\mathbf{x} = \mathbf{0}$" to "so A is invertible" without saying why. That step is **not** obvious. It is exactly Theorem T2 (4 ⇒ 1) of Lecture 6, p.31. Equivalently, use Proposition P4 of Lecture 4 to get "nonsingular" and then Theorem T1 of Lecture 6, "nonsingular ⇔ invertible". It only works because A is square. Name the theorem and you lose nothing. Its Step 2 ($BAA^{-1} = I_nA^{-1} \Rightarrow B = A^{-1}$) also uses associativity silently. Fine, but writing $B(AA^{-1}) = (BA)A^{-1}$ shows it.

Why the word 'square' matters

Lecture 6, Example 1 (p.5–6): $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$ give $AB = I_2$ but $BA = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \neq I_3$. A one-sided inverse of a non-square matrix is **not** an inverse. In the proof, Theorem T2 is where "square" is used, so say it there.

(b) Is μ_A surjective when $\operatorname{rank} A = m$?

MIDTERM 2025 Q4(B) · 2025 MIDTERM P.5 Prove or disprove: μ_A is surjective.

1. **Answer: true.** Say first what must be shown: μ_A is surjective means for every $\mathbf{b} \in \mathbb{R}^m$ there is $\mathbf{x} \in \mathbb{R}^n$ with $A\mathbf{x} = \mathbf{b}$, i.e. $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$.
2. Sizes: always $\operatorname{rank} A \leq \min(m, n)$ (Lecture 4, p.34). So $m = \operatorname{rank} A \leq n$, and since $n \neq m$, $m < n$.
3. Let $\mathbf{b} \in \mathbb{R}^m$ be arbitrary. $[A \mid \mathbf{b}]$ has m rows, so its rank is at most m. Adding a column never lowers the rank, so $m = \operatorname{rank} A \leq \operatorname{rank}[A \mid \mathbf{b}] \leq m$. Hence $\operatorname{rank}[A \mid \mathbf{b}] = \operatorname{rank} A$.
4. By Proposition P1 (Rouché–Capelli, Lecture 4 p.37), $A\mathbf{x} = \mathbf{b}$ is consistent. (This is exactly the slide's "In particular: if $\operatorname{rank}(A) = m$, $A\mathbf{x} = \mathbf{b}$ has at least one solution for every $\mathbf{b}$".) Another way to say it: the echelon form of A has no zero row, so a row $(0 \cdots 0 \mid c \neq 0)$ can never appear.
5. So every $\mathbf{b} \in \mathbb{R}^m$ is $\mu_A(\mathbf{x})$ for some $\mathbf{x}$: μ_A is surjective. ■

ANSWER
True: $\operatorname{rank} A = m \Rightarrow \operatorname{rank}[A \mid \mathbf{b}] = m$ for every $\mathbf{b} \Rightarrow A\mathbf{x} = \mathbf{b}$ always consistent $\Rightarrow \mu_A$ is onto. (Also $m < n$.)

Gaps in the official solution

The verdict ("True") and the outline are right. The official solution has three weak spots. (1) "Since $\operatorname{rank}(A) = m$ and $n \neq m$, it follows that $m < n$" needs its reason: $\operatorname{rank} A \leq \min(m, n)$, so $m \leq n$. (2) "Because A has full row rank, $A\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$" is the key step and should cite Rouché–Capelli or the zero-row argument. (3) The sentence about free variables and infinitely many solutions is true ($n - m \geq 1$ free variables) but plays no role in surjectivity. Also, the hypothesis $n \neq m$ is not needed for surjectivity: $\operatorname{rank} A = m$ alone is enough. It only rules out the square case, and with $m = n$ the map would even be bijective.

Where students lose points

- Mixing up the two rank conditions: $\operatorname{rank} A = m$ (rows) ⇒ **existence** for every $\mathbf{b}$, i.e. surjective. $\operatorname{rank} A = n$ (columns) ⇒ **uniqueness**. Lecture 6 p.15 states both.
- Arguing "$n > m$ so there are free variables, so it is onto". Free variables give *many* solutions, not *existence*. A 2×3 matrix of rank 1 has free variables and is not onto.
- In (a), "$BA = I$, so B is the inverse by definition". Definition D2 (Lecture 6 p.18) needs **both** $AB = I$ and $BA = I$. Proving the other side is the whole point.
- In (a), writing $B = A^{-1}$ before you have shown A^{-1} exists.
- "Disprove" answers without a concrete counterexample. If the claim is true (as here), you must prove it for **every** $\mathbf{b}$, not check one example.

Question 5 (20 points): inverse by Gaussian elimination

Use Gaussian elimination to find A^{-1}, if it exists, for $A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 1 & 4 \end{bmatrix}$. *If A is invertible, verify your answer by checking that $AA^{-1} = I_4$ and $A^{-1}A = I_4$.*

How to recognise it

"Find A^{-1} using Gaussian elimination, if it exists." This is the method of Lecture 6, p.32: reduce $[A \mid I_4]$ to $[I_4 \mid A^{-1}]$. This matrix is almost upper triangular, so it is short. One operation clears below the diagonal, then you clear upwards column by column. **Time budget:** about 15 minutes, including the check.

MIDTERM 2025 Q5 · 2025 MIDTERM P.6–7 Find A^{-1} and verify.

1. $[A \mid I_4] = \left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 4 & 0 & 0 & 0 & 1 \end{array}\right]$.
2. $R_4 \to R_4 - R_3$: $\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 3 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array}\right]$. Four pivots ⇒ rank 4 ⇒ A is invertible.
3. Clear column 4: $R_3 \to R_3 - 3R_4$, $R_2 \to R_2 - R_4$, $R_1 \to R_1 - R_4$: $\left[\begin{array}{cccc|cccc} 1 & 1 & 1 & 0 & 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 0 & 0 & 0 & 4 & -3 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array}\right]$.
4. Clear column 3: $R_2 \to R_2 - R_3$, $R_1 \to R_1 - R_3$: $\left[\begin{array}{cccc|cccc} 1 & 1 & 0 & 0 & 1 & 0 & -3 & 2 \\ 0 & 1 & 0 & 0 & 0 & 1 & -3 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 4 & -3 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array}\right]$.
5. Clear column 2: $R_1 \to R_1 - R_2$: $\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 1 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & -3 & 2 \\ 0 & 0 & 1 & 0 & 0 & 0 & 4 & -3 \\ 0 & 0 & 0 & 1 & 0 & 0 & -1 & 1 \end{array}\right]$.
6. Verify AA^{-1}. Row i of A times the columns of A^{-1}: row 1 $(1,1,1,1) \to (1,0,0,0)$; row 2 $(0,1,1,1) \to (0,1,0,0)$; row 3 $(0,0,1,3) \to (0,0,4-3,-3+3) = (0,0,1,0)$; row 4 $(0,0,1,4) \to (0,0,4-4,-3+4) = (0,0,0,1)$. So $AA^{-1} = I_4$.
7. Verify $A^{-1}A$. Row 1 $(1,-1,0,0) \to (1,0,0,0)$; row 2 $(0,1,-3,2) \to (0,1,1-3+2,1-9+8) = (0,1,0,0)$; row 3 $(0,0,4,-3) \to (0,0,1,0)$; row 4 $(0,0,-1,1) \to (0,0,0,1)$. So $A^{-1}A = I_4$.

ANSWER

$$A^{-1} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 4 & -3 \\ 0 & 0 & -1 & 1 \end{bmatrix}, \text{ and } AA^{-1} = A^{-1}A = I_4.$$

The verification in the official solution

Every intermediate matrix in the official solution is correct (each one was replayed with sympy). But its verification is one sentence: "Multiplying A by A^{-1} yields the identity matrix". It shows no product and does not mention $A^{-1}A$ at all, although the question asks for **both**. On your paper, show both products, at least row by row as in steps 6–7. That part of the 20 points is easy to collect.

Where students lose points

- Doing an operation on the left block only. Every row operation acts on the **whole** row of $[A \mid I]$.
- Forgetting the "if it exists": say why it exists (4 pivots, or the left block reached I_4).
- Showing only $AA^{-1} = I_4$. The question explicitly asks for $A^{-1}A$ too.
- Copying errors in a 4×8 matrix. Write each new matrix in full, and box the final right block.

In one minute

- Q1: span = system with the vectors as columns $\Rightarrow a = -1, c = -2, d = 2$, $\mathbf{b} = -\mathbf{u} - 2\mathbf{v} + 2\mathbf{w}$. Check by multiplying out.
- Q2: echelon form first (b is not!), then rank = nonzero rows, then Rouché–Capelli, then r vs n. (a) infinitely many, (b) none, (c) unique $(3, -\frac{1}{2}, -1)$.
- Q3: reduce $[A \mid \mathbf{b}]$ with letters $\Rightarrow 5b_1 - b_2 - 2b_3 - 2b_4 = 0$; $\mathbf{b} = (1, 0, 2, 3)$ gives $-5 \neq 0 \Rightarrow$ no solution.
- Q4(a): $A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = B(A\mathbf{x}) = \mathbf{0} \Rightarrow$ invertible by **Theorem T2** $\Rightarrow B = (BA)A^{-1} = A^{-1}$.
- Q4(b): true. $\operatorname{rank} A \leq \min(m, n)$ gives $m < n$; $\operatorname{rank} A = m \Rightarrow$ consistent for every $\mathbf{b}$ (P1) $\Rightarrow$ onto.
- Q5: $[A \mid I_4] \to [I_4 \mid A^{-1}]$, and show **both** AA^{-1} and $A^{-1}A$.
- The official numbers are all correct; the points are lost on missing reasons, so always name the theorem.

Cheat sheet

Read this the night before the exam.

Linear systems & notation

- **Linear eq.** $a_1x_1 + \cdots + a_nx_n = b \iff \mathbf{a} \cdot \mathbf{x} = b$: a **hyperplane** in $\mathbb{R}^n$, normal $\mathbf{a}$.
- **Solution** of a system: satisfies **all** m equations = point in the intersection of the m hyperplanes. **General solution** = complete parametric description.
- A is $m \times n$ (m rows = equations, n columns = variables); a_{ij} = row i, column j. $A\mathbf{x} = \mathbf{b}$: $\mathbf{A}_i \cdot \mathbf{x} = b_i$, $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$.
- **Equal matrices**: same shape and $a_{ij} = b_{ij}$ for all i, j.
- **Augmented** $[A|\mathbf{b}]$: coefficients | constants. Missing variable = 0.
- Ex. 1: $(4, -1, 0) + x_3(-1, 1, 1)$. Ex. 2: $\left(\frac{4}{3}, \frac{5}{3}\right)$.

Row operations & Theorem 1

- (i) $R_i \leftrightarrow R_j$ · (ii) $R_i \to rR_i$, $r \neq 0$ · (iii) $R_i \to R_i + rR_j$.
- Inverses: same swap · $R_i \to \frac{1}{r}R_i$ · $R_i \to R_i - rR_j$ (same type).
- **Thm 1**: $A\mathbf{x} = \mathbf{b} \to C\mathbf{x} = \mathbf{d}$ by (i)–(iii) $\Rightarrow$ same solution set. Proof: $\mathbf{u}$ solves old $\Rightarrow (r\mathbf{A}_k) \cdot \mathbf{u} = rb_k$, $(r\mathbf{A}_k + \mathbf{A}_l) \cdot \mathbf{u} = rb_k + b_l$; converse via inverse operations.
- p.10–13: $(2, 1, 0, 0) + x_3(0, 1, 1, 0) + x_4(-1, 3, 0, 1)$; pivot x_1, x_2, free x_3, x_4.

Echelon forms

- **D1 echelon**: leading entries move right; zeros below them; zero rows at bottom.
- **D2 reduced**: + every leading entry 1 + zeros **above** it (non-pivot columns: anything).
- **D3 pivot** = leading entry (in echelon form). Pivot columns $\to$ pivot vars; other columns $\to$ free vars. Constants column is **not** a variable.
- p.35 echelon not reduced · p.36 reduced · p.37, 42, 43 not echelon · p.41 echelon, free x_1 · p.44 reduced, pivot x_1, x_3, x_4, free x_2, x_5.

Gaussian elimination

- Swap a good pivot up, clear **below** with (iii), move right, repeat; back-substitute. Gauss–Jordan: scale pivots to 1, clear **above**, read off.
- **None**: row $[0 \cdots 0 \,|\, c \neq 0]$ (inconsistent, D4; p.49: $0 = 8$). **One**: no free variables. **Infinitely many**: ≥ 1 free variable.
- Parametric form: $\mathbf{x} = \mathbf{p} + x_{f_1}\mathbf{v}_1 + \ldots$, one vector per free variable; check $\mathbf{p}$.
- Ex. 3: $(9, 0, 0, 0) + x_2(-2, 1, 0, 0)$. p.28: $\left(\frac{27}{10}, \frac{1}{2}, -\frac{3}{10}\right)$.
- $\mathbf{b} \in \mathrm{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} \iff [\mathbf{u}\ \mathbf{v}\ \mathbf{w} \,|\, \mathbf{b}]$ consistent; solution = (a, c, d).

Ax = b: two views, consistency

- **Row view:** $\mathbf{c}$ solves $\iff \mathbf{A}_i \cdot \mathbf{c} = b_i$ for all rows: intersection of hyperplanes.
- **Column view:** $A\mathbf{x} = x_1\mathbf{v}_1 + \cdots + x_n\mathbf{v}_n$. Solvable $\iff \mathbf{b} \in \mathrm{Span}(\text{columns})$; $\mathbf{x}$ = the coefficients. Span test: vectors as **columns**, $\mathbf{b}$ after the bar.
- **D1:** consistent = at least one solution; inconsistent = none.
- Inconsistent $\iff$ echelon form of $[A|\mathbf{b}]$ has a row $[0 \cdots 0 \,|\, c]$, $c \neq 0$.
- **P1 (Rouché–Capelli):** consistent $\iff \mathrm{rank}\, A = \mathrm{rank}[A|\mathbf{b}]$ (else $\mathrm{rank}[A|\mathbf{b}] = \mathrm{rank}\, A + 1$). $r = m \Rightarrow$ solvable for every $\mathbf{b}$.

Constraint equations (Q3)

1. $[A|\mathbf{b}]$ with $b_1, \ldots, b_m$ as letters.
2. Row-reduce; each operation acts on the b-column too. Divide only by numbers.
3. Each zero row of the A-part: (its b-entry) $= 0$. Clear fractions.
4. Test $\mathbf{b}$: substitute. $0 \Rightarrow$ consistent, $\neq 0 \Rightarrow$ no solution.

- # constraints $= m - r$ (zero rows of echelon form of A). $r = m$: none ("if any").
- Self-check: every column of A satisfies the constraint. Multiples are equally correct (Remark p.27).
- Ex. 3: $b_4 - 7b_2 - 9b_3 + 13b_1 = 0$; Ex. 4: $-b_1 + b_2 - b_3 + b_4 = 0$; Ex. 5: $b_3 - 2b_1 - b_2 = 0$.
- Shifted plane (Ex. 6): $(2, -1, 0) + s(1, 3, 2) + t(-2, 1, 4) \Rightarrow 10x - 8y + 7z = 28$.

Rank and classifying a system (Q2)

- **D2:** $\mathrm{rank}\, A$ = # nonzero rows (= pivots) in **any** echelon form; $r \leq \min(m, n)$.
- Q2 order: reduce to echelon **first** · i) last column $= \mathbf{0}$? homogeneous · ii) $\mathrm{rank}\, A$ (left of bar), $\mathrm{rank}[A|\mathbf{b}]$ (whole) · iii) pivot x_j = pivot columns, rest free (inconsistent: "—") · iv) table.

ranks	r vs n	solutions
differ	any	none
equal	$r = n$	exactly one
equal	$r < n$	infinitely many ($n - r$ free)

- n = columns of A, not rows. $n > m \Rightarrow$ never unique.

Homogeneous, T1, uniqueness

- **D3:** $\mathbf{b} \neq \mathbf{0}$: inhomogeneous; $A\mathbf{x} = \mathbf{0}$: associated homogeneous system. Always consistent ($\mathbf{x} = \mathbf{0}$).
- **P2:** $A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y}$. **T1:** $A\mathbf{u}_1 = \mathbf{b} \Rightarrow$ all solutions $\mathbf{u}_1 + \mathbf{v}$, $A\mathbf{v} = \mathbf{0}$ (proof: $A(\mathbf{u} - \mathbf{u}_1) = \mathbf{0}$).
- $A\mathbf{x} = \mathbf{0}$: r pivot, $n - r$ free variables; unique (trivial) $\iff r = n$; $n > m \Rightarrow$ infinitely many.
- **P3:** consistent $+ r = n \iff$ unique. Unique for every $\mathbf{b} \iff r = m = n$.
- **P4** ($n \times n$): nonsingular ($r = n$) $\iff$ RREF $= I_n \iff A\mathbf{x} = \mathbf{0}$ only trivial $\iff$ unique solution $\forall \mathbf{b}$. Singular: $r < n$.

Matrix basics (Lecture 5)

- $A = [a_{ij}]$ is $m \times n$: m rows, n columns; a_{ij} = row i, column j. Rows $\in \mathbb{R}^n$, columns $\in \mathbb{R}^m$.
- $A = B \iff$ same size and $a_{ij} = b_{ij}$ for all i, j. O: all zeros. $\mathbb{M}_n(\mathbb{R})$: square.
- Diagonal: $a_{ij} = 0$ for $i \neq j$. Upper $\triangle$: $a_{ij} = 0$ for $i > j$ (zeros below). Lower $\triangle$: $a_{ij} = 0$ for $i < j$.
- $(cA)_{ij} = ca_{ij}$; $(A + B)_{ij} = a_{ij} + b_{ij}$, **same size only** ($2 \times 3$ + 3×2 undefined).
- Prop 1.1: $A + B = B + A$, assoc., $O + A = A$, $A + (-A) = O$, $c(dA) = (cd)A$, $c(A + B) = cA + cB$, $(c + d)A = cA + dA$, $1A = A$. Proof: compare the (i, j)-entries.
- $m \times n$ matrix = vector in $\mathbb{R}^{mn}$.

Matrix multiplication

- Size check: $(m \times n)(n \times p) = m \times p$. Columns of A = rows of B.
- $(AB)_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}$ = row i of $A \cdot$ column j of B.
- Ex.2: $\begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -7 \end{bmatrix}$; $AB = \begin{bmatrix} 3 & 4 \\ 10 & 6 \end{bmatrix}$, BA is 3×3.
- Ex.4: $\begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ 1 & 4 \end{bmatrix}$, reversed $= \begin{bmatrix} 3 & 6 \\ 1 & 6 \end{bmatrix}$.
- I_n: 1 on diagonal. Prop 1.2: $AI_n = A = I_m A$; $(A + A')B = AB + A'B$, $A(B + B') = AB + AB'$; $(cA)B = c(AB) = A(cB)$; $(AB)C = A(BC)$.
- Assoc. proof: column j: $A(Bc^{(j)}) = \sum c_k A b_k = (AB)c^{(j)}$.

Traps (prove or disprove)

- $AB \neq BA$: one undefined / different sizes / different entries.
- $(A + B)^2 = A^2 + AB + BA + B^2$, **not** $A^2 + 2AB + B^2$ (unless $AB = BA$).
- $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} = O$: $AB = O \not\Rightarrow A = O$ or $B = O$.
- $AB = AC \not\Rightarrow B = C$: $A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, B, C with the same first row.
- One counterexample = full disproof.

Matrices as functions

- A is $m \times n$: $\mu_A : \mathbb{R}^n \to \mathbb{R}^m$, $\mu_A(x) = Ax$. Linear: $\mu_A(x + y) = \mu_A(x) + \mu_A(y)$, $\mu_A(cx) = c\mu_A(x)$.
- $\mu_A(e_j)$ = column j; $\mu_A = \mu_B \Rightarrow A = B$. $\mu_{I_n}(x) = x$.
- $\mu_{AB} = \mu_A \circ \mu_B$: $\mathbb{R}^p \xrightarrow{\mu_B} \mathbb{R}^n \xrightarrow{\mu_A} \mathbb{R}^m$ (B first). Explains $(AB)C = A(BC)$.
- Onto $\iff \operatorname{rank} A = m$ (pivot in every row, no $(0 \cdots 0 \mid c \neq 0)$). 1-1 $\iff \operatorname{rank} A = n$ (no free var.). $\operatorname{rank} \leq \min(m, n)$.
- Midterm Q4(b): rank $= m \Rightarrow \mu_A$ onto: **true**. Rank $= n < m \Rightarrow$ not onto.

Left & right inverses

- A is $m \times n$. **Right:** $B_{n \times m}$, $AB = I_m$. **Left:** $C_{n \times m}$, $CA = I_n$. Symmetry: $AB = I \Rightarrow A$ is a left inverse of B.
- $AB = I \not\Rightarrow BA = I$: $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = I_2$, reverse product $\neq I_3$.
- Right inverse $\Rightarrow$ **existence**: $\mathbf{x} = B\mathbf{b}$ solves $A\mathbf{x} = \mathbf{b}$ for every $\mathbf{b}$.
- Left inverse $\Rightarrow$ **uniqueness** only: a solution, if any, is $C\mathbf{b}$. Check $A(C\mathbf{b}) = \mathbf{b}$; may fail.
- **L1:** right $\Rightarrow \operatorname{rank} = m$; left $\Rightarrow \operatorname{rank} = n$; both $\Rightarrow$ square, full rank.
- $C = CI = C(AB) = (CA)B = IB = B$.

Invertible = nonsingular

- **D2:** $A \in M_n$ invertible if $\exists B$: $AB = I_n = BA$; $B = A^{-1}$, $A = B^{-1}$. Unique.
- **T1:** nonsingular $\iff$ invertible. **T2:** invertible $\iff \operatorname{rank} A = n \iff$ RREF $= I_n \iff A\mathbf{x} = \mathbf{0}$ only $\mathbf{x} = \mathbf{0} \iff A\mathbf{x} = \mathbf{b}$ unique $\forall \mathbf{b}$ ($\mathbf{x} = A^{-1}\mathbf{b}$).
- **P1:** $(AB)^{-1} = B^{-1}A^{-1}$: show $(AB)(B^{-1}A^{-1}) = I = (B^{-1}A^{-1})(AB)$.
- $BA = I_n \Rightarrow B = A^{-1}$: $A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = (BA)\mathbf{x} = B\mathbf{0} = \mathbf{0} \Rightarrow A$ invertible (T2) $\Rightarrow B = (BA)A^{-1} = A^{-1}$. Don't use A^{-1} before step 1!

Gauss–Jordan for A^{-1}

- $[\,A \mid I_n\,] \to [\,I_n \mid A^{-1}\,]$. Ops on **whole** rows: $R_i \leftrightarrow R_j$, $R_i \to cR_i$ ($c \neq 0$), $R_i \to R_i + cR_j$.
- Down (clear below pivots, swap if pivot 0) $\to$ scale pivots to 1 $\to$ up (clear above).
- **Zero row in left block** $\Rightarrow$ singular, not invertible. Stop.
- Verify $AA^{-1} = I$ and $A^{-1}A = I$; factor out a denominator first.
- $\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 1 & 0 & 1 \end{bmatrix}^{-1} = \frac{1}{6} \begin{bmatrix} 1 & -2 & 5 \\ 4 & -2 & -4 \\ -1 & 2 & 1 \end{bmatrix}$.
- 2×2: $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$, invertible $\iff ad - bc \neq 0$.
- Wide A, $\operatorname{rank} = m < n$: reduce $[\,A \mid I_m\,]$, solve $A\mathbf{x} = e_j$; free vars $\Rightarrow \infty$ right inverses.

The 5 midterm question types at a glance

- **Span**: $\mathbf{b} \in \mathrm{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\} \Leftrightarrow [\mathbf{u}\,\mathbf{v}\,\mathbf{w} \mid \mathbf{b}]$ consistent (vectors as **columns**). '25: $\mathbf{b} = -\mathbf{u} - 2\mathbf{v} + 2\mathbf{w}$
- **Classify**: echelon form **first**; rank = nonzero rows; $r_A < r_{[A|b]} \Rightarrow$ none; $r_A = r_{[A|b]} = n \Rightarrow$ one; $= r < n \Rightarrow \infty$ ($n - r$ free). $\mathbf{b} = \mathbf{0} \Rightarrow$ homogeneous, always consistent
- **Constraint**: reduce $[A \mid \mathbf{b}]$ with letters; $m - r$ zero rows $\Rightarrow$ right sides $= 0$; check on columns of A. '25: $5b_1 - b_2 - 2b_3 - 2b_4 = 0$; $(1, 0, 2, 3) \to -5 \Rightarrow$ no sol.
- **Proof**: definition $\to A\mathbf{x} = \mathbf{0}$ / $A\mathbf{x} = \mathbf{b} \to$ theorem. Square + $A\mathbf{x} = \mathbf{0}$ only trivially $\Rightarrow$ invertible (T2, L6). $\mathrm{rank}\, A = m \Rightarrow$ solvable for all $\mathbf{b} \Rightarrow \mu_A$ onto (P1, L4). $\mathrm{rank}\, A = n \Rightarrow$ uniqueness only
- **Inverse**: $[A \mid I] \to [I \mid A^{-1}]$; zero row on left $\Rightarrow$ singular; show $AA^{-1} = I$ **and** $A^{-1}A = I$. 2×2: $\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$
- $\mathrm{rank}\, A \leq \min(m, n)$; one-sided inverse of a non-square matrix is not an inverse (L6 Ex.1)

Answer templates

- **Span**: system/augmented matrix $\to$ labelled row ops $\to$ "no row $(0 \cdots 0 \mid c \neq 0)$, consistent, so $\mathbf{b} \in$ Span" $\to$ back-substitute $\to$ combination + check
- **Classify** (per matrix): type ($\mathbf{b} = \ldots$) · echelon form · $\mathrm{rank}\, A$, $\mathrm{rank}[A|b]$ · pivot/free x_i · # solutions + reason
- **Constraint**: reduce with letters $\to$ "row k: $(0 \cdots 0 \mid \ldots)$, consistent iff $= 0$" $\to$ boxed integer constraint $\to$ plug in $\mathbf{b}$
- **Proof**: claim + definition · hypotheses (sizes!) · one step per line with [reason] · conclusion ■. Disprove: concrete matrix + vector
- **Inverse**: full $[A \mid I]$ each step $\to$ "n pivots $\Rightarrow$ invertible" $\to$ boxed $A^{-1} \to$ both products row by row
- BA=I proof: $A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = B(A\mathbf{x}) = \mathbf{0} \Rightarrow$ T2 $\Rightarrow B = (BA)A^{-1} = A^{-1}$

CHAPTER 20

Self-test

Circle your answers, then check the answers chapter. No peeking.

1 **Geometrically, the solution set of a single linear equation $\mathbf{a} \cdot \mathbf{x} = b$ in $\mathbb{R}^n$ (with $\mathbf{a} \neq \mathbf{0}$) is**

A a line in $\mathbb{R}^n$, for every n

B the whole space $\mathbb{R}^n$

C a hyperplane in $\mathbb{R}^n$

D a single point of $\mathbb{R}^n$

---

2 **A solution of a system of m linear equations in n variables corresponds to**

A a point in the union of the m hyperplanes

B one of the normal vectors of the hyperplanes

C a point lying on at least one of the m hyperplanes

D a point in the intersection of the m hyperplanes

---

3 **Ex. 2: what is the unique solution of $x_1 + x_2 = 3,\ 2x_1 - x_2 = 1$?**

A $\left(\frac{4}{3}, \frac{5}{3}\right)$ **B** $\left(\frac{5}{3}, \frac{4}{3}\right)$

C $(2, 1)$ **D** $(1, 2)$

---

4 **Which of the following is not an elementary row operation?**

A Multiply a row by $-\frac{1}{5}$ **B** Add 5 times row 1 to row 3

C Swap row 1 and row 3 **D** Multiply a row by 0

---

5 **Apply $R_2 \to R_2 - 3R_1$ to $\left[\begin{array}{cccc|c} 1 & 1 & -1 & -2 & 3 \\ 3 & -2 & 2 & 9 & 4 \end{array}\right]$. The new second row is**

A $\left[\begin{array}{cccc|c} 6 & 1 & -1 & 3 & 13 \end{array}\right]$

B $\left[\begin{array}{cccc|c} 0 & -5 & 5 & 15 & -5 \end{array}\right]$

C $\left[\begin{array}{cccc|c} 0 & -5 & 5 & 15 & 13 \end{array}\right]$

D $\left[\begin{array}{cccc|c} 0 & -5 & 5 & 3 & -5 \end{array}\right]$

---

6 **Theorem 1 states that if $A\mathbf{x} = \mathbf{b}$ is transformed into $C\mathbf{x} = \mathbf{d}$ by elementary row operations, then**

A both systems have the same set of solutions

B both systems have the same coefficient matrix

C the new system has at least as many solutions as the old one

D the new system is in echelon form

---

7 **Which matrix is in reduced echelon form?**

A $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

B $\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

C $\begin{bmatrix} 2 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{bmatrix}$

D $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

---

8 **Midterm Q2(b) gives** $\left[\begin{array}{cccc|c} 1 & 1 & 2 & 3 & 3 \\ 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 3 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 \end{array}\right]$**. What must you do before reading off the ranks?**

A Nothing: count the nonzero rows, rank $A = 4$

B Swap the last column to the front so that $\mathbf{b}$ becomes a pivot column

C Divide every row by its first entry so that all leading entries are 1

D Row reduce: it is not in echelon form (nonzero entries below the leading -1 in column 3)

---

9 **After elimination a row reads** $\left[\begin{array}{cccc|c} 0 & 0 & 0 & 0 & 8 \end{array}\right]$**. What can you conclude?**

A The row is redundant and can be ignored

B The system has infinitely many solutions

C $x_4 = 8$

D The system is inconsistent: it has no solution

---

10 **A consistent system has 3 equations in 5 unknowns. How many solutions does it have?**

A None

B Exactly two

C Exactly one

D Infinitely many

---

11 **Row view of** $A\mathbf{x} = \mathbf{b}$**: a vector** $\mathbf{c}$ **is a solution precisely when ...**

A $\mathbf{c}$ lies on at least one hyperplane $\mathbf{A}_i \cdot \mathbf{x} = b_i$

B $\mathbf{c}$ is a linear combination of the rows of A

C $\mathbf{c}$ lies in the intersection of all the hyperplanes $\mathbf{A}_i \cdot \mathbf{x} = b_i$

D $\mathbf{c}$ is orthogonal to every row of A

---

12 **Column interpretation: $A\mathbf{x} = \mathbf{b}$ is consistent if and only if ...**

A $\mathbf{b}$ is orthogonal to the columns of A

B $\mathbf{b} \in \text{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$, the span of the columns of A

C $\mathbf{b}$ is one of the columns of A

D the columns of A are linearly independent

---

13 **Find the constraint equation for $A\mathbf{x} = \mathbf{b}$ to be consistent, $A = \begin{bmatrix} 1 & 2 \\ 3 & 7 \\ 2 & 4 \end{bmatrix}$.**

A $b_1 - 2b_3 = 0$

B $2b_1 + b_3 = 0$

C $-2b_1 + b_3 = 0$

D $-3b_1 + b_2 = 0$

---

14 **The constraint equation of a system is $4b_1 + b_2 - 2b_3 + b_4 = 0$. For which $\mathbf{b}$ is $A\mathbf{x} = \mathbf{b}$ consistent?**

A $\mathbf{b} = (1, 1, 1, 1)$

B $\mathbf{b} = (2, 0, 1, 0)$

C $\mathbf{b} = (0, 1, 0, 1)$

D $\mathbf{b} = (1, 0, 2, 0)$

---

15 **A is a 4×6 matrix. Which value is impossible for $\text{rank}(A)$?**

A 5

B 3

C 0

D 4

---

16 **A is 3×5 with $\text{rank}(A) = 3$. Then $A\mathbf{x} = \mathbf{b}$ ...**

A has only the trivial solution

B has a unique solution for every $\mathbf{b}$

C is inconsistent for some $\mathbf{b}$

D has at least one solution for every $\mathbf{b} \in \mathbb{R}^3$

---

17 **A homogeneous system $A\mathbf{x} = \mathbf{0}$ has 3 equations and 5 unknowns. How many free variables does its general solution have?**

A none

B at least 2

C exactly 5

D exactly 3

---

18 **Why is a homogeneous system $A\mathbf{x} = \mathbf{0}$ always consistent?**

A because the trivial solution $\mathbf{x} = \mathbf{0}$ always satisfies it

B because A is square

C because $\text{rank } A = n$

D because it always has free variables

---

19 **Suppose $A\mathbf{x} = \mathbf{b}$ is consistent, A is $m \times n$ with rank r. It has a unique solution if and only if ...**

A $m = n$

B $r = n$

C $r = m$

D $\mathbf{b} = \mathbf{0}$

---

20 **A system $A\mathbf{x} = \mathbf{b}$ has 3 equations and 5 unknowns. Which is impossible?**

A consistent for every $\mathbf{b}$
B no solution
C infinitely many solutions
D exactly one solution

---

21 **What is the size of $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$?**

A 2×2
B 2×3
C 3×2
D 6×1

---

22 **With $B = \begin{bmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}$ (Example 1), what is $4B$?**

A $\begin{bmatrix} 28 & 8 & 9 \\ 40 & 11 & 12 \end{bmatrix}$
B $\begin{bmatrix} 28 & 32 & 36 \\ 40 & 44 & 48 \end{bmatrix}$
C $\begin{bmatrix} 28 & 32 & 36 \\ 10 & 11 & 12 \end{bmatrix}$
D $\begin{bmatrix} 11 & 12 & 13 \\ 14 & 15 & 16 \end{bmatrix}$

---

23 **Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \in \mathbb{M}_{2\times 3}(\mathbb{R})$ and $B = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \in \mathbb{M}_{3\times 1}(\mathbb{R})$. Which is correct?**

A $AB = \begin{bmatrix} -2 \\ -2 \\ 0 \end{bmatrix}$ and BA is undefined
B $AB = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$ and BA is undefined
C $AB = \begin{bmatrix} -2 \\ -2 \end{bmatrix}$ and BA is 3×3
D AB is undefined and BA is 3×3

---

24 **A is a 2×3 matrix. Which products are both equal to A?**

A AI_2 and I_3A
B AI_2 and I_2A
C AI_3 and I_3A
D AI_3 and I_2A

---

25 **Let $A = \begin{bmatrix} 2 & -1 & 0 \\ 1 & 3 & 4 \end{bmatrix}$. The function $\mu_A(x) = Ax$ is a map**

A $\mu_A : \mathbb{R}^3 \to \mathbb{R}^2$
B $\mu_A : \mathbb{R}^3 \to \mathbb{R}^3$
C $\mu_A : \mathbb{R}^6 \to \mathbb{R}^2$
D $\mu_A : \mathbb{R}^2 \to \mathbb{R}^3$

---

26 For $A \in \mathbb{M}_{m\times n}(\mathbb{R})$ and $B \in \mathbb{M}_{n\times p}(\mathbb{R})$, the function μ_{AB} equals

A $\mu_B \circ \mu_A$

B $\mu_A \circ \mu_B$

C $\mu_A \cdot \mu_B$ (pointwise product)

D $\mu_A + \mu_B$

---

27 Can a 4×2 matrix have a right inverse?

A No: a right inverse would force $\operatorname{rank}(A) = 4$, but the rank is at most 2

B Yes, infinitely many, because there are free variables

C Yes, exactly one if its columns are independent

D Only if A also has a left inverse

---

28 Square A has a right inverse B and a left inverse C. In $C = CI_n = C(AB) = (CA)B = I_nB = B$, the step $C(AB) = (CA)B$ uses ...

A commutativity of matrix multiplication

B associativity of matrix multiplication

C the distributive law

D uniqueness of the inverse

---

29 By Definition D2, $A \in M_n(\mathbb{R})$ is invertible if there is $B \in M_n(\mathbb{R})$ with ...

A $AB = BA$

B $AB = I_n$ only

C $AB = I_n = BA$

D $\operatorname{rank}(B) = n$

---

30 Which matrix is not invertible?

A $\begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix}$

B $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

C $\begin{bmatrix} 2 & 4 \\ 1 & 2 \end{bmatrix}$

D $\begin{bmatrix} 3 & 1 \\ -1 & 0 \end{bmatrix}$

Mock midterm (90 minutes)

Same shape as the real exam: 5 questions × 20 points. Solve on paper.

1 Let $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 0 \\ -1 \\ -2 \end{bmatrix}$.

(a) Determine whether $\mathbf{b} \in \operatorname{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$.

(b) If it does, find scalars $a, c, d \in \mathbb{R}$ such that $\mathbf{b} = a\mathbf{u} + c\mathbf{v} + d\mathbf{w}$. **(20 pts)**

2 For each linear system below, given by its augmented matrix, determine: i.) whether the system is homogeneous or inhomogeneous, ii.) the rank of the coefficient matrix A and of the augmented matrix $[A|\mathbf{b}]$, iii.) the pivot variables and free variables, iv.) the number of solutions if it exists.

(a) $\left[\begin{array}{cccc|c} 1 & -1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$ (b) $\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & -4 & 0 & -2 \end{array}\right]$ (c) $\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right]$ **(20 pts)**

3 Consider the linear system $A\mathbf{x} = \mathbf{b}$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & -1 \\ 2 & 3 & 0 \\ 1 & -2 & 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} \in \mathbb{R}^4$.

(a) Determine a constraint equation that the entries b_1, b_2, b_3, b_4 must satisfy in order for the system to be consistent.

(b) Let $\mathbf{b} = (2, 0, 2, 2)$. Determine whether $A\mathbf{x} = \mathbf{b}$ has a solution by using the constraint equation you found in part (a). If it does, find it. **(20 pts)**

4 Let A be an $m \times n$ matrix with $n \neq m$, and assume that $\text{rank}(A) = n$. Consider the linear transformation $\mu_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by $\mu_A(x) = Ax$. Prove or disprove that μ_A is surjective. **(20 pts)**

5 Consider $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$. Use Gaussian elimination to find A^{-1}, if it exists. If A is invertible, verify your answer by checking $AA^{-1} = I_3$ and $A^{-1}A = I_3$. **(20 pts)**

CHAPTER 21

Answers

Got one wrong? Read why, then go back to that chapter.

1 **C** Slide 3: the equation represents a hyperplane in $\mathbb{R}^n$; $\mathbf{x}$ is a solution iff the point lies on it. It is a line only when $n = 2$. (Lecture 3 p.3)

2 **D** A solution must satisfy **all** m equations, so it lies on every hyperplane at once: the intersection (slide 5). (Lecture 3 p.5)

3 **A** Adding the equations gives $3x_1 = 4$, so $x_1 = \frac{4}{3}$ and $x_2 = 3 - \frac{4}{3} = \frac{5}{3}$. $(2, 1)$ satisfies the first equation but gives $2 \cdot 2 - 1 = 3 \neq 1$ in the second. (Lecture 3 p.8)

4 **D** Type (ii) requires a **nonzero** real number. Multiplying by 0 would trivialize the equation and cannot be undone (slide 32). (Lecture 3 p.9, 32)

5 **B** Entry by entry: $3 - 3 = 0,\ -2 - 3 = -5,\ 2 + 3 = 5,\ 9 + 6 = 15,\ 4 - 9 = -5$. The row $(6, 1, -1, 3 \mid 13)$ is what $R_2 + 3R_1$ would give. (Lecture 3 p.25)

6 **A** Row operations change the matrix but never the solution set (proved case by case for (i), (ii), (iii), then reversed with inverse operations). (Lecture 3 p.29)

7 **B** It is in echelon form, its leading entries are 1, and the pivot columns 1 and 2 have zeros above and below. The others: a 2 above a pivot; a zero row that is not at the bottom; a leading entry 2. (Lecture 3 p.34)

8 **D** Ranks can only be counted in an echelon form. Here $R_3 + 3R_2$ and $R_4 + R_2$ give $(0\,0\,0\,0 \mid 1)$ twice, so $\operatorname{rank} A = 2$, $\operatorname{rank}[A \mid b] = 3$. Counting the 4 nonzero rows as given would be wrong. (Lecture 3 p.33; 2025 midterm p.3)

9 **D** The row says $0 = 8$, which is absurd (slide 50, D4). A zero row $[0 \cdots 0|0]$ would be the harmless redundant one. (Lecture 3 p.49–50)

10 **D** At most 3 pivots for 5 variable columns, so at least 2 free variables. Consistent + free variables = infinitely many. (Lecture 3 p.40)

11 **C** Each row gives one equation $\mathbf{A}_i \cdot \mathbf{c} = b_i$; all of them must hold, so $\mathbf{c}$ is in the intersection. 'At least one' would satisfy only some equations. (Lecture 4 p.5)

12 **B** Consistent precisely when $\mathbf{b} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$ for some scalars, i.e. $\mathbf{b}$ lies in the span of the columns. Being one of the columns is sufficient, not necessary. (Lecture 4 p.19)

13 **C** $R_2 \leftarrow R_2 - 3R_1$ gives $[0\ 1 \mid b_2 - 3b_1]$ (a pivot row, no condition); $R_3 \leftarrow R_3 - 2R_1$ gives $[0\ 0 \mid b_3 - 2b_1]$. Only the zero row gives a constraint: $-2b_1 + b_3 = 0$. (Lecture 4 p.21–24)

14 **D** Substitute: $4 + 0 - 4 + 0 = 0$ for $(1, 0, 2, 0)$. The others give 4, 2, 6, all nonzero, so those systems have no solution. (Lecture 4 p.21)

15 **A** $\operatorname{rank}(A) \leq \min(m, n) = \min(4, 6) = 4$: every pivot needs its own row and there are only 4 rows. (Lecture 4 p.34)

16 **D** Full row rank $r = m = 3$: by P1 ('in particular') it is consistent for every $\mathbf{b}$. It is not unique, since $n - r = 2$ free variables remain. (Lecture 4 p.37)

17 **B** $r \leq m = 3$, and free variables $= n - r = 5 - r \geq 2$. The exact number depends on the rank, so 'at least 2'; in particular infinitely many solutions. (Lecture 4 p.47, 49)

18 **A** $A\mathbf{0} = \mathbf{0}$ for every matrix (Note p.47). Rank, free variables and squareness matter only for *how many* solutions there are. (Lecture 4 p.47)

19 **B** Proposition P3: unique $\iff$ $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\iff$ $r = n$ (no free variables). $r = m$ is about existence for every $\mathbf{b}$. (Lecture 4 p.50)

20 **D** Uniqueness needs $r = n = 5$, but $r \leq m = 3$. So the system has either no solution or infinitely many. (Consistent for every $\mathbf{b}$ happens when $r = 3 = m$.) (Lecture 4 p.49–50)

21 **B** Size is always rows × columns: 2 rows, 3 columns. 3×2 swaps the order. (Lecture 5 p.3)

22 **B** Scalar multiplication multiplies **every** entry by 4: $(4B)_{ij} = 4b_{ij}$. Adding 4, or scaling only one row or column, is not scalar multiplication. (Lecture 5 p.12)

23 **B** $(2 \times 3)(3 \times 1) = 2 \times 1$: $AB = (1 - 3,\ 4 - 6)^T = (-2, -2)^T$. BA is $(3 \times \boxed{1})(\boxed{2} \times 3)$; inner sizes $1 \neq 2$, so undefined (Example 3). (Lecture 5 p.23)

24 **D** $AI_n = A = I_m A$ with A of size $m \times n$ (Prop. 1.2 (1)). On the right you need I_3 (3 = columns of A), on the left I_2 (2 = rows). AI_2 and $I_3 A$ fail the size check. (Lecture 5 p.26)

25 **A** A is 2×3, so Ax needs $x \in \mathbb{R}^3$ (number of columns) and gives a vector in $\mathbb{R}^2$ (number of rows): $\mu_A : \mathbb{R}^n \to \mathbb{R}^m$. (Lecture 5 p.34)

26 **B** $\mu_{AB}(c) = (AB)c = A(Bc) = \mu_A(\mu_B(c))$, so $\mu_{AB} = \mu_A \circ \mu_B$: first $\mu_B : \mathbb{R}^p \to \mathbb{R}^n$, then $\mu_A : \mathbb{R}^n \to \mathbb{R}^m$. $\mu_B \circ \mu_A$ is not even defined in general. (Lecture 5 p.35)

27 **A** Lemma L1: a right inverse needs $\operatorname{rank}(A) = m = 4$. A matrix with 2 columns has rank ≤ 2. (Lecture 6 p.16)

28 **B** Only regrouping happens (p.17: 'Using the associativity...'). Matrix multiplication is not commutative. (Lecture 6 p.17)

29 **C** The definition requires both products to be I_n. (For square matrices one implies the other, but that is a theorem, not the definition.) (Lecture 6 p.18)

30 **C** $ad - bc = 2 \cdot 2 - 4 \cdot 1 = 0$ (Example 10). The others give -2, 1, -2. (Lecture 6 p.39–40, 45)

Mock midterm: full solutions

QUESTION 1 Let $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, $\mathbf{v} = \begin{bmatrix} 1 \\ 3 \\ 0 \end{bmatrix}$, $\mathbf{w} = \begin{bmatrix} 0 \\ 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 0 \\ -1 \\ -2 \end{bmatrix}$.

(a) Determine whether $\mathbf{b} \in \operatorname{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$.

(b) If it does, find scalars $a, c, d \in \mathbb{R}$ such that $\mathbf{b} = a\mathbf{u} + c\mathbf{v} + d\mathbf{w}$.

Write $a\mathbf{u} + c\mathbf{v} + d\mathbf{w} = \mathbf{b}$ as a system: the augmented matrix is $[\mathbf{u}\ \mathbf{v}\ \mathbf{w} \mid \mathbf{b}]$. Eliminate; $\mathbf{b}$ is in the span iff there is no row $[0\ 0\ 0 \mid c \neq 0]$.

1. We want scalars with $a\mathbf{u} + c\mathbf{v} + d\mathbf{w} = \mathbf{b}$, i.e. the system $2a + c = 0,\ -a + 3c + 2d = -1,\ a - d = -2$, with augmented matrix $\left[\begin{array}{ccc|c} 2 & 1 & 0 & 0 \\ -1 & 3 & 2 & -1 \\ 1 & 0 & -1 & -2 \end{array}\right]$.
2. Step 1 (eliminate column 1): $R_2 \to R_2 + \frac{1}{2}R_1$, $R_3 \to R_3 - \frac{1}{2}R_1$: $\left[\begin{array}{ccc|c} 2 & 1 & 0 & 0 \\ 0 & \frac{7}{2} & 2 & -1 \\ 0 & -\frac{1}{2} & -1 & -2 \end{array}\right]$.
3. Step 2 (eliminate the second entry of row 3): $R_3 \to R_3 + \frac{1}{7}R_2$: $\left[\begin{array}{ccc|c} 2 & 1 & 0 & 0 \\ 0 & \frac{7}{2} & 2 & -1 \\ 0 & 0 & -\frac{5}{7} & -\frac{15}{7} \end{array}\right]$. Echelon form with no row $[0\ 0\ 0 \mid c \neq 0]$: the system is consistent.
4. From row 3: $-\frac{5}{7}d = -\frac{15}{7} \Rightarrow d = 3$.
5. From row 2: $\frac{7}{2}c + 2 \cdot 3 = -1 \Rightarrow \frac{7}{2}c = -7 \Rightarrow c = -2$.
6. From row 1: $2a + c = 0 \Rightarrow 2a - 2 = 0 \Rightarrow a = 1$.
7. Check: $\mathbf{u} - 2\mathbf{v} + 3\mathbf{w} = (2 - 2 + 0,\ -1 - 6 + 6,\ 1 - 0 - 3) = (0, -1, -2) = \mathbf{b}$ ✓.

ANSWER
(a) Yes, $\mathbf{b} \in \mathrm{Span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$. (b) $a = 1,\ c = -2,\ d = 3$: $\mathbf{b} = \mathbf{u} - 2\mathbf{v} + 3\mathbf{w}$.

QUESTION 2 For each linear system below, given by its augmented matrix, determine: i.) whether the system is homogeneous or inhomogeneous, ii.) the rank of the coefficient matrix A and of the augmented matrix $[A|\mathbf{b}]$, iii.) the pivot variables and free variables, iv.) the number of solutions if it exists.

(a) $\left[\begin{array}{cccc|c} 1 & -1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$ (b) $\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & -4 & 0 & -2 \end{array}\right]$ (c) $\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right]$

Homogeneous = zero last column. System (b) is not in echelon form: reduce before counting ranks.

1. **(a)** $n = 4$ unknowns, $m = 3$ rows. Already in echelon form.
2. i) homogeneous (last column all zero). ii) $\mathrm{rank}\, A = 2$, $\mathrm{rank}[A|\mathbf{b}] = 2$. iii) pivot variables x_1, x_3; free variables x_2, x_4. iv) Consistent and $r = 2 < n = 4$: **infinitely many solutions** (2 free variables).

3 **(b)** $n = 4$ unknowns, $m = 4$ rows. Not in echelon form yet, so reduce first:

$$\left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 2 & 0 & 1 \\ 0 & 0 & -4 & 0 & -2 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 + 4R_2]{R_3 \leftarrow R_3 - 2R_2} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 & 10 \end{array}\right] \xrightarrow{R_4 \leftarrow R_4 + 2R_3} \left[\begin{array}{cccc|c} 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & -5 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

4 i) inhomogeneous ($\mathbf{b} \neq \mathbf{0}$). ii) $\operatorname{rank} A = 2$, $\operatorname{rank}[A|\mathbf{b}] = 3$. iii) — (inconsistent; the echelon form of A has pivots in columns 1, 3). iv) A row $[0 \cdots 0 \,|\, c \neq 0]$ appears, ranks differ: **no solution**.

5 **(c)** $n = 3$ unknowns, $m = 4$ rows. Already in echelon form.

6 i) inhomogeneous ($\mathbf{b} \neq \mathbf{0}$). ii) $\operatorname{rank} A = 3$, $\operatorname{rank}[A|\mathbf{b}] = 3$. iii) pivot variables x_1, x_2, x_3; no free variables. iv) Consistent and $r = 3 = n$: **unique solution**: $x_3 = 2,\ x_2 = -2,\ x_1 = -3$.

ANSWER

(a) homogeneous, ranks 2 and 2, pivot x_1, x_3, free x_2, x_4, infinitely many solutions

(b) inhomogeneous, ranks 2 and 3, pivot/free: —, no solution

(c) inhomogeneous, ranks 3 and 3, pivot x_1, x_2, x_3, no free variable, exactly one solution (unique)

QUESTION 3 Consider the linear system $A\mathbf{x} = \mathbf{b}$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 3 & -1 \\ 2 & 3 & 0 \\ 1 & -2 & 1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{bmatrix} \in \mathbb{R}^4$

.

(a) Determine a constraint equation that the entries b_1, b_2, b_3, b_4 must satisfy in order for the system to be consistent.

(b) Let $\mathbf{b} = (2, 0, 2, 2)$. Determine whether $A\mathbf{x} = \mathbf{b}$ has a solution by using the constraint equation you found in part (a). If it does, find it.

Carry $b_1, \ldots, b_4$ through the elimination; the zero row of the A-part gives the constraint.

1. Form the augmented matrix with $b_1, \ldots, b_4$ as letters and row-reduce; every operation also acts on the last column:

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 1 & 3 & -1 & b_2 \\ 2 & 3 & 0 & b_3 \\ 1 & -2 & 1 & b_4 \end{array}\right] \xrightarrow[R_4 \leftarrow R_4 - R_1]{\substack{R_2 \leftarrow R_2 - R_1 \\ R_3 \leftarrow R_3 - 2R_1}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 2 & -2 & -b_1 + b_2 \\ 0 & 1 & -2 & -2b_1 + b_3 \\ 0 & -3 & 0 & -b_1 + b_4 \end{array}\right]$$

$$\xrightarrow[R_4 \leftarrow R_4 + 3R_2]{\substack{R_2 \leftrightarrow R_3 \\ R_3 \leftarrow R_3 - 2R_2}} \left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 1 & -2 & -2b_1 + b_3 \\ 0 & 0 & 2 & 3b_1 + b_2 - 2b_3 \\ 0 & 0 & -6 & -7b_1 + 3b_3 + b_4 \end{array}\right]$$

$$\xrightarrow{R_4 \leftarrow R_4 + 3R_3} \left[\begin{array}{ccc|c} 1 & 1 & 1 & b_1 \\ 0 & 1 & -2 & -2b_1 + b_3 \\ 0 & 0 & 2 & 3b_1 + b_2 - 2b_3 \\ 0 & 0 & 0 & 2b_1 + 3b_2 - 3b_3 + b_4 \end{array}\right]$$

2. The A-part of the last row is zero, so that row reads $0 = 2b_1 + 3b_2 - 3b_3 + b_4$. The other three rows have pivots ($\operatorname{rank} A = 3$, $m - r = 1$ constraint).
3. Constraint equation: $2b_1 + 3b_2 - 3b_3 + b_4 = 0$. Self-check with the columns of A (each column is a consistent right-hand side): $(1, 1, 2, 1)$: $2 + 3 - 6 + 1 = 0$; $(1, 3, 3, -2)$: $2 + 9 - 9 - 2 = 0$; $(1, -1, 0, 1)$: $2 - 3 + 0 + 1 = 0$ ✓.
4. (b) Substitute $\mathbf{b} = (2, 0, 2, 2)$ into the constraint: $4 + 0 - 6 + 2 = 0$. The constraint holds, so the system is **consistent**.
5. Back-substitution in the echelon form (with these b_i) gives $\mathbf{x} = (1, 0, 1)$. Check: $A\mathbf{x} = (2, 0, 2, 2) = \mathbf{b}$ ✓. It is unique since $r = 3 = n$.

ANSWER
(a) $2b_1 + 3b_2 - 3b_3 + b_4 = 0$ (any nonzero multiple is equally correct). (b) Consistent, the constraint gives 0; the solution is $\mathbf{x} = (1, 0, 1)$.

QUESTION 4 Let A be an $m \times n$ matrix with $n \neq m$, and assume that $\operatorname{rank}(A) = n$. Consider the linear transformation $\mu_A : \mathbb{R}^n \to \mathbb{R}^m$ defined by $\mu_A(x) = Ax$. Prove or disprove that μ_A is surjective.

Compare with Midterm Q4(b), where the rank equals the number of *rows*. Here it equals the number of columns. Which rows of the echelon form are zero?

1. **False**, in general. Since $\operatorname{rank} A \leq \min(m, n)$ and $\operatorname{rank} A = n \neq m$, we have $n < m$.
2. An echelon form of A has only n pivots for $m > n$ rows, so its last $m - n \geq 1$ rows are zero. For a suitable b the augmented matrix gets a row $(0 \cdots 0 \mid c)$ with $c \neq 0$, and then $Ax = b$ is inconsistent.

3. Concrete counterexample (Midterm Q3): $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \\ 1 & 2 & 3 \end{bmatrix}$, $m = 4$, $n = 3$, $\operatorname{rank} A = 3 = n$.

4. Row reducing $[A \mid b]$ ($R_2 - R_1,\ R_3 - R_1,\ R_4 - R_1$, then $R_2 \to R_2/(-2)$, $R_4 - R_2$, $R_4 + R_3$) gives the last row $\left(0\ 0\ 0 \mid b_4 + b_3 - \frac{5}{2}b_1 + \frac{1}{2}b_2\right)$. So $Ax = b$ is consistent iff $5b_1 - b_2 - 2b_3 - 2b_4 = 0$.

5. Take $b = (1, 0, 2, 3)^T \in \mathbb{R}^4$: $5 \cdot 1 - 0 - 2 \cdot 2 - 2 \cdot 3 = -5 \neq 0$. So $\operatorname{rank}[A \mid b] = 4 \neq 3 = \operatorname{rank} A$, and there is no $x \in \mathbb{R}^3$ with $\mu_A(x) = b$.

6. Hence μ_A is not surjective. (It is one-to-one, though: rank $= n$ means no free variables.)

ANSWER

Disproved: if $\operatorname{rank} A = n < m$, μ_A is **not** surjective. Counterexample: the 4×3 matrix above with $b = (1, 0, 2, 3)^T \notin$ image.

QUESTION 5 Consider $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$. Use Gaussian elimination to find A^{-1}, if it exists. If A is invertible, verify your answer by checking $AA^{-1} = I_3$ and $A^{-1}A = I_3$.

Row-reduce $[\,A \mid I_3\,]$ to $[\,I_3 \mid A^{-1}\,]$: clear column 1 with R_1, make pivots 1, then clear upward.

1. Form $[\,A \mid I_3\,]$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & -1 & 3 & 0 & 1 & 0 \\ 4 & 1 & 8 & 0 & 0 & 1 \end{array}\right]$$

2. $R_2 \to R_2 - 2R_1,\ R_3 \to R_3 - 4R_1$ (clear column 1 below the pivot):

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & -1 & -1 & -2 & 1 & 0 \\ 0 & 1 & 0 & -4 & 0 & 1 \end{array}\right]$$

3. $R_2 \to -R_2$ (make the second pivot 1):

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 1 & 0 & -4 & 0 & 1 \end{array}\right]$$

4. $R_3 \to R_3 - R_2$ (clear column 2 below the pivot):

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & -1 & -6 & 1 & 1 \end{array}\right]$$

5. $R_3 \to -R_3$ (make the third pivot 1):

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & 2 & -1 & 0 \\ 0 & 0 & 1 & 6 & -1 & -1 \end{array}\right]$$

6 $R_2 \to R_2 - R_3,\ R_1 \to R_1 - 2R_3$ (clear column 3 above the pivot; column 2 of row 1 is already 0, so the left block is I_3):

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -11 & 2 & 2 \\ 0 & 1 & 0 & -4 & 0 & 1 \\ 0 & 0 & 1 & 6 & -1 & -1 \end{array}\right]$$

7 The left block is I_3, so A is invertible (Theorem T2) and the right block is A^{-1}.

8 Verify: row 1 of A times column 1 of A^{-1}: $1(-11) + 0(-4) + 2(6) = 1$; row 1 times column 2: $2 + 0 - 2 = 0$; row 3 times column 3: $4 \cdot 2 + 1 \cdot 1 + 8(-1) = 1$; all nine entries give $AA^{-1} = I_3$. Likewise $A^{-1}A = I_3$, e.g. row 1 of A^{-1} times column 1 of A: $-11 + 4 + 8 = 1$.

ANSWER

$A^{-1} = \begin{bmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{bmatrix}$, and $AA^{-1} = A^{-1}A = I_3$.